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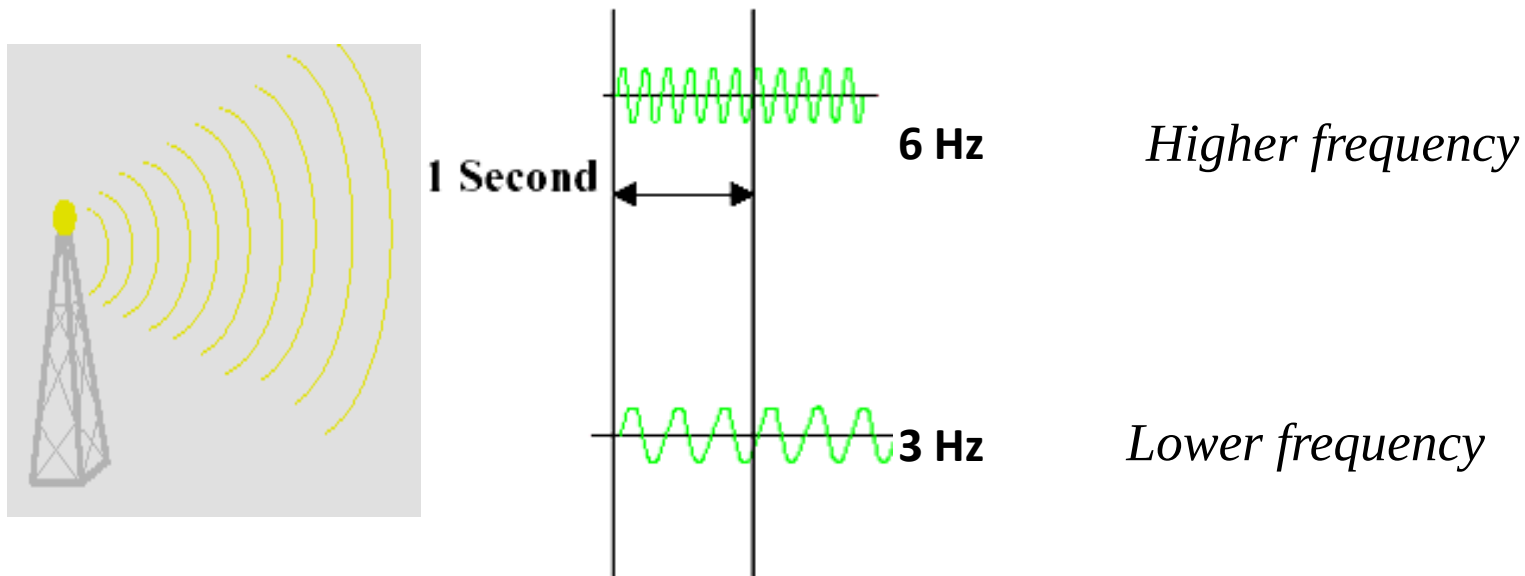
Learning Objectives of This Lecture

1. Understand the background, process, and new applications of the **frequency-domain analysis** method.
2. Proficiently master the **Fourier series analysis** methods for **periodic signals** in trigonometric and exponential forms.
3. Proficiently use the relationship between waveform symmetry and harmonic characteristics to obtain Fourier series.
4. Be familiar with the characteristics of Fourier series of typical periodic signals such as **rectangular waves**.

3.1 Introduction



- **Time-domain analysis:** The independent variable of the signal or system model is time (Chapter 2).
- **Frequency-domain analysis:** The independent variable is frequency (Chapter 3).
- **Complex frequency-domain analysis:** The independent variable is complex frequency-its real part is frequency (Chapter 4).
- **Frequency** is the number of periodic changes completed per unit time (unit: cycles per second or **Hertz Hz**), which describes the speed of signal change.
- The frequency of a sinusoidal signal can be directly observed from the time domain.



3.1 Introduction



- In 1807, *Joseph Fourier*, the French mathematician and physicist, proposed the principle of **expanding periodic functions into a series of sine and cosine waves**, while studying heat conduction theory, laying the foundation for frequency-domain analysis.
- *Siméon Poisson*, *Carl Gauss* and other scientists applied this achievement to electrical engineering.
- In the late 19th century, capacitors for practical engineering applications were invented.
- Since the 20th century, the solution to a series of specific problems such as resonant circuits, filters, and sinusoidal oscillators opened up a broad prospect for the further application of sinusoidal functions and Fourier analysis.
- The **Fourier Transform** has many advantages in the theoretical research and engineering application of electrical, communication and control systems.

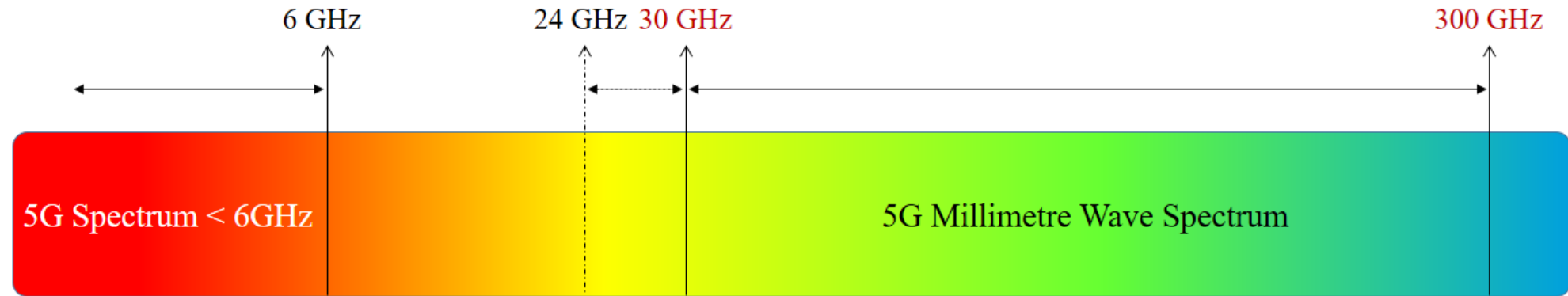


Joseph Fourier
(1768-1830)

3.1 Introduction



Fourier analysis serves as the basis of spectrum allocation.



Frequencies used in radio communication											
3–30 Hz	30–300 Hz	300–3000 Hz	3–30 kHz	30–300 kHz	300 kHz – 3 MHz	3–30 MHz	30–300 MHz	300 MHz – 3 GHz	3–30 GHz	30–300 GHz	300 GHz – 3 THz
10^5 – 10^4 km	10^4 – 10^3 km	10^3 –100 km	100–10 km	10–1 km	1 km – 100 m	100–10 m	10–1 m	1 m – 10 cm	10–1 cm	1 cm – 1 mm	1 mm – 0.1 mm
ELF	SLF	ULF	VLF	LF	MF	HF	VHF	UHF	SHF	EHF	THF

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- Frequency-domain analysis reveals the inherent frequency characteristics of signals and the close relationship between the time characteristics and frequency characteristics of signals, thus deriving important concepts such as signal spectrum, bandwidth, filtering, modulation, and frequency division multiplexing.
- Frequency-domain analysis is carried out through Fourier transform (Fourier analysis). The signal is decomposed orthogonally, that is, decomposed into a combination of trigonometric functions or complex exponential functions.

Fourier Analysis

Fourier series expansion: **Periodic signals** can be expressed as a **weighted sum** of **sinusoidal signals with harmonic relationships**.

Fourier transform: **Aperiodic signals** can be expressed as a **weighted integral** of **sinusoidal signals over all frequency components** from 0 to infinitely high frequencies.

Any periodic signal can be decomposed into a linear superposition of sinusoidal functions. This statement is ().

- ☐ A True
- ☒ B False

提交

3.2 Fourier Series Analysis of Periodic Signals

- In 1807, *Fourier* proposed that any periodic signal could be expanded into a superposition of sinusoidal signals with harmonic relationships.
- *Lagrange* disapproved the claim because there were breakpoints in actual signals.
- In 1822, *Fourier* published *Théorie analytique de la chaleur* (*The Analytical Theory of Heat*).
- In 1829, *Dirichlet* proved that periodic signals could be expanded into Fourier series only when certain conditions are met.
- **Dirichlet conditions (sufficient but not necessary conditions for the existence of Fourier series):**
 - (1) Within one period, the signal is **continuous** or has only **a finite number of first-kind discontinuities** (the function has finite left and right limits at the discontinuity, e.g., $x(t)=|\sin(t)|/\sin(t)$ at $t=0$).
 - (2) Within one period, the signal has **a finite number of maximum and minimum values**.
 - (3) Within one period, the signal is **absolutely integrable**, i.e., $\int_0^T |f(t)| dt < \infty$.
- **Signals in engineering all satisfy Dirichlet conditions.**

3.2 Fourier Series Analysis of Periodic Signals

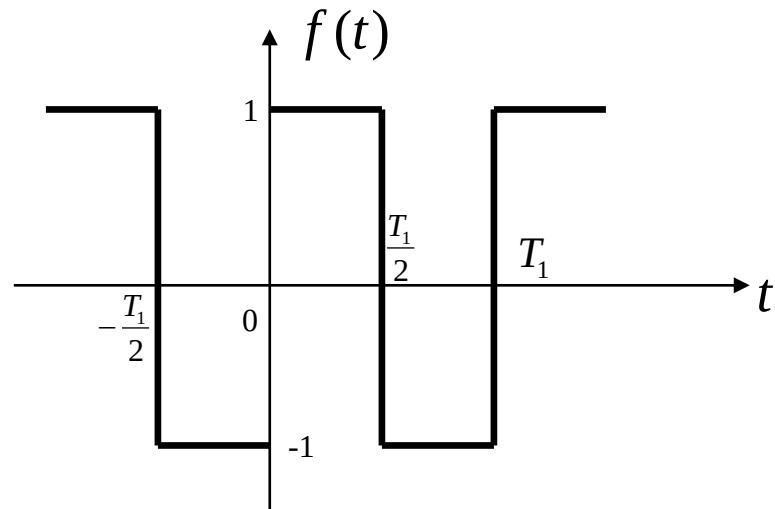


- Any periodic function that satisfies the Dirichlet conditions can be expanded into an infinite series of orthogonal functions.
- If the orthogonal function set is a **trigonometric** function set or a **complex exponential** function set, the series expanded by the periodic function is called "Fourier series".
- Why choose trigonometric functions (**sinusoidal waves**) as the basic signals of Fourier series?
 - **Sinusoidal waves have fidelity**: After a sinusoidal signal passes through an LTI system, the output is still a sinusoidal signal—only the amplitude and phase may change, while the frequency and waveform remain unchanged.
 - **The integral and derivative of a trigonometric (or exponential) function are still trigonometric (or exponential) functions.**

3.2 Fourier Series Analysis of Periodic Signals

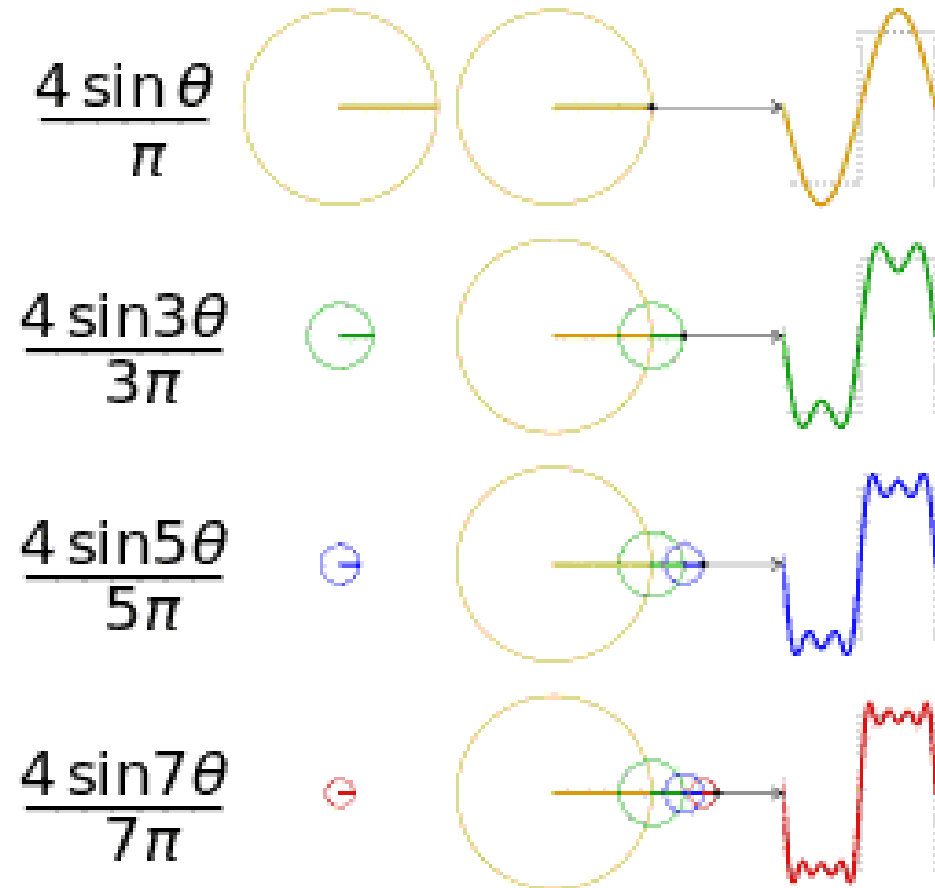


Generation of Periodic Rectangular Signals



How many sinusoidal waves need to be superimposed to form a rectangular signal?

Answer: Infinite.



3.2 Fourier Series Analysis of Periodic Signals



3.2.1 Fourier Series in Trigonometric Form

Let $f(t)$ be a periodic signal with period T_1 .

Its **fundamental angular frequency** is $\omega_1 = 2\pi f_1 = \frac{2\pi}{T_1}$.

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t) \quad (1)$$

DC component: $a_0 = \frac{1}{T_1} \int_{t_0}^{t_0+T_1} f(t) dt$

Amplitude of cosine component : $a_n = \frac{2}{T_1} \int_{t_0}^{t_0+T_1} f(t) \cos n\omega_1 t dt$

Amplitude of sine component: $b_n = \frac{2}{T_1} \int_{t_0}^{t_0+T_1} f(t) \sin n\omega_1 t dt$

The integral limits in the above formulas are generally set as: $0 \sim T_1$ or $-\frac{T_1}{2} \sim \frac{T_1}{2}$

3.2 Fourier Series Analysis of Periodic Signals



The Fourier series in trigonometric form can also be expressed as:

$$f(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_1 t + \varphi_n) \quad (2)$$

where $c_0 = a_0$ $c_n^2 = a_n^2 + b_n^2$ $\varphi_n = \arctan\left(-\frac{b_n}{a_n}\right)$

Proof:

$$\begin{aligned} f(t) &= c_0 + \sum_{n=1}^{\infty} c_n \left[\frac{a_n}{c_n} \cos(n\omega_1 t) + \frac{b_n}{c_n} \sin(n\omega_1 t) \right] \\ &= c_0 + \sum_{n=1}^{\infty} c_n [\cos(n\omega_1 t) \cos(-\varphi_n) + \sin(n\omega_1 t) \sin(-\varphi_n)] \\ &= c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_1 t + \varphi_n) \end{aligned}$$

3.2 Fourier Series Analysis of Periodic Signals



3.2.2 Fourier Series in Exponential Form

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t) \quad (1)$$

Based on Euler's formula,

$$\cos(n\omega_1 t) = \frac{1}{2}(e^{jn\omega_1 t} + e^{-jn\omega_1 t}), \quad \sin(n\omega_1 t) = \frac{1}{2j}(e^{jn\omega_1 t} - e^{-jn\omega_1 t})$$

Substitute into (1),

$$f(t) = a_0 + \sum_{n=1}^{\infty} \left(\frac{a_n - jb_n}{2} e^{jn\omega_1 t} + \frac{a_n + jb_n}{2} e^{-jn\omega_1 t} \right)$$

$$\text{Let } F(n\omega_1) = \frac{1}{2}(a_n - jb_n)$$

$$F(-n\omega_1) = \frac{1}{2}(a_n + jb_n)$$

$$f(t) = a_0 + \sum_{n=1}^{\infty} F(n\omega_1) e^{jn\omega_1 t} + \sum_{n=1}^{\infty} F(-n\omega_1) e^{-jn\omega_1 t} = \sum_{n=-\infty}^{\infty} F(n\omega_1) e^{jn\omega_1 t}$$

3.2 Fourier Series Analysis of Periodic Signals



$$f(t) = \sum_{n=-\infty}^{\infty} F(n\omega_1) e^{jn\omega_1 t}$$

Exponential Form:

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t} \quad (3)$$

where $F_n = \frac{1}{T_1} \int_{t_0}^{t_0+T_1} f(t) e^{-jn\omega_1 t} dt$ -----Complex amplitude

$$F_0 = a_0 = c_0$$

$$F_n = |F_n| e^{j\varphi_n} = \frac{1}{2} (a_n - jb_n)$$

$$|F_n| = \frac{1}{2} \sqrt{a_n^2 + b_n^2} = \frac{1}{2} c_n \quad \varphi_n = \arctan \left(-\frac{b_n}{a_n} \right)$$

3.2 Fourier Series Analysis of Periodic Signals



3.2.3 Spectrum of Periodic Signals and Its Characteristics

1. Spectrum of Periodic Signals

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t) \quad (1)$$

$$f(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_1 t + \varphi_n) \quad (2)$$

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t} \quad (3)$$

To clearly indicate which frequency components a signal contains and the proportion of each component, a **spectrum diagram** can be drawn for intuitive representation.

If frequency is taken as the horizontal axis and amplitude/phase as the vertical axis, the relative magnitude and phase of each frequency component can be observed directly. Such diagrams are called the **amplitude spectrum** and **phase spectrum** of the signal in trigonometric form.

3.2 Fourier Series Analysis of Periodic Signals



Spectrum Diagram of Periodic Rectangular Signals



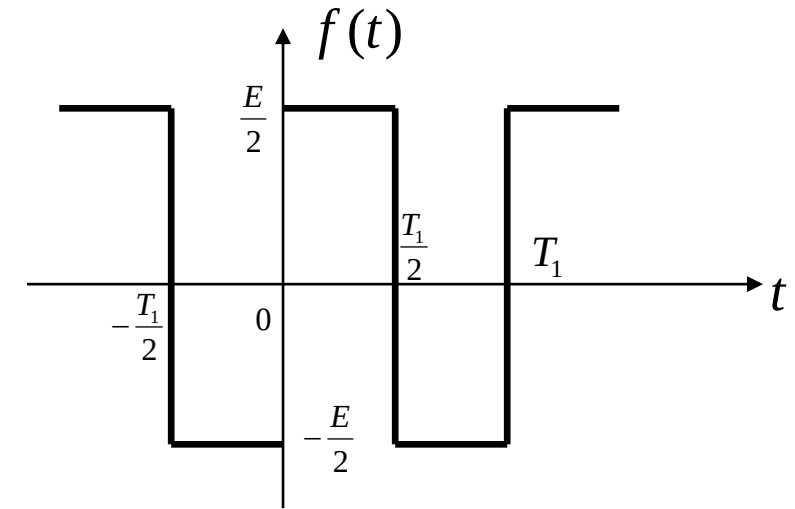
https://www.bilibili.com/video/BV1Gz411z7Sx?p=1&share_medium=iphone&share_plat=ios&share_session_id=91DCCAAB-FE86-4054-B1D8-C56A685066FE&share_source=WEIXIN&share_tag=s_i×tamp=1648135144&unique_k=CIVMRwB&share_times=1

3.2 Fourier Series Analysis of Periodic Signals

Example 3-1: Find the Fourier series in trigonometric and exponential forms of the periodic rectangular signal shown in the figure, and draw their respective spectrum diagrams.

Solution: The expression of $f(t)$ within one period is:

$$f(t) = \begin{cases} \frac{E}{2} & 0 \leq t < \frac{T_1}{2} \\ -\frac{E}{2} & -\frac{T_1}{2} \leq t < 0 \end{cases} \quad (\text{Odd function})$$



$$a_0 = \frac{1}{T_1} \int_{-T_1/2}^{T_1/2} f(t) dt = 0 \quad a_n = \frac{2}{T_1} \int_{-T_1/2}^{T_1/2} f(t) \cos n\omega_1 t dt = 0$$

$$b_n = \frac{2}{T_1} \int_{-T_1/2}^{T_1/2} f(t) \sin n\omega_1 t dt = \frac{E}{n\pi} [1 - \cos(n\pi)] = \begin{cases} \frac{2E}{n\pi} & n = 1, 3, 5 \dots \\ 0 & n = 2, 4, 6 \dots \end{cases}$$

3.2 Fourier Series Analysis of Periodic Signals



$$c_0 = 0$$

$$c_n = b_n = \begin{cases} \frac{2E}{n\pi} & n = 1, 3, 5 \dots \\ 0 & n = 2, 4, 6 \dots \end{cases}$$

$$\varphi_n = \arctan\left(-\frac{b_n}{a_n}\right) = -\frac{\pi}{2} \quad (n = 1, 3, 5 \dots)$$

So,
$$f(t) = \frac{2E}{\pi} \sum_{n=1,3,5\dots}^{\infty} \frac{1}{n} \sin n\omega_1 t$$

$$= \frac{2E}{\pi} \left(\sin \omega_1 t + \frac{1}{3} \sin 3\omega_1 t + \frac{1}{5} \sin 5\omega_1 t + \dots \right)$$

or
$$f(t) = \frac{2E}{\pi} \sum_{n=1,3,5\dots}^{\infty} \frac{1}{n} \cos\left(n\omega_1 t - \frac{\pi}{2}\right)$$

3.2 Fourier Series Analysis of Periodic Signals



$$F_n = \frac{1}{2}(a_n - jb_n) = -j\frac{b_n}{2} = \begin{cases} -\frac{jE}{n\pi} & n = \pm 1, \pm 3, \pm 5 \dots \\ 0 & n = \pm 2, \pm 4, \pm 6 \dots \end{cases}$$

$$f(t) = \left(\frac{jE}{\pi} e^{-j\omega_1 t} - \frac{jE}{\pi} e^{j\omega_1 t} \right) + \left(\frac{jE}{3\pi} e^{-j3\omega_1 t} - \frac{jE}{3\pi} e^{j3\omega_1 t} \right) + \dots$$

$$|F_n| = \frac{E}{|n|\pi} \quad (n = \pm 1, \pm 3, \pm 5 \dots)$$

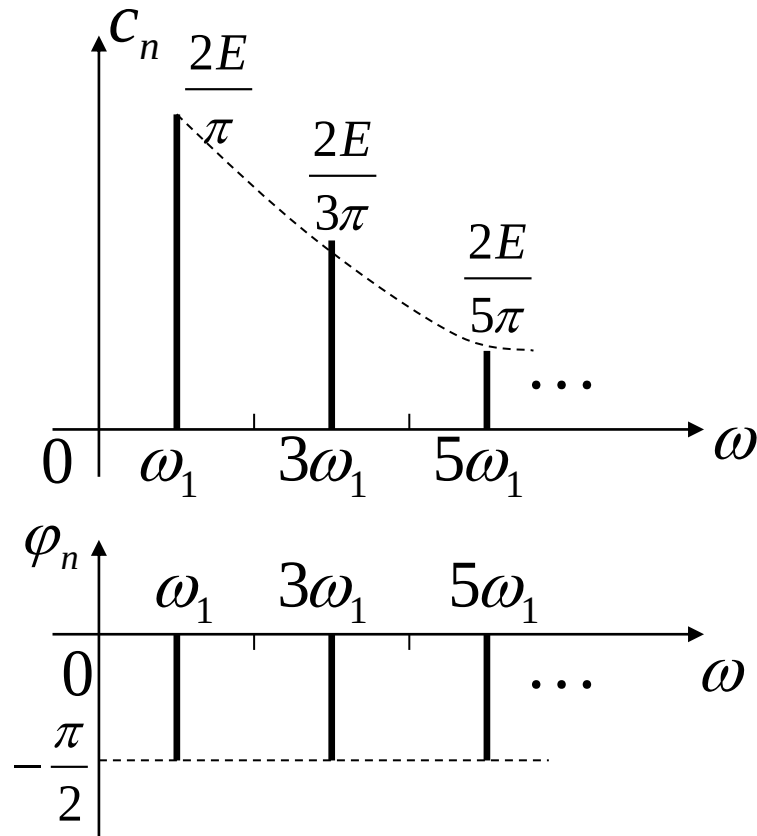
$$\varphi_n = \begin{cases} -\frac{\pi}{2} & (n = 1, 3, 5 \dots) \\ \frac{\pi}{2} & (n = -1, -3, -5 \dots) \end{cases}$$

3.2 Fourier Series Analysis of Periodic Signals



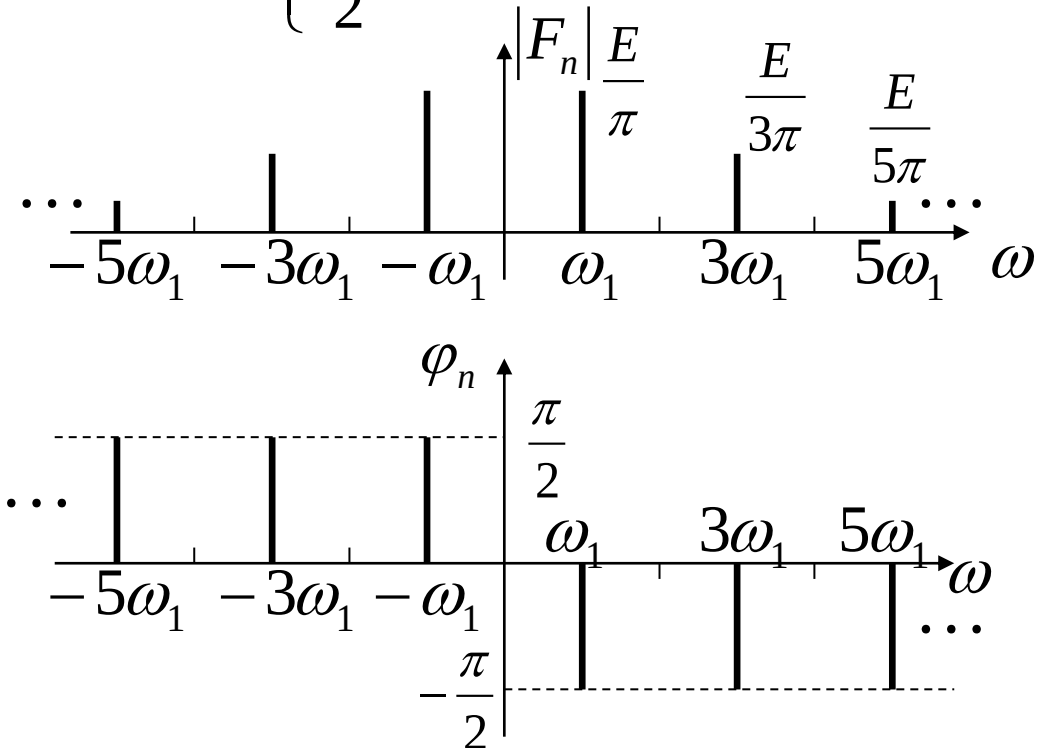
$$c_n = \begin{cases} \frac{2E}{n\pi} & n = 1, 3, 5 \dots \\ 0 & n = 2, 4, 6 \dots \end{cases}$$

$$\varphi_n = -\frac{\pi}{2} \quad (n = 1, 3, 5 \dots)$$



$$|F_n| = \frac{E}{|n|\pi} \quad (n = \pm 1, \pm 3, \pm 5 \dots)$$

$$\varphi_n = \begin{cases} -\frac{\pi}{2} & (n = 1, 3, 5 \dots) \\ \frac{\pi}{2} & (n = -1, -3, -5 \dots) \end{cases}$$



2. Characteristics of Periodic Signal Spectrum

- (1) **Discreteness:** The spectrum is discrete (not continuous), which is called a discrete spectrum.
- (2) **Harmonicity:** Spectral lines appear at integer multiples of the fundamental frequency ω_1 .
- (3) **Convergence:** The amplitude of spectral lines in the amplitude spectrum gradually decays to zero as n increases ($n \rightarrow \infty$).

3.2 Fourier Series Analysis of Periodic Signals

3.2.4 Relationship Between Waveform Symmetry and Harmonic Characteristics

When a signal $f(t)$ is expanded into a Fourier series, if $f(t)$ is a real function and its waveform satisfies a certain **symmetry**, some terms in the Fourier series disappear, and the expressions of the remaining coefficients become simpler.

There are two types of waveform symmetry:

- (1) **symmetry about the entire period;**
- (2) **symmetry about half the period.**

3.2 Fourier Series Analysis of Periodic Signals

1. Even Function

$$f(t) = f(-t)$$

$$b_n = \frac{2}{T_1} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) \sin n\omega_1 t dt = 0$$

$$a_0 = \frac{1}{T_1} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) dt = \frac{2}{T_1} \int_0^{\frac{T_1}{2}} f(t) dt$$

$$a_n = \frac{2}{T_1} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) \cos n\omega_1 t dt = \frac{4}{T_1} \int_0^{\frac{T_1}{2}} f(t) \cos n\omega_1 t dt$$

The Fourier series of an even function **does not contain sine components**, but may contain DC and cosine components.

2. Odd Function

$$f(t) = -f(-t)$$

$$a_0 = \frac{1}{T_1} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) dt = 0$$

$$a_n = \frac{2}{T_1} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) \cos n\omega_1 t dt = 0$$

$$b_n = \frac{2}{T_1} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) \sin n\omega_1 t dt = \frac{4}{T_1} \int_0^{\frac{T_1}{2}} f(t) \sin n\omega_1 t dt$$

The Fourier series of an odd function **does not contain DC and cosine components**, but only sine components.

After adding a DC component to an odd function, the Fourier series still contains sine components. This statement is ().

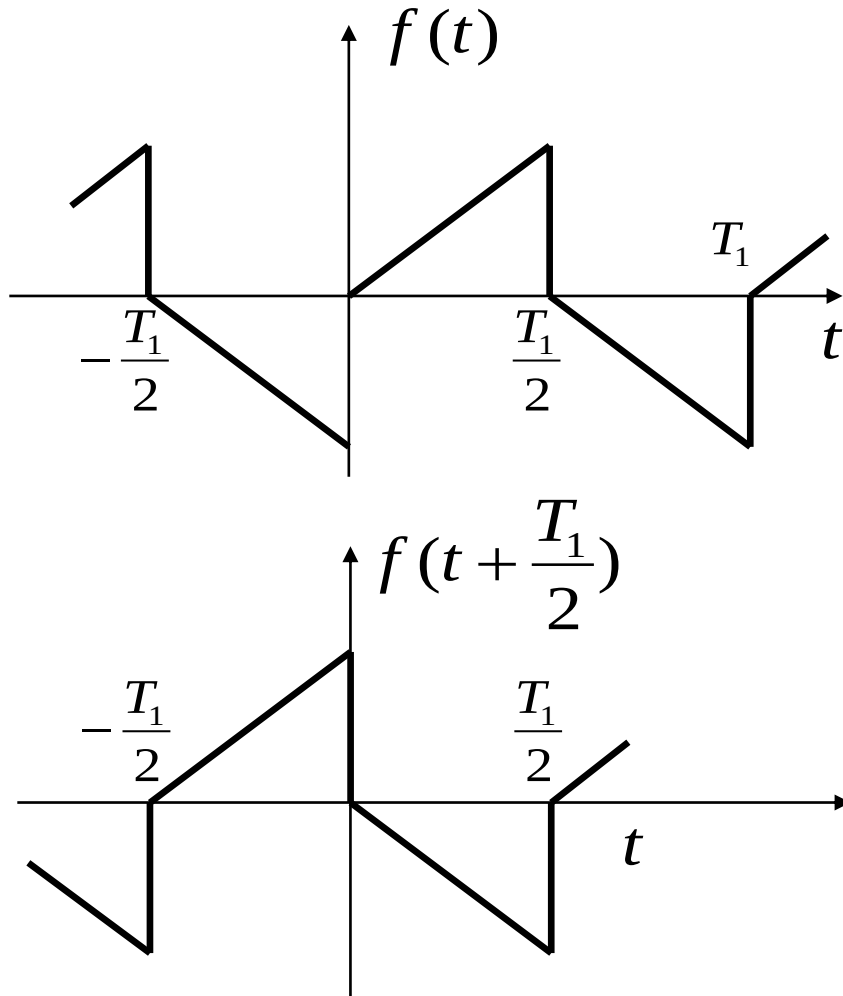
☒ A True

☐ B False

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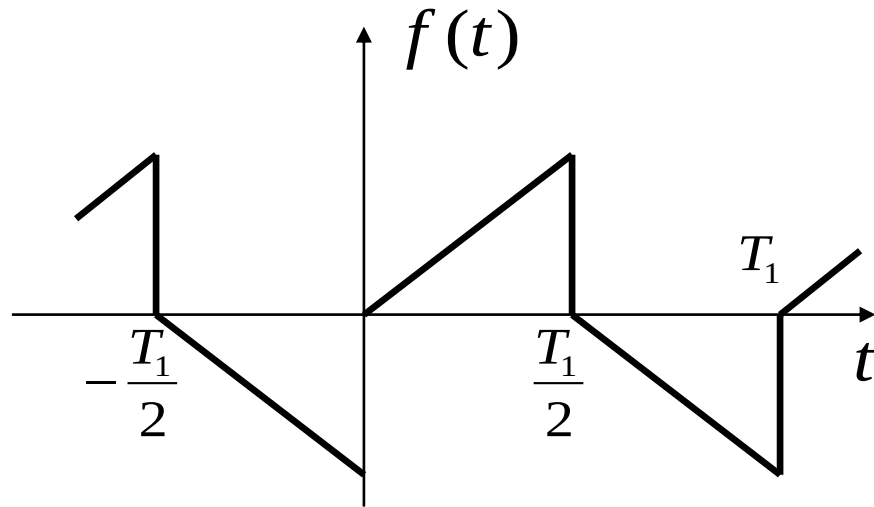
3. Odd Harmonic Function (Half-Wave Odd Symmetry)

$$f\left(t \pm \frac{T_1}{2}\right) = -f(t)$$



$$\begin{aligned} a_n &= \frac{2}{T_1} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) \cos(n\omega_1 t) dt \\ &= \frac{2}{T_1} \left[\int_0^{\frac{T_1}{2}} f(t) \cos(n\omega_1 t) dt + \int_{-\frac{T_1}{2}}^0 f(t) \cos(n\omega_1 t) dt \right] \\ &= \frac{2}{T_1} \left[\int_0^{\frac{T_1}{2}} f(t) \cos(n\omega_1 t) dt - \int_{-\frac{T_1}{2}}^0 f\left(t + \frac{T_1}{2}\right) \cos(n\omega_1 t) dt \right] \\ &= \frac{2}{T_1} \left[\int_0^{\frac{T_1}{2}} f(t) \cos(n\omega_1 t) dt - \int_0^{\frac{T_1}{2}} f(\tau) \cos[n\omega_1(\tau - T_1/2)] d\tau \right] \\ &= \frac{2}{T_1} \left[\int_0^{\frac{T_1}{2}} f(t) \cos(n\omega_1 t) dt - \int_0^{\frac{T_1}{2}} f(t) \cos(n\omega_1 t - n\pi) dt \right] \\ &= \begin{cases} 0 & (n = 0, 2, 4, 6 \dots) \\ \frac{4}{T_1} \int_0^{\frac{T_1}{2}} f(t) \cos n\omega_1 t dt & (n = 1, 3, 5 \dots) \end{cases} \end{aligned}$$

3.2 Fourier Series Analysis of Periodic Signals



$$a_0 = 0$$

$$a_n = \begin{cases} 0 & (n = 2, 4, 6 \dots) \\ \frac{4}{T_1} \int_0^{\frac{T_1}{2}} f(t) \cos n\omega_1 t dt & (n = 1, 3, 5 \dots) \end{cases}$$

$$b_n = \begin{cases} 0 & (n = 2, 4, 6 \dots) \\ \frac{4}{T_1} \int_0^{\frac{T_1}{2}} f(t) \sin n\omega_1 t dt & (n = 1, 3, 5 \dots) \end{cases}$$

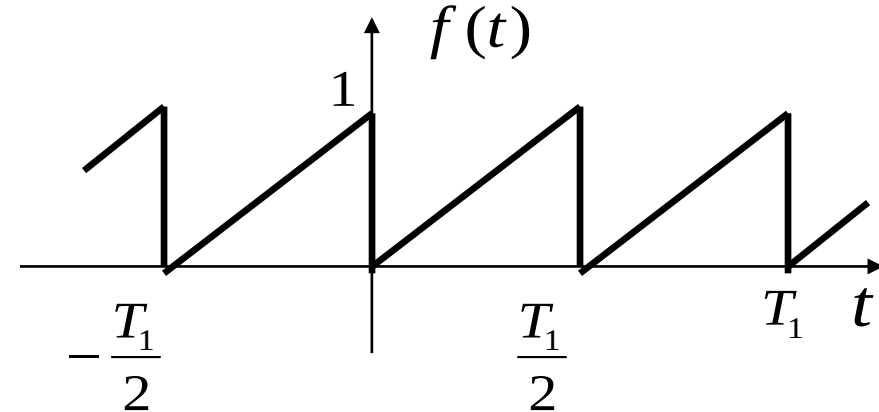
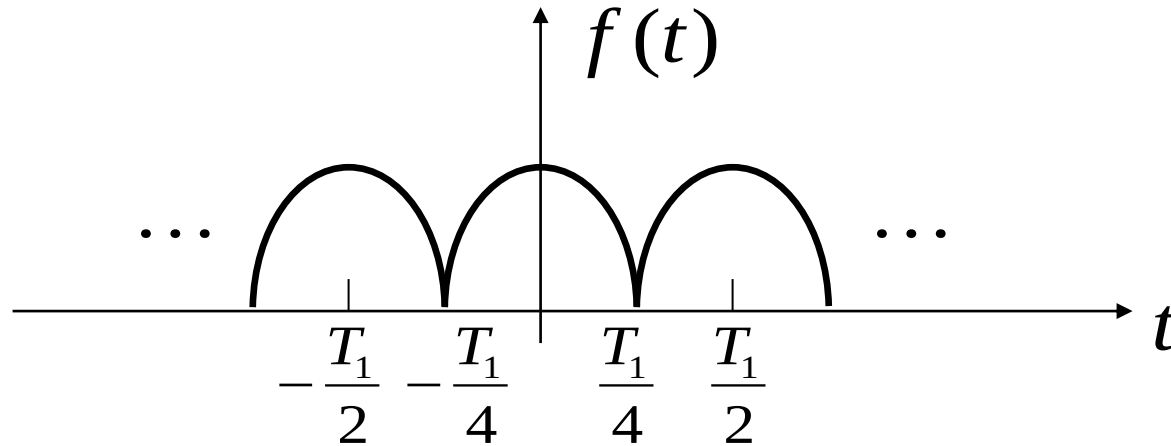
The Fourier series of a half-wave odd symmetric function contains only sine and cosine components of the fundamental wave and odd harmonics, and **does not contain DC components and even harmonic components**—thus it is called an **odd harmonic function**.

3.2 Fourier Series Analysis of Periodic Signals



4. Even Harmonic Function (Half-Wave Even Symmetry)

$$f(t \pm \frac{T_1}{2}) = f(t)$$



The Fourier series of a half-wave even symmetric function contains only sine and cosine components of even harmonics (as well as DC components), and **does not contain odd harmonic components**—thus it is called an **even harmonic function**.

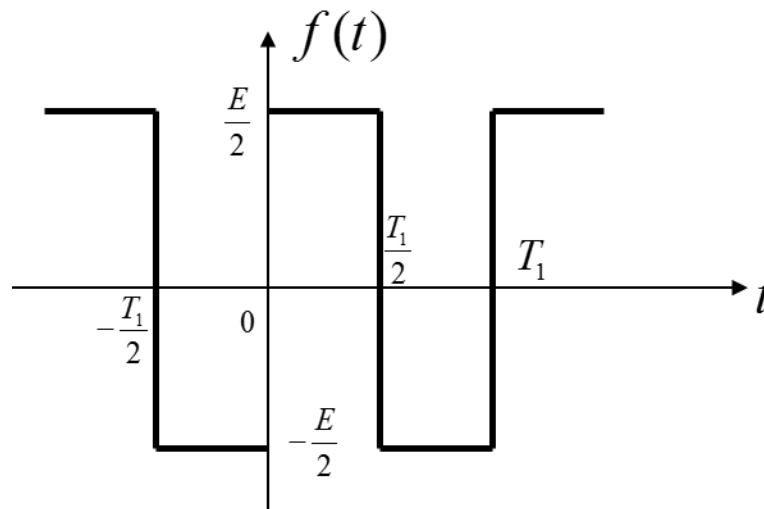
The zero components of the Fourier series in trigonometric form for the periodic rectangular signal shown in the figure are ().

☒ A $a_n = 0 \quad (n = 0, 2, 4 \dots)$

☒ B $a_n = 0 \quad (n = 1, 3, 5 \dots)$

☐ C $b_n = 0 \quad (n = 1, 3, 5 \dots)$

☒ D $b_n = 0 \quad (n = 2, 4, 6 \dots)$



提交

3.2 Fourier Series Analysis of Periodic Signals



Which components of the Fourier series in trigonometric form for the periodic rectangular signal shown in the figure are zero?

This is an **odd function**, so the DC component and cosine components are zero.

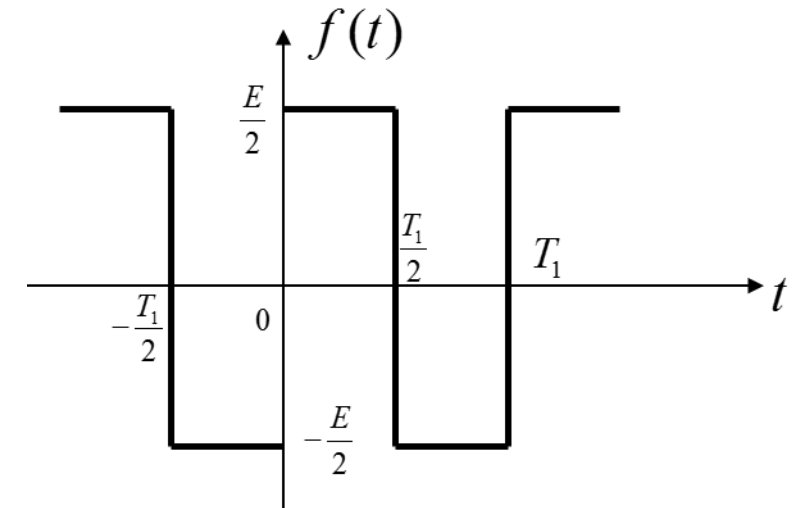
$$a_n = 0 \quad (n = 0, 2, 4 \dots)$$

$$a_n = 0 \quad (n = 1, 3, 5 \dots)$$

It is also an **odd harmonic function** (half-wave odd symmetry), so the DC and even harmonic components are zero.

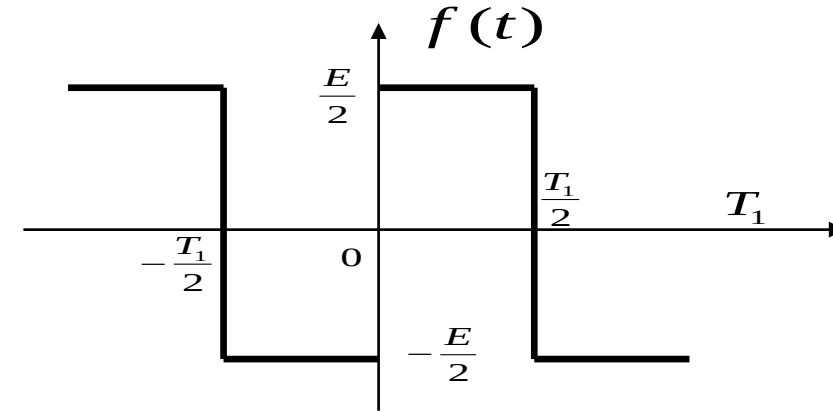
$$a_n = 0 \quad (n = 0, 2, 4 \dots)$$

$$b_n = 0 \quad (n = 2, 4, 6 \dots)$$



3.2 Fourier Series Analysis of Periodic Signals

3.2.5 Gibbs Phenomenon



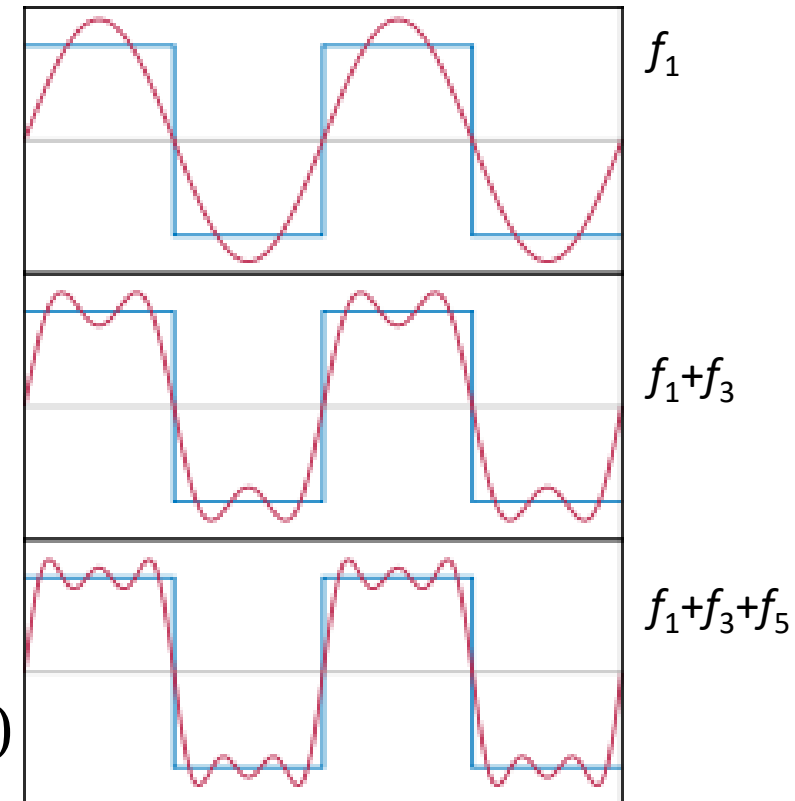
$n=1:$ $f(t) \approx \frac{2E}{\pi} \sin \omega_1 t$

$n=3:$

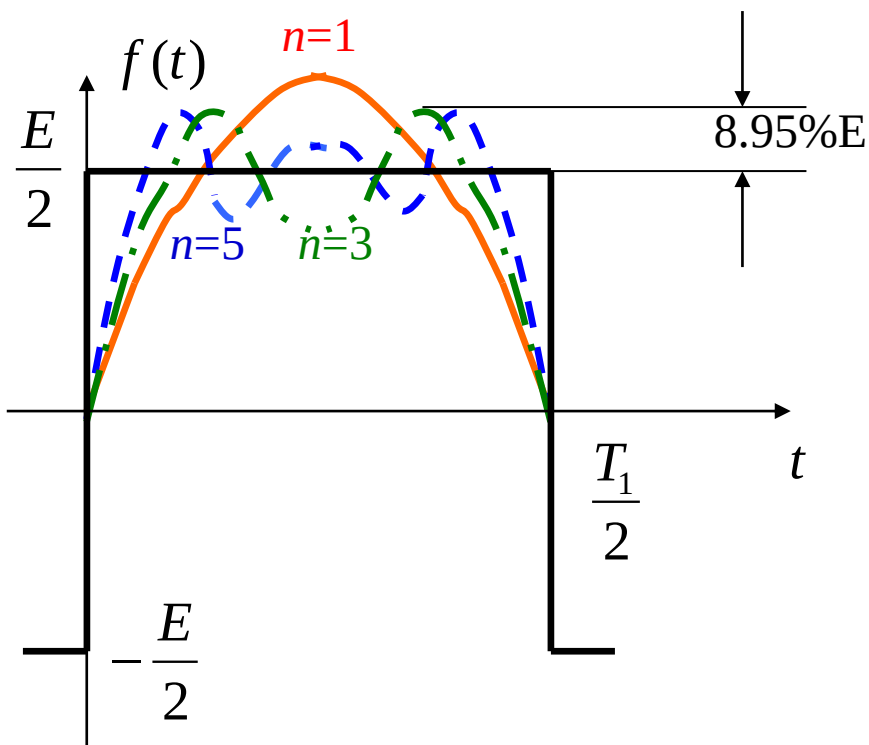
$$f(t) \approx \frac{2E}{\pi} \left(\sin \omega_1 t + \frac{1}{3} \sin 3\omega_1 t \right)$$

$n=5:$

$$f(t) \approx \frac{2E}{\pi} \left(\sin \omega_1 t + \frac{1}{3} \sin 3\omega_1 t + \frac{1}{5} \sin 5\omega_1 t \right)$$



3.2 Fourier Series Analysis of Periodic Signals



When the number of terms N is large, the peak value tends to a constant (**approximately 9% of the total jump value**) and gradually decays in the form of undulating oscillations starting from the discontinuous point.

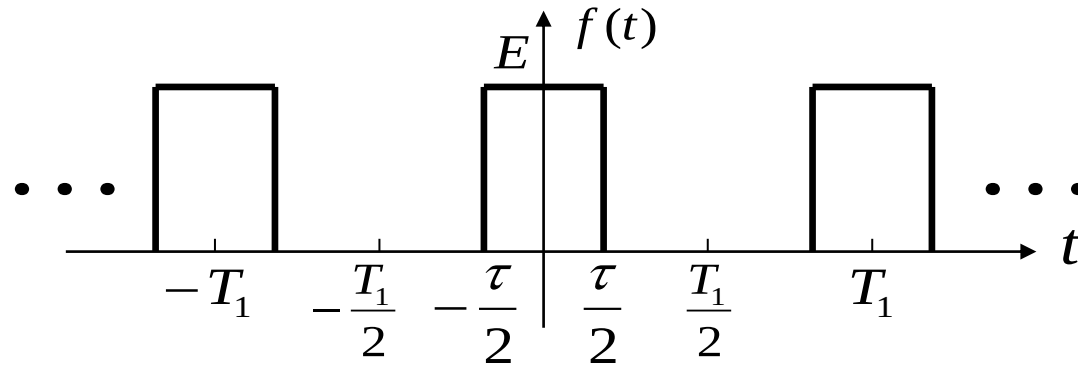
High-frequency components mainly affect the **transition edge** of the pulse.
Low-frequency components mainly affect the **top** of the pulse.

The more drastic the waveform change of $f(t)$, the richer the high-frequency components it contains; the slower the change, the richer the low-frequency components it contains.

3.3 Fourier Series of Typical Periodic Signals



Fourier Series of Periodic Rectangular Pulse Signals



$b_n = 0$ (The Fourier series of an even function does not contain sine terms)

$$a_0 = \frac{2}{T_1} \int_0^{\frac{T_1}{2}} f(t) dt = \frac{2}{T_1} \int_0^{\frac{\tau}{2}} E dt = \frac{E\tau}{T_1}$$

$$\begin{aligned} a_n &= \frac{4}{T_1} \int_0^{\frac{T_1}{2}} f(t) \cos n\omega_1 t dt = \frac{4}{T_1} \int_0^{\frac{\tau}{2}} E \cos n\omega_1 t dt = \frac{4E}{n\omega_1 T_1} \sin\left(\frac{n\omega_1 \tau}{2}\right) = \frac{2E\tau}{T_1} \cdot \frac{\sin\left(\frac{n\omega_1 \tau}{2}\right)}{\frac{n\omega_1 \tau}{2}} \\ &= \frac{2E\tau}{T_1} \text{Sa}\left(\frac{n\omega_1 \tau}{2}\right) = c_n \end{aligned}$$

3.3 Fourier Series of Typical Periodic Signals



Fourier series of a periodic rectangular pulse signals in trigonometric form:

$$f(t) = \frac{E\tau}{T_1} + \frac{2E\tau}{T_1} \sum_{n=1}^{\infty} \text{Sa}\left(\frac{n\omega_1\tau}{2}\right) \cos n\omega_1 t$$

Since $F_n = \frac{1}{2}(a_n - jb_n) = \frac{1}{2}a_n = \frac{E\tau}{T_1} \text{Sa}\left(\frac{n\omega_1\tau}{2}\right)$

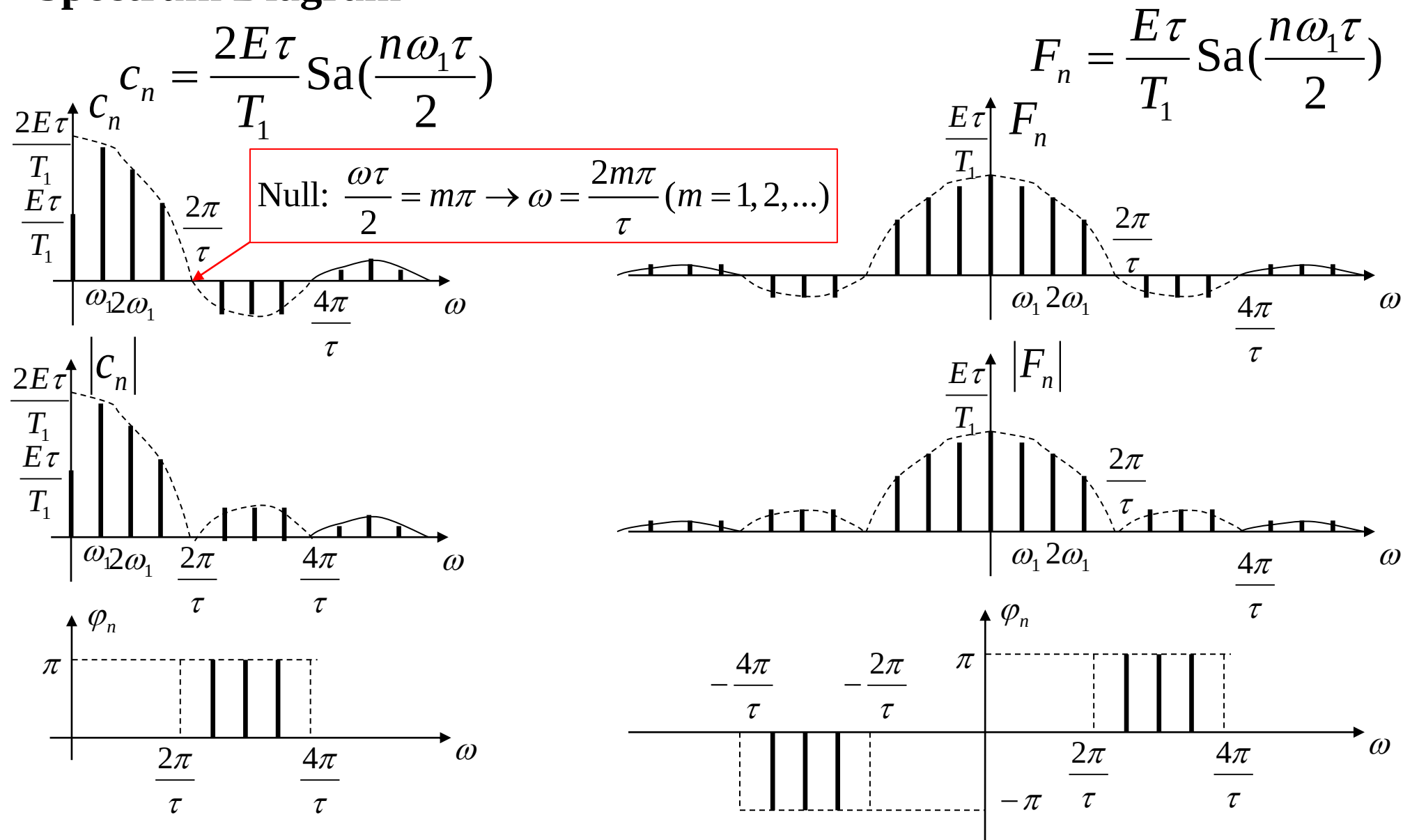
Fourier series in exponential form:

$$f(t) = \frac{E\tau}{T_1} \sum_{n=-\infty}^{\infty} \text{Sa}\left(\frac{n\omega_1\tau}{2}\right) e^{jn\omega_1 t}$$

3.3 Fourier Series of Typical Periodic Signals



Spectrum Diagram



- Repeat the examples in the lecture notes.
- 3.50(a)(d) in the textbook.

Review of Last Lecture

3.1 Introduction

3.2 Fourier Series Analysis of Periodic Signals

3.3 Fourier Series of Typical Periodic Signals

3.2 Fourier Series Analysis of Periodic Signals



Fourier Series Expansion: **Periodic signals** can be expressed as a **weighted sum** of **sinusoidal signals with harmonic relationships**.

- **Dirichlet conditions** (**sufficient but unnecessary conditions** for the existence of Fourier series):
 - (1) Within one period, the signal is **continuous** or has only **a finite number of first-kind discontinuities** (the function has finite left and right limits at the discontinuity, e.g., $x(t)=|\sin(t)|/\sin(t)$ at $t=0$).
 - (2) Within one period, the signal has **a finite number of maximum and minimum values**.
 - (3) Within one period, the signal is **absolutely integrable**, i.e., $\int_0^T |f(t)| dt < \infty$.
- **Signals in engineering all satisfy Dirichlet conditions.**

3.2 Fourier Series Analysis of Periodic Signals



3.2.1 Fourier Series in Trigonometric Form

Let $f(t)$ be a periodic signal with period T_1 .

Its **fundamental angular frequency** is $\omega_1 = 2\pi f_1 = \frac{2\pi}{T_1}$.

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t) \quad (1)$$

DC component: **Amplitude of cosine component :** **Amplitude of sine component:**

$$a_0 = \frac{1}{T_1} \int_{t_0}^{t_0+T_1} f(t) dt \qquad a_n = \frac{2}{T_1} \int_{t_0}^{t_0+T_1} f(t) \cos n\omega_1 t dt \qquad b_n = \frac{2}{T_1} \int_{t_0}^{t_0+T_1} f(t) \sin n\omega_1 t dt$$

The integral limits in the above formulas are generally set as: $0 \sim T_1$ or $-\frac{T_1}{2} \sim \frac{T_1}{2}$

The Fourier series in trigonometric form can also be expressed as:

$$f(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_1 t + \varphi_n) \quad (2)$$

where $c_0 = a_0$ $c_n^2 = a_n^2 + b_n^2$ $\varphi_n = \arctan(-\frac{b_n}{a_n})$

3.2 Fourier Series Analysis of Periodic Signals



3.2.2 Fourier Series in Exponential Form

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t} \quad (3)$$

where $F_n = \frac{1}{T_1} \int_{t_0}^{t_0+T_1} f(t) e^{-jn\omega_1 t} dt$ -----Complex amplitude

$$F_0 = a_0 = c_0$$

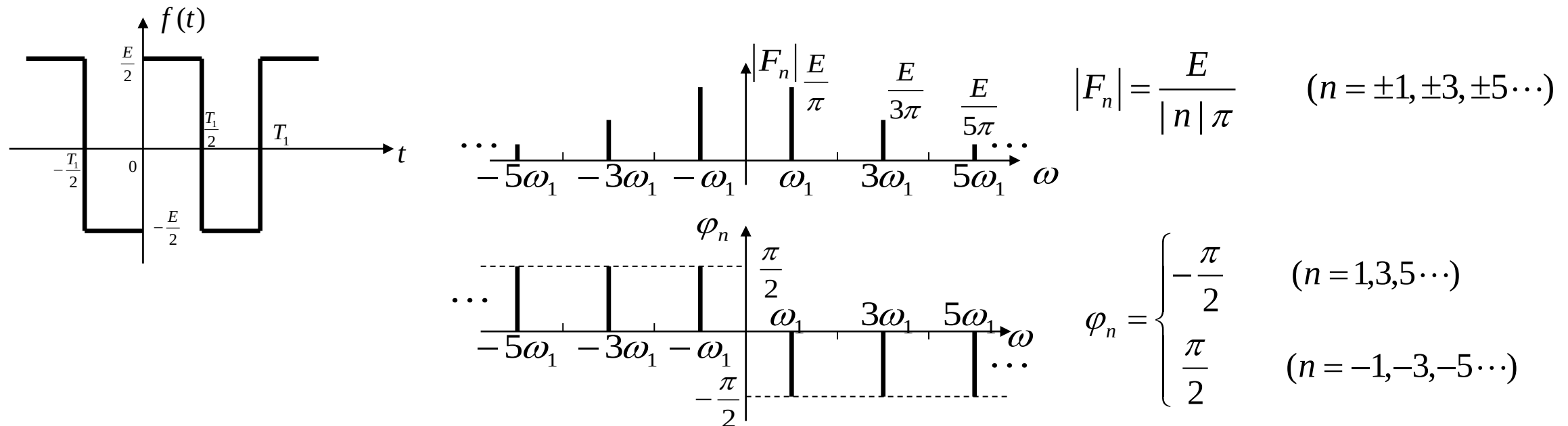
$$F_n = |F_n| e^{j\varphi_n} = \frac{1}{2} (a_n - jb_n)$$

$$|F_n| = \frac{1}{2} \sqrt{a_n^2 + b_n^2} = \frac{1}{2} c_n \quad \varphi_n = \arctan \left(-\frac{b_n}{a_n} \right)$$

3.2 Fourier Series Analysis of Periodic Signals

3.2.3 Characteristics of Periodic Signal Spectrum

- (1) **Discreteness:** The spectrum is discrete.
- (2) **Harmonicity:** Spectral lines appear at integer multiples of the fundamental frequency ω_1 .
- (3) **Convergence:** The amplitude of spectral lines in the amplitude spectrum gradually decays to zero as n increases ($n \rightarrow \infty$).



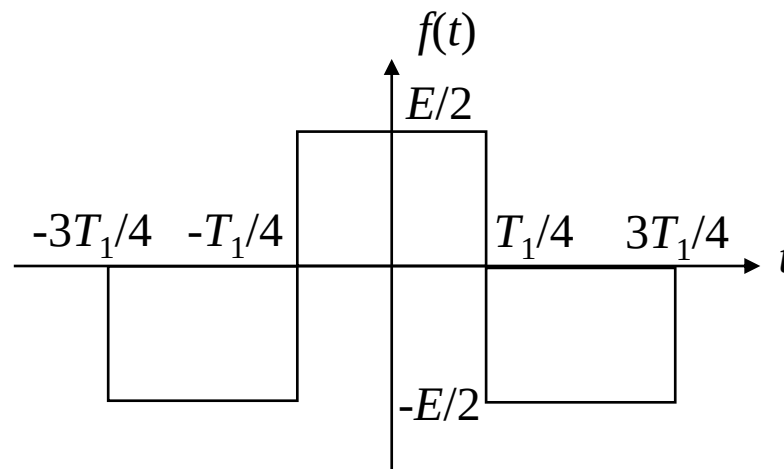
The zero components of the Fourier series in trigonometric form for the periodic rectangular signal shown in the figure are ().

☒ A $a_n = 0 \quad (n = 0, 2, 4 \dots)$

☐ B $a_n = 0 \quad (n = 1, 3, 5 \dots)$

☒ C $b_n = 0 \quad (n = 1, 3, 5 \dots)$

☒ D $b_n = 0 \quad (n = 2, 4, 6 \dots)$



提交

3.2 Fourier Series Analysis of Periodic Signals

3.2.4 Relationship Between Waveform Symmetry and Harmonic Characteristics

If $f(t)$ is a real function and its waveform satisfies a certain **symmetry**, some terms in the Fourier series disappear.

Waveform Symmetry	Frequency Components
Even Function $f(t) = f(-t)$	No sine components. $b_n = 0 \ (n = 1, 2, 3 \dots)$
Odd Function $f(t) = -f(-t)$	No DC and cosine components. $a_n = 0 \ (n = 0, 1, 2, 3 \dots)$
Odd Harmonic Function (Half-Wave Odd Symmetry) $f(t \pm \frac{T_1}{2}) = -f(t)$	No DC and even harmonic components. $a_n = 0 \ (n = 0, 2, 4, 6 \dots)$ $b_n = 0 \ (n = 2, 4, 6 \dots)$
Even Harmonic Function (Half-Wave Even Symmetry) $f(t \pm \frac{T_1}{2}) = f(t)$	No odd harmonic components. $a_n = 0 \ (n = 1, 3, 5 \dots)$ $b_n = 0 \ (n = 1, 3, 5 \dots)$

3.2 Fourier Series Analysis of Periodic Signals



Which components of the Fourier series in trigonometric form for the periodic rectangular signal shown in the figure are zero?

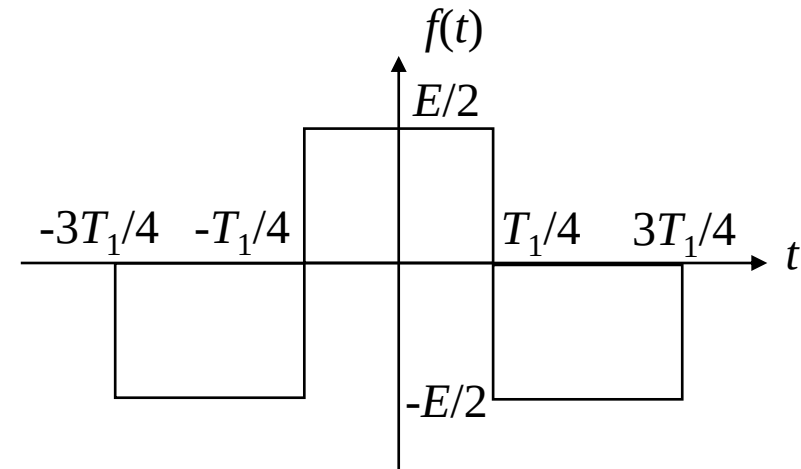
This is an **even function**, so the sine components are zero.

$$b_n = 0 \quad (n = 1, 2, 3, \dots)$$

It is also an **odd harmonic function** (half-wave odd symmetry), so the DC and even harmonic components are zero.

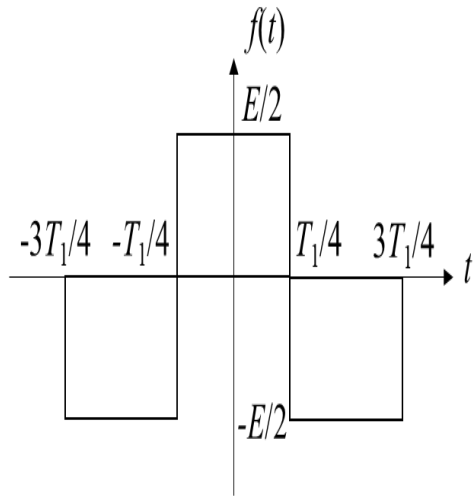
$$a_n = 0 \quad (n = 0, 2, 4, \dots)$$

$$b_n = 0 \quad (n = 2, 4, 6, \dots)$$



3.2 Fourier Series Analysis of Periodic Signals

Compared with the example in the last lecture, the two periodic rectangular signals have a time difference of $T_1/4$, and the corresponding trigonometric function phase difference is $\pi/2$.



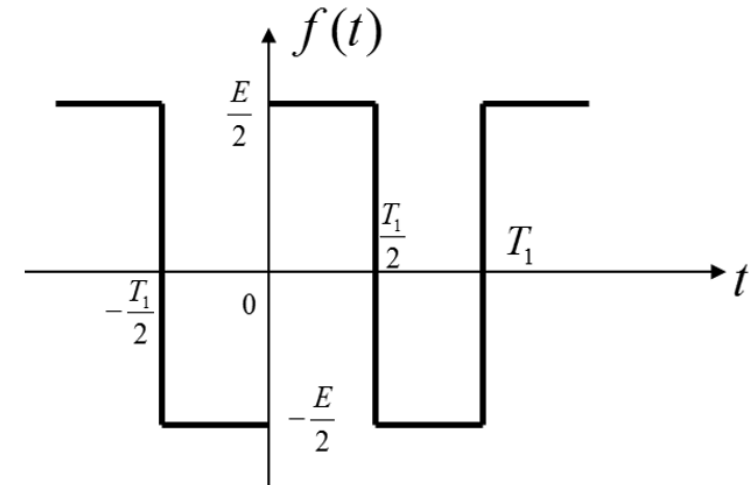
The zero components of the Fourier series in trigonometric form for the periodic rectangular signal shown in the figure are ().

A $a_n = 0 \quad (n = 0, 2, 4 \dots)$

B $a_n = 0 \quad (n = 1, 3, 5 \dots)$

C $b_n = 0 \quad (n = 1, 3, 5 \dots)$

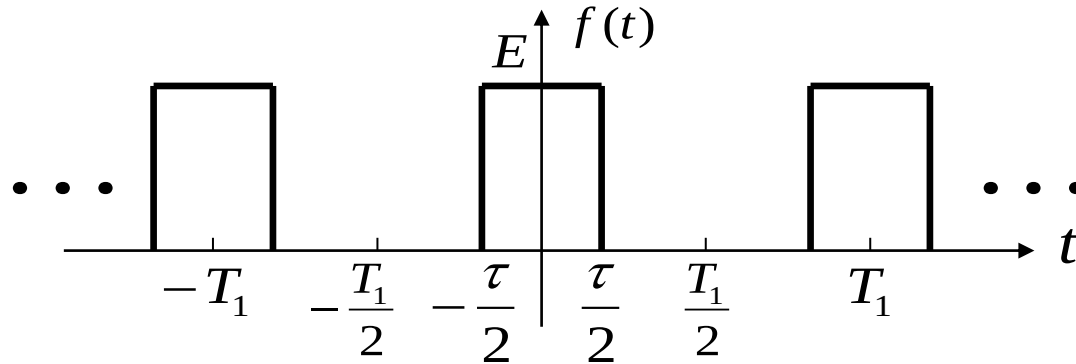
D $b_n = 0 \quad (n = 2, 4, 6 \dots)$



3.3 Fourier Series of Typical Periodic Signals



Fourier Series of Periodic Rectangular Pulse Signals



$b_n = 0$ (The Fourier series of an even function does not contain sine terms)

$$a_0 = \frac{2}{T_1} \int_0^{\frac{T_1}{2}} f(t) dt = \frac{E\tau}{T_1} \quad a_n = \frac{4}{T_1} \int_0^{\frac{T_1}{2}} f(t) \cos n\omega_1 t dt = \frac{2E\tau}{T_1} \text{Sa}\left(\frac{n\omega_1\tau}{2}\right) = c_n$$

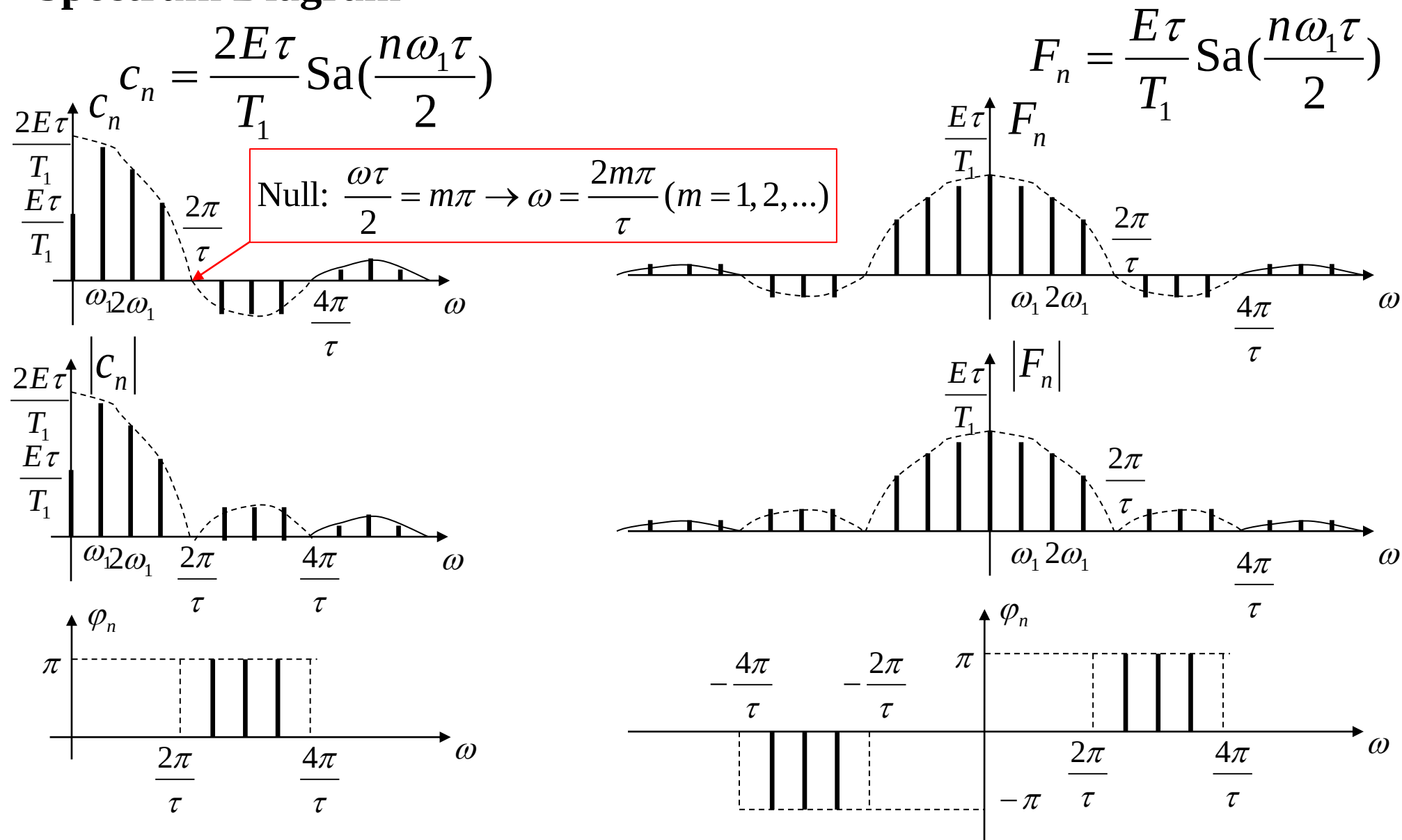
$$f(t) = \frac{E\tau}{T_1} + \frac{2E\tau}{T_1} \sum_{n=1}^{\infty} \text{Sa}\left(\frac{n\omega_1\tau}{2}\right) \cos n\omega_1 t$$

$$f(t) = \frac{E\tau}{T_1} \sum_{n=-\infty}^{\infty} \text{Sa}\left(\frac{n\omega_1\tau}{2}\right) e^{jn\omega_1 t} \quad F_n = \frac{1}{2}(a_n - jb_n) = \frac{1}{2}a_n = \frac{E\tau}{T_1} \text{Sa}\left(\frac{n\omega_1\tau}{2}\right)$$

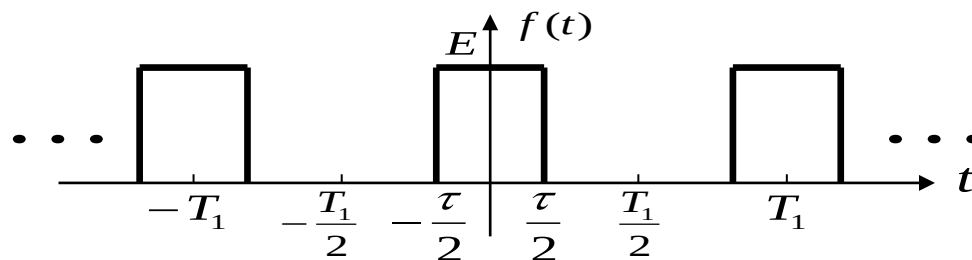
3.3 Fourier Series of Typical Periodic Signals



Spectrum Diagram



For the periodic rectangular signal waveform shown in the figure, which of the following statements are correct? ()



- ☒ A If T_1 is doubled, the spectral line spacing of the signal spectrum is halved, and the amplitude is halved.
- ☐ B If T_1 is doubled, the signal bandwidth is halved, and the amplitude is doubled.
- ☐ C If τ is halved, the spectral line spacing of the signal spectrum is halved, and the amplitude is doubled.
- ☒ D If τ is halved, the signal bandwidth is doubled, and the amplitude is halved.

提交

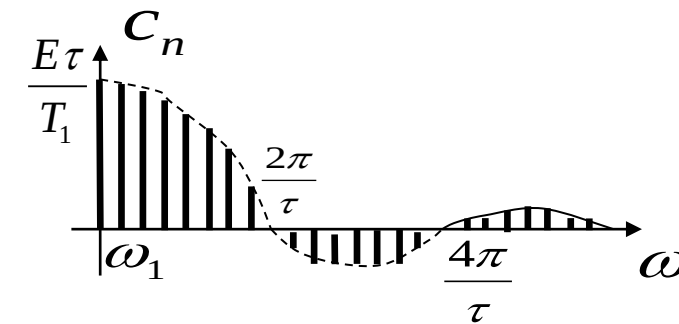
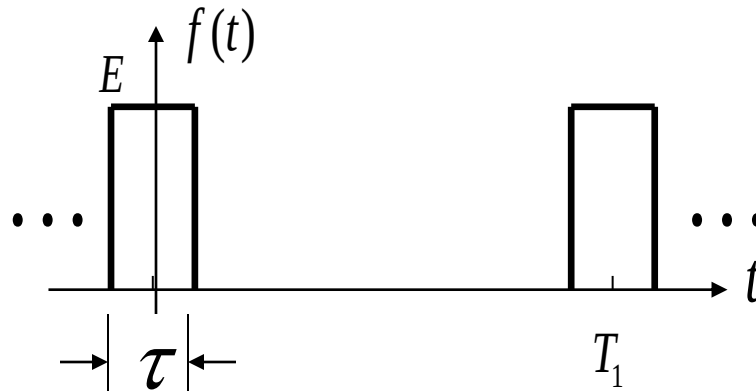
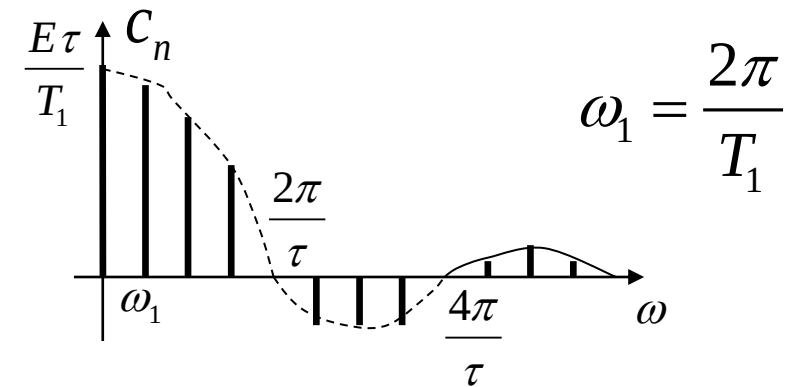
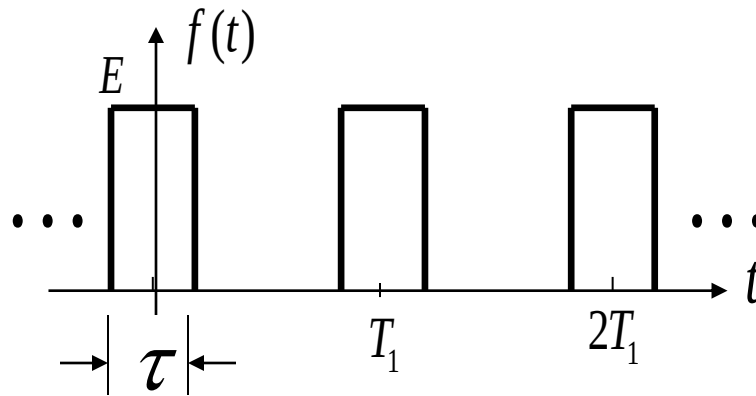
3.3 Fourier Series of Typical Periodic Signals



Relationship Between Spectrum Structure and Waveform Parameters (T_1, τ)

a) If τ remains unchanged and T_1 is doubled:

- The spectral line spacing of the amplitude spectrum is halved;
- The amplitude is halved.

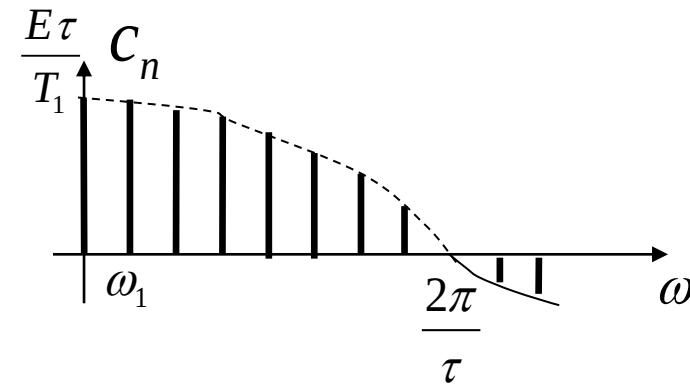
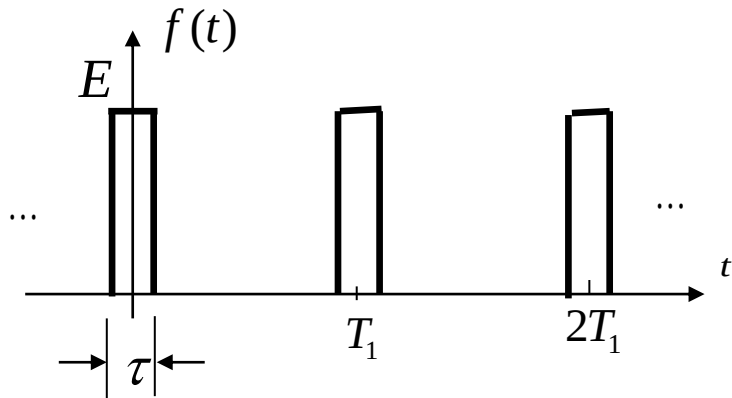
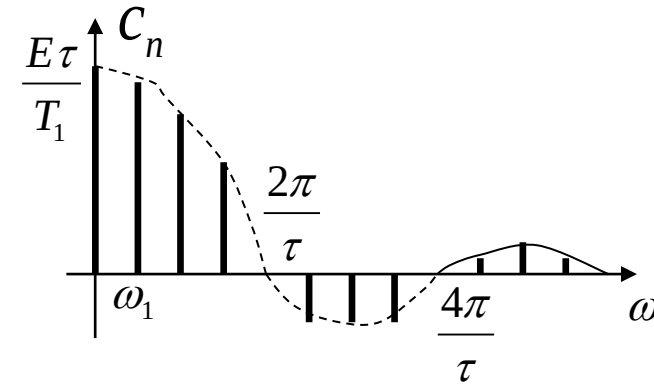
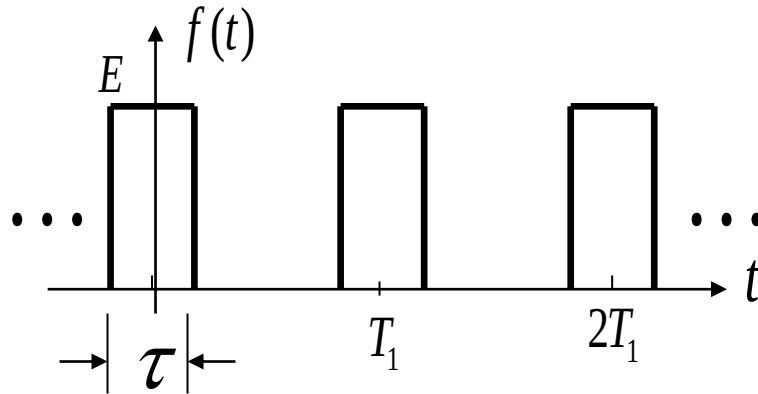


3.3 Fourier Series of Typical Periodic Signals



b) If T_1 remains unchanged and τ is halved:

- The signal bandwidth is doubled.
- The amplitude of the amplitude spectrum is halved.



Summary

1. **Spectral line spacing** $\omega_1 (= \frac{2\pi}{T_1})$: Only related to T_1 , inversely proportional to T_1 .
2. **Cross-zero frequency**: Only related to τ , inversely proportional to τ .
3. **Spectral line amplitude**: Related to both T_1 and τ , inversely proportional to T_1 , directly proportional to τ .

Contents of This Lecture

3.4 Fourier Transform

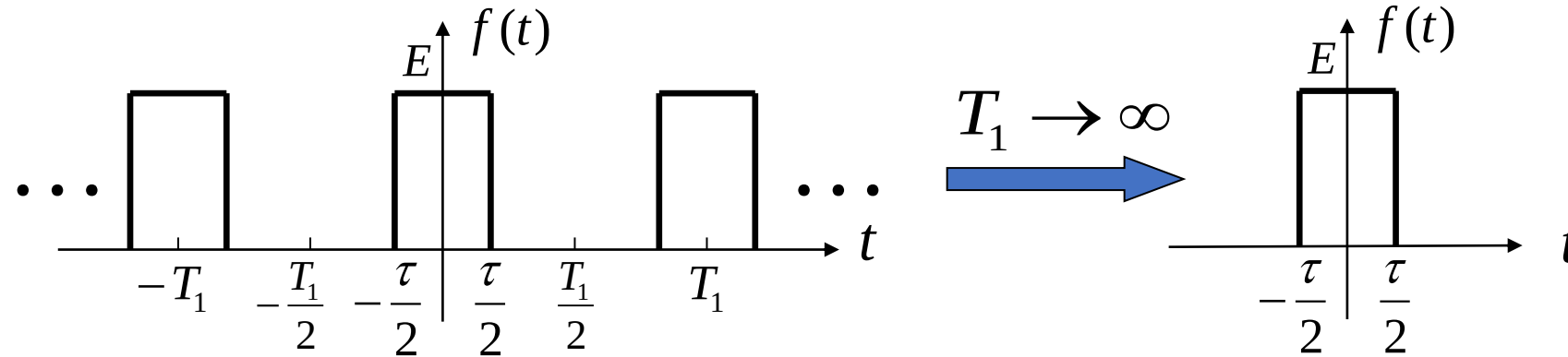
3.5 Fourier Transform of Typical Aperiodic Signals

3.6 Fourier Transform of Impulse and Step Functions

Learning Objectives of This Lecture

1. Gain an in-depth understanding of the connection and difference between **Fourier transform** and **Fourier series**.
2. Proficiently master the Fourier transform of **typical aperiodic signals, impulse functions, and step functions**.

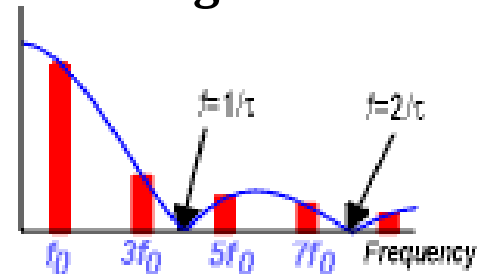
3.4 Fourier Transform



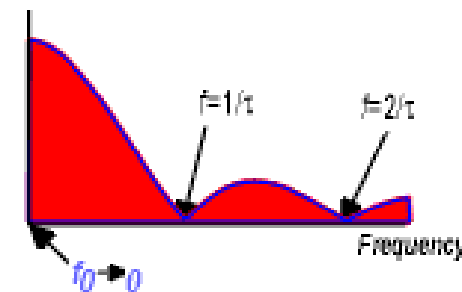
$T_1 \uparrow \longrightarrow \omega_1 = \frac{2\pi}{T_1} \downarrow \longrightarrow$ Spectral line spacing (proportional to ω_1) $\downarrow$

$T_1 \rightarrow \infty \longrightarrow \omega_1 = \frac{2\pi}{T_1} \rightarrow 0 \longrightarrow$ Spectral line spacing $\rightarrow 0$

Discrete Spectrum of Periodic Signals



Continuous Spectrum of Aperiodic Signals



3.4 Fourier Transform



Due to $T_1 \rightarrow \infty$, $F_n = \frac{1}{T_1} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) e^{-jn\omega_1 t} dt \rightarrow 0$

Spectral Density Function: $\lim_{T_1 \rightarrow \infty} F_n T_1 = \lim_{T_1 \rightarrow \infty} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) e^{-jn\omega_1 t} dt$

When $T_1 \rightarrow \infty$, discrete frequency $n\omega_1 \rightarrow$ continuous frequency ω .

Then $\lim_{T_1 \rightarrow \infty} F_n T_1 = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$

Let **$F(\omega)$** denote the **Fourier Transform of aperiodic signal $f(t)$** .

$$F(\omega) = \mathcal{F} [f(t)] = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

Inverse Fourier Transform

$$f(t) = \mathcal{F}^{-1} [F(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

$$F(\omega) = |F(\omega)| e^{j\varphi(\omega)}$$

$|F(\omega)|$: **Amplitude Spectrum**

$\varphi(\omega)$: **Phase Spectrum**

3.4 Fourier Transform



Physical Meaning of Fourier Transform

It can be proved,

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega = \int_0^{\infty} \underbrace{\frac{|F(\omega)|}{\pi}}_{\text{Amplitude}} d\omega \cos[\omega t + \varphi(\omega)] = \lim_{\Delta\omega \rightarrow 0} \sum_{n=0}^{\infty} \underbrace{\frac{|F(n\Delta\omega)|}{\pi}}_{\text{Amplitude}} \Delta\omega \cos[n\Delta\omega t + \varphi(n\Delta\omega)]$$

$f(t)$ is the sum of infinitely many continuous cosine signals with infinitesimal amplitude, $\left(\frac{1}{\pi}|F(\omega)|d\omega\right)$ covering frequencies from 0 to ∞ .

Aperiodic signals can be expressed as a **weighted integral of sinusoidal signals**, containing all **continuous frequency components from zero to infinitely high frequencies**.

For energy-limited signals (e.g., single-pulse signals), the amplitude of each frequency component, $\left(\frac{1}{\pi}|F(\omega)|d\omega\right)$ tends to be infinitely small. Thus, the spectrum can no longer be represented by amplitude, but by the **spectral density function** instead.

Sufficient but Not Necessary Condition for the Existence of Fourier Transform

$$\int_{-\infty}^{\infty} |f(t)| dt < \infty \quad \text{i.e., } f(t) \text{ is absolutely integrable.}$$

Proof: $\int_{-\infty}^{\infty} |f(t)e^{-j\omega t}| dt = \int_{-\infty}^{\infty} |f(t)| dt < \infty$

Most **energy signals** satisfy this condition.

If a signal does not satisfy the above condition, Fourier transform may still exist, and may be conducted by introducing the concept of **impulse function** $\delta(t)$.

Relationship Between Fourier Series & Transform

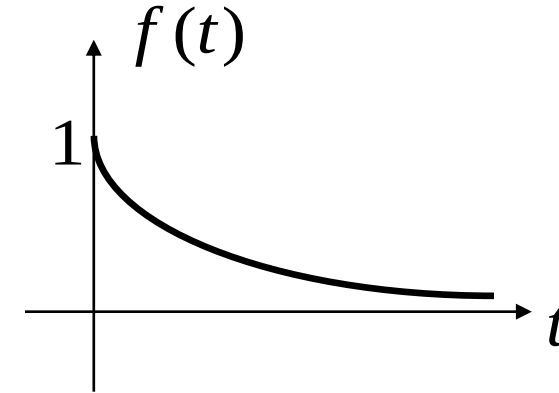
Feature	Fourier Series	Fourier Transform
Expression	$f(t) = \sum_{n=-\infty}^{+\infty} F_n e^{jn\omega_1 t}$	$f(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\omega) e^{j\omega t} d\omega$
Spectrum Type	Discrete Spectrum $F_n = \frac{1}{T_1} \int_{-T_1/2}^{T_1/2} f(t) e^{-jn\omega_1 t} dt$ $F_n = F_n e^{j\phi_n}$	Continuous Spectrum (Spectral Density) $F(\omega) = \int_{-\infty}^{+\infty} f(t) e^{-j\omega t} dt$ $F(\omega) = F(\omega) e^{j\phi(\omega)}$
Existence Condition	Dirichlet conditions within one period $\int_0^T f(t) dt < \infty$	Dirichlet conditions over $(-\infty, \infty)$ $\int_{-\infty}^{\infty} f(t) dt < \infty$

3.5 Fourier Transform of Typical Aperiodic Signals



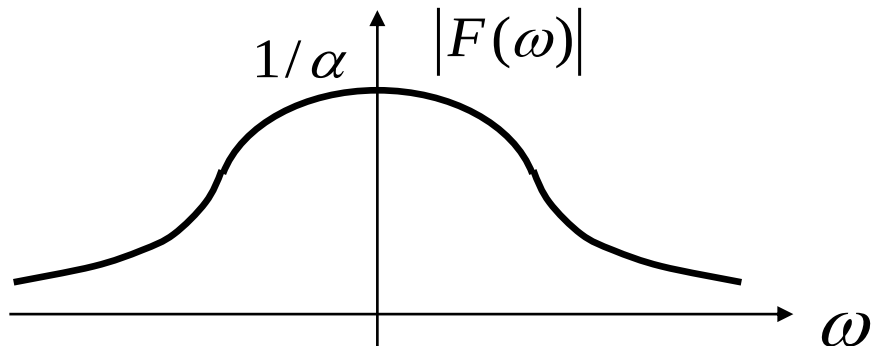
1. Single-sided Exponential Signal

$$f(t) = \begin{cases} e^{-\alpha t} & t > 0 \\ 0 & t < 0 \end{cases} = e^{-\alpha t} u(t)$$

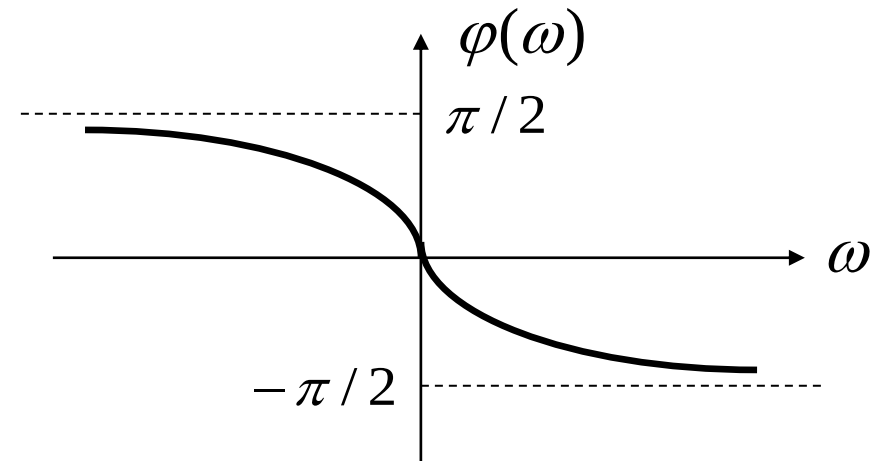


$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt = \int_0^{\infty} e^{-\alpha t} e^{-j\omega t} dt = \frac{1}{\alpha + j\omega}$$

$$|F(\omega)| = \frac{1}{\sqrt{\alpha^2 + \omega^2}}$$



$$\varphi(\omega) = -\arctan\left(\frac{\omega}{\alpha}\right)$$



The inverse Fourier transform of the spectrum function $X(\omega) = \frac{1}{j\omega + 2}$ is ().

☒ A $e^{-2t}u(t)$

☐ B $te^{-2t}u(t)$

☐ C $e^{-0.5t}u(t)$

☐ D $te^{2t}u(t)$

提交

2. Odd-symmetric Two-sided Exponential Signal

$$f(t) = e^{-\alpha t} u(t) - e^{\alpha t} u(-t)$$

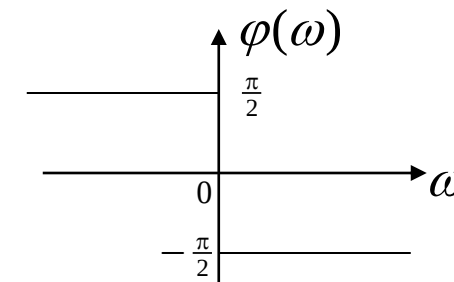
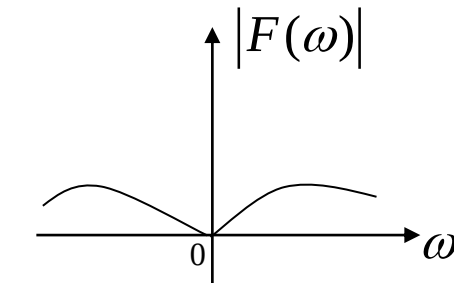
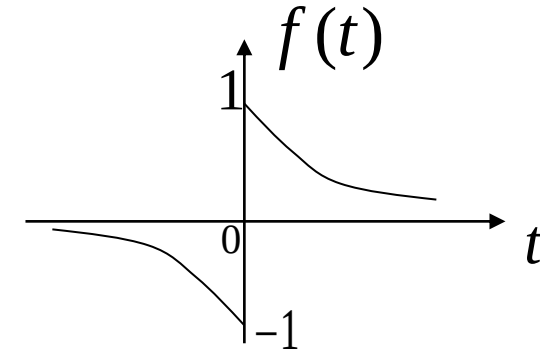
$$= f_1(t) + f_2(t)$$

$$F_1(\omega) = \frac{1}{\alpha + j\omega}$$

$$F_2(\omega) = \int_{-\infty}^{\infty} f_2(t) e^{-j\omega t} dt = \int_{-\infty}^0 -e^{\alpha t} e^{-j\omega t} dt = \frac{-1}{\alpha - j\omega}$$

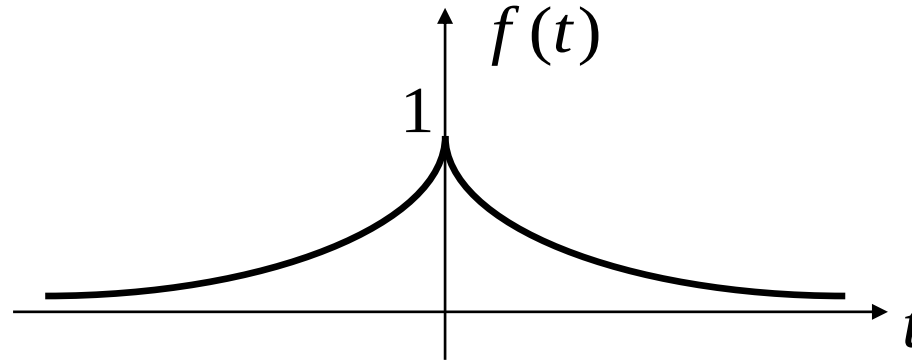
$$F(\omega) = F_1(\omega) + F_2(\omega) = \frac{-2j\omega}{\alpha^2 + \omega^2}$$

$$|F(\omega)| = \frac{2|\omega|}{\alpha^2 + \omega^2}, \quad \varphi(\omega) = \begin{cases} \pi/2 & \omega < 0 \\ -\pi/2 & \omega > 0 \end{cases}$$

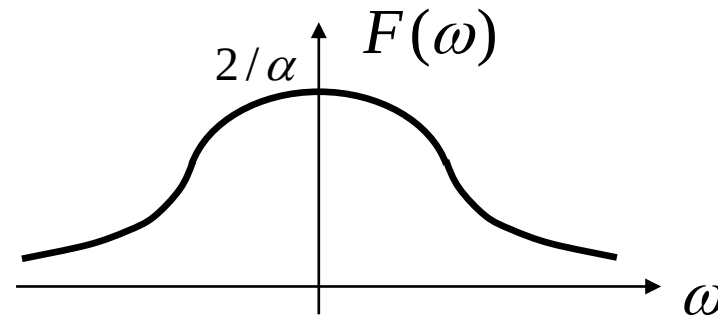


3. Even-symmetric Two-sided Exponential Signal

$$f(t) = e^{-\alpha|t|} \quad (\alpha > 0)$$



$$F(\omega) = \frac{2\alpha}{\alpha^2 + \omega^2}$$

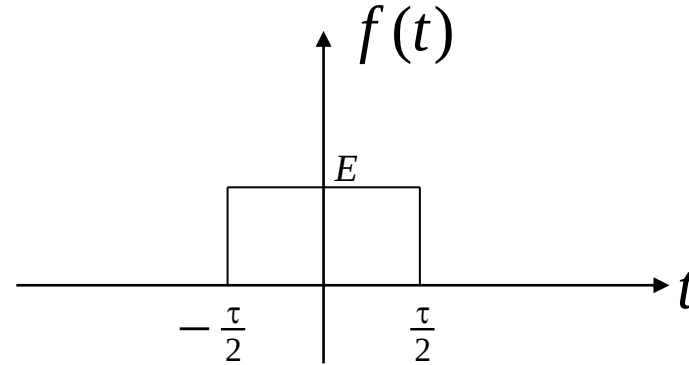


$$|F(\omega)| = \frac{2\alpha}{\alpha^2 + \omega^2}$$

$$\varphi(\omega) = 0$$

4. Rectangular Pulse Signal (Gate Function)

$$f(t) = E[u(t + \frac{\tau}{2}) - u(t - \frac{\tau}{2})]$$

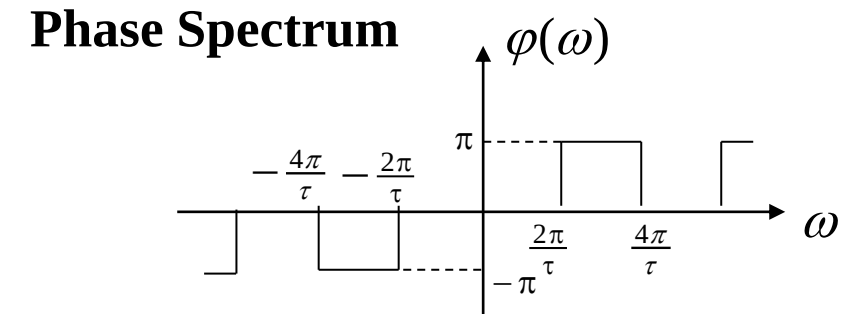
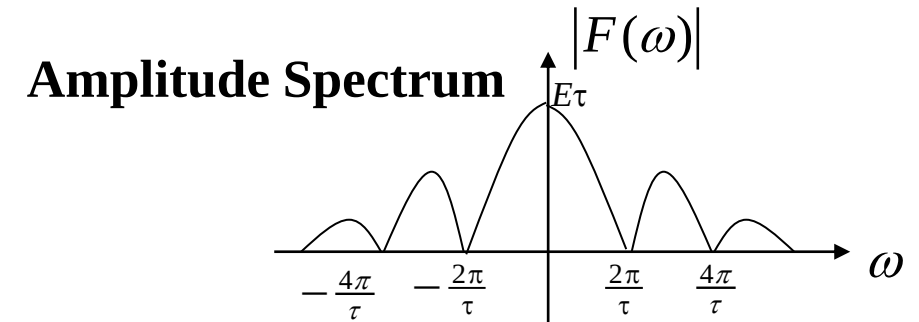
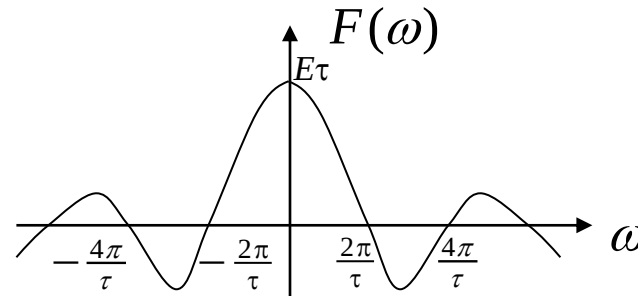
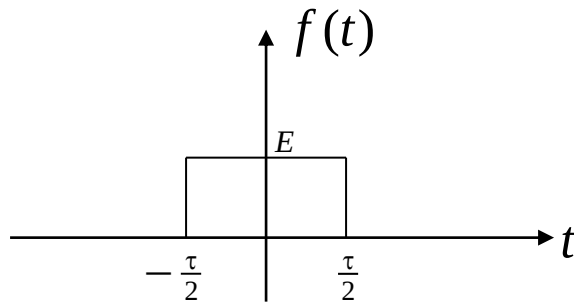


$$\begin{aligned} F(\omega) &= \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt = \int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} Ee^{-j\omega t} dt = E \frac{1}{-j\omega} e^{-j\omega t} \Big|_{-\frac{\tau}{2}}^{\frac{\tau}{2}} \\ &= E \frac{e^{-j\omega \frac{\tau}{2}} - e^{j\omega \frac{\tau}{2}}}{-j\omega} = 2E \frac{\sin\left(\frac{\tau\omega}{2}\right)}{\omega} = E\tau \text{Sa}\left(\frac{\tau\omega}{2}\right) \end{aligned}$$

3.5 Fourier Transform of Typical Aperiodic Signals



$$E[u(t + \frac{\tau}{2}) - u(t - \frac{\tau}{2})] \xleftrightarrow{FT} E\tau \text{Sa}(\omega \frac{\tau}{2})$$



Equivalent Bandwidth

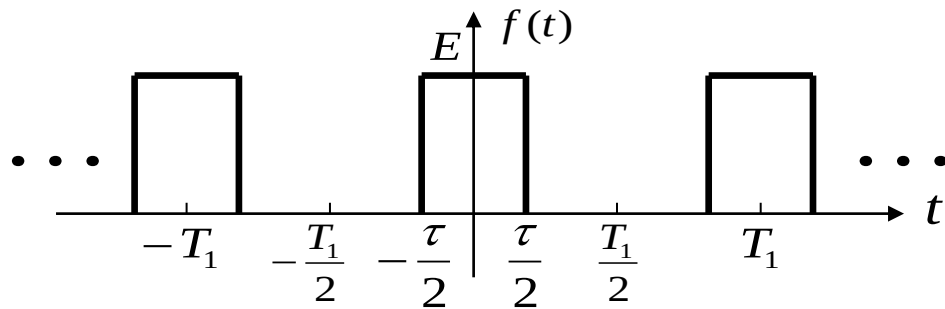
It can be seen from the above spectrum diagram of the rectangular wave that the signal energy is mainly concentrated within the frequency range $0 \sim 2\pi / \tau$. This range, $B\omega$ is called the **equivalent bandwidth** of the rectangular wave.

$$B_{\omega} = \frac{2\pi}{\tau} \text{ (rad/s) or } \frac{1}{\tau} \text{ (Hz)}$$

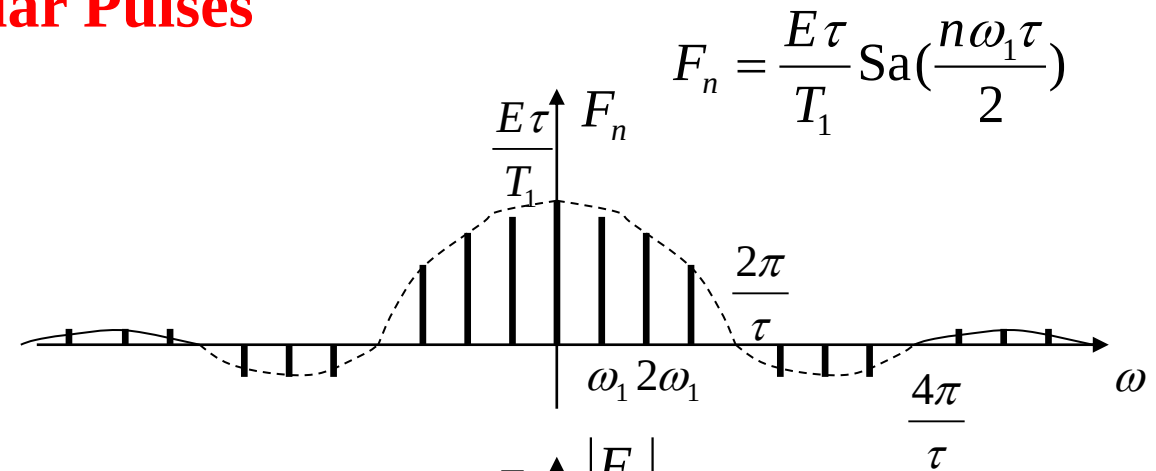
3.5 Fourier Transform of Typical Aperiodic Signals



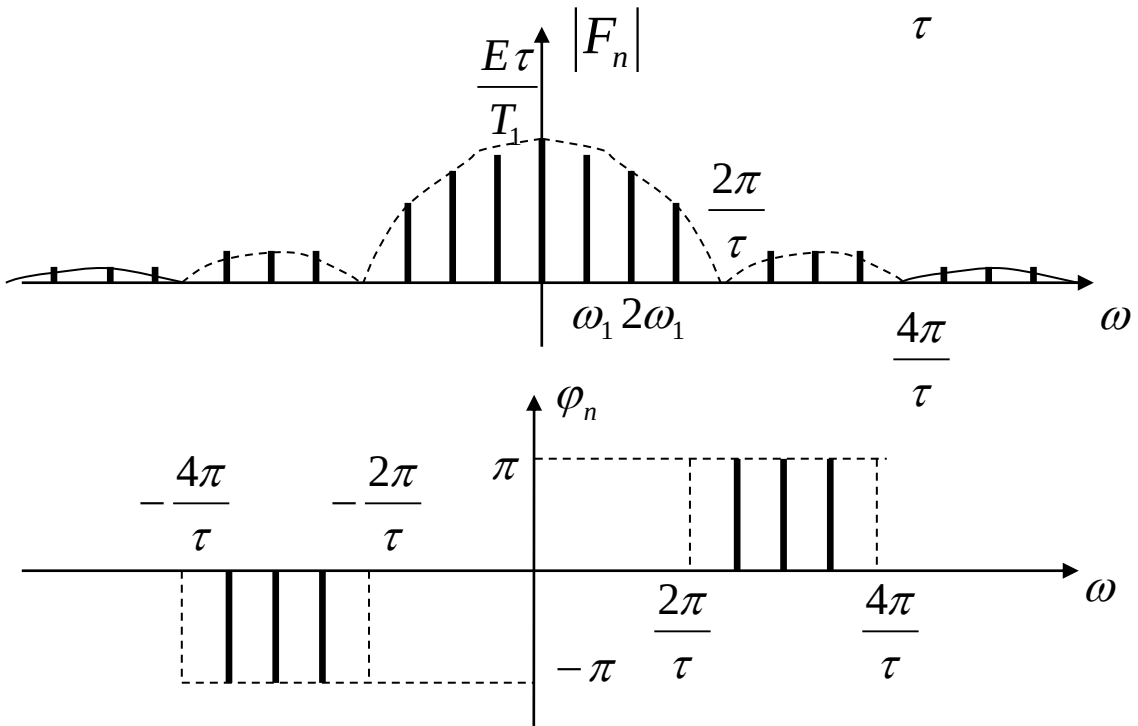
Review: Spectrum of Periodic Rectangular Pulses



Amplitude
Spectrum



Phase
Spectrum



3.5 Fourier Transform of Typical Aperiodic Signals



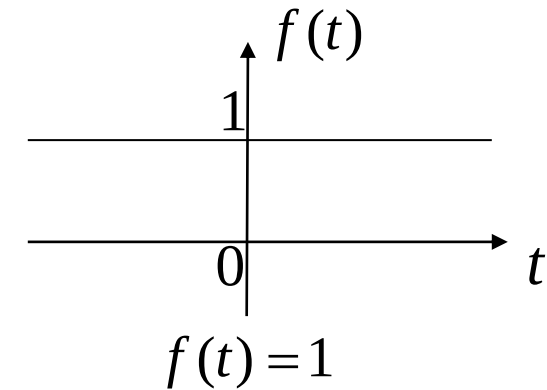
5. DC Signal

A DC signal is a constant (set to 1 here).

It does not satisfy the absolutely integrable condition, so its Fourier transform cannot be directly obtained by integration—we use the **impulse function $\delta(\omega)$** to derive it.

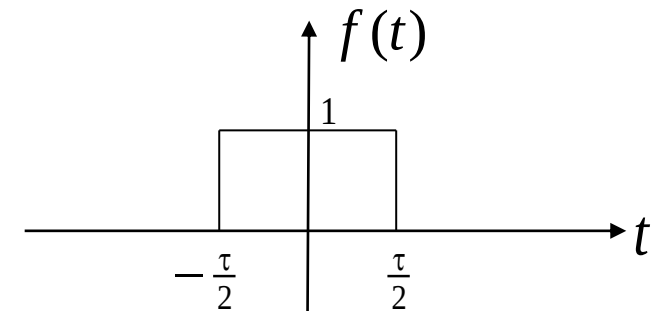
This result can be obtained by taking the limit of τ to ∞ for a rectangular pulse with amplitude 1 and width τ :

$$f(t) = 1 = \lim_{\tau \rightarrow \infty} \left[u\left(t + \frac{\tau}{2}\right) - u\left(t - \frac{\tau}{2}\right) \right]$$

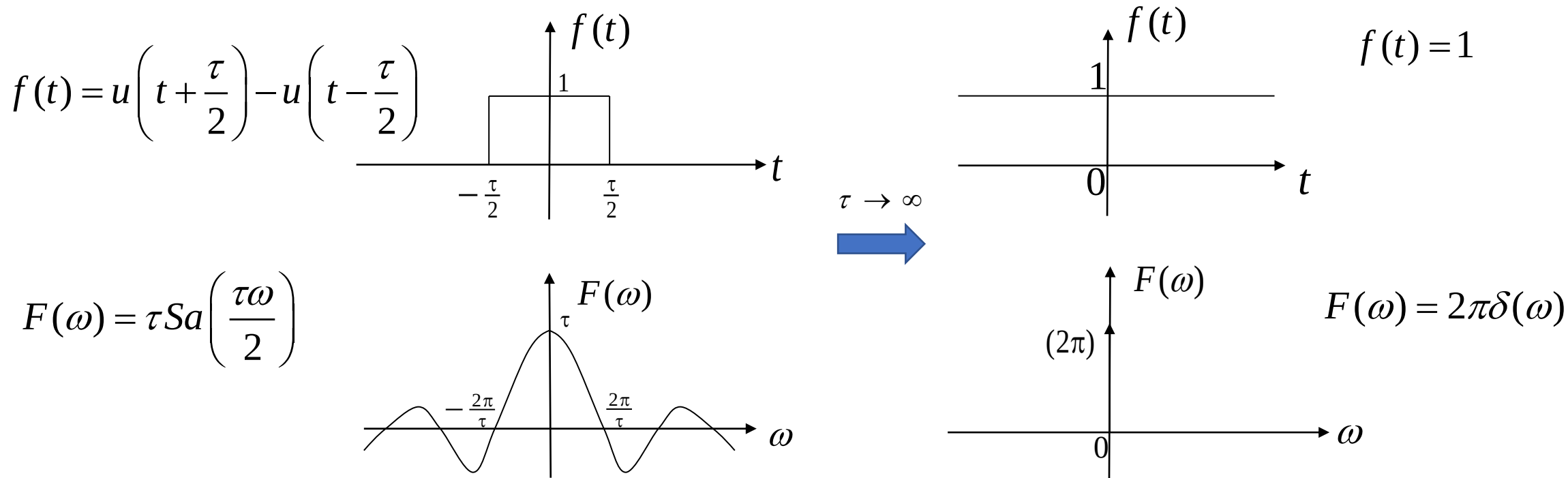


$\tau \rightarrow \infty$ 

$$f(t) = u\left(t + \frac{\tau}{2}\right) - u\left(t - \frac{\tau}{2}\right)$$



3.5 Fourier Transform of Typical Aperiodic Signals



When $\tau \rightarrow \infty$, the spectrum width tends to 0, and the amplitude tends to infinity (forming an **impulse function**), the area of $F(\omega)$ (strength of the impulse function) is:

$$\int_{-\infty}^{\infty} \tau \text{Sa}\left(\frac{\tau\omega}{2}\right) d\omega = 2 \int_{-\infty}^{\infty} \text{Sa}\left(\frac{\tau\omega}{2}\right) d\left(\frac{\tau\omega}{2}\right) = 2\pi \quad \left(\int_{-\infty}^{\infty} \text{Sa}(x) dx = \pi \right)$$

So

$$1 \xleftrightarrow{FT} 2\pi\delta(\omega)$$

$$\int_{-\infty}^{\infty} e^{-j\omega t} dt = 2\pi\delta(\omega)$$

6. Signum Function Signal

$$f(t) = \text{sgn}(t)$$

The signum function **does not satisfy the absolutely integrable condition.**

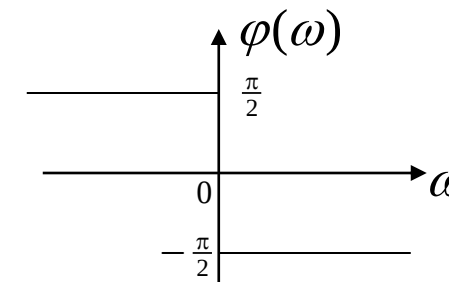
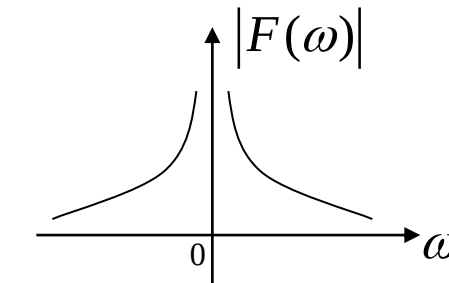
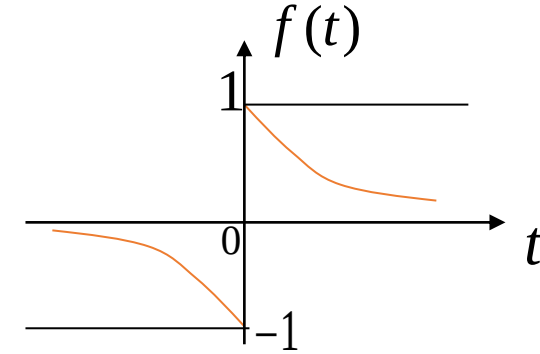
Its Fourier transform is derived by taking $\alpha \rightarrow 0$ for the odd-symmetric two-sided exponential signal:

$$f(t) = \text{sgn}(t) = \lim_{\alpha \rightarrow 0} [-e^{\alpha t} u(-t) + e^{-\alpha t} u(t)]$$

So

$$F(\omega) = \lim_{\alpha \rightarrow 0} \frac{-2j\omega}{\alpha^2 + \omega^2} = -j \frac{2}{\omega} = \frac{2}{j\omega}$$

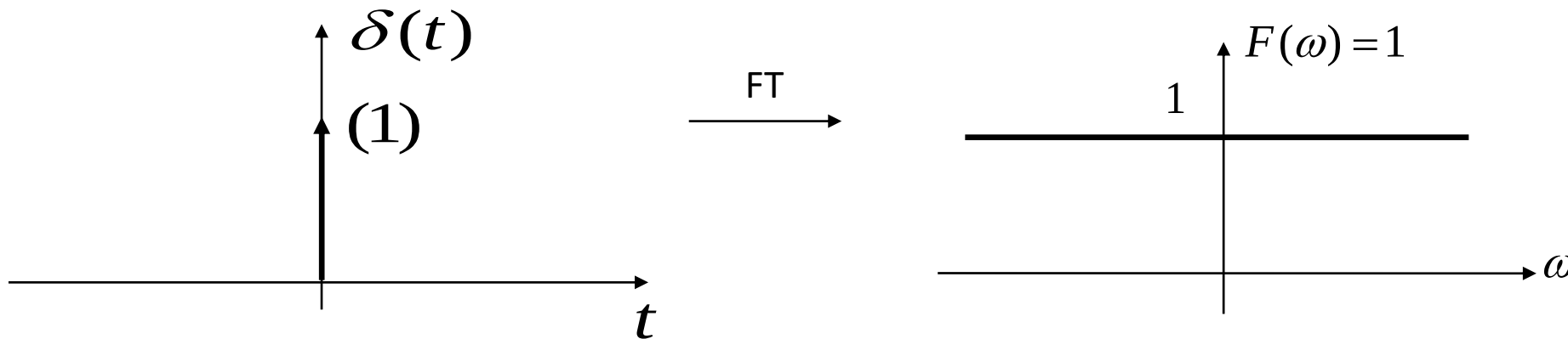
$$|F(\omega)| = \frac{2}{|\omega|}, \quad \varphi(\omega) = \begin{cases} \pi/2 & \omega < 0 \\ -\pi/2 & \omega > 0 \end{cases}$$



1. Fourier Transform of Impulse Function

$$F(\omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} \delta(t) dt = 1$$

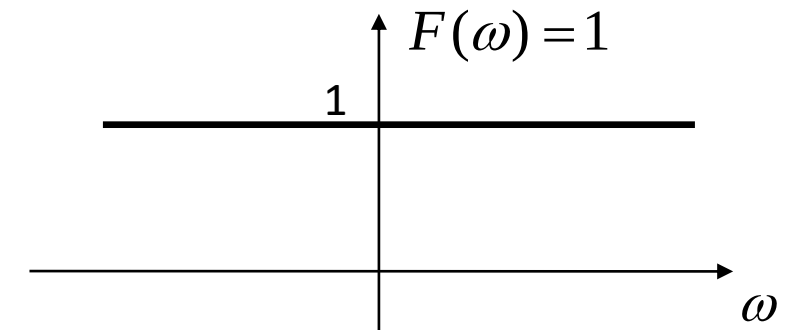
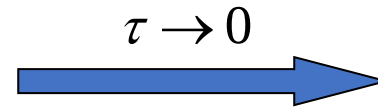
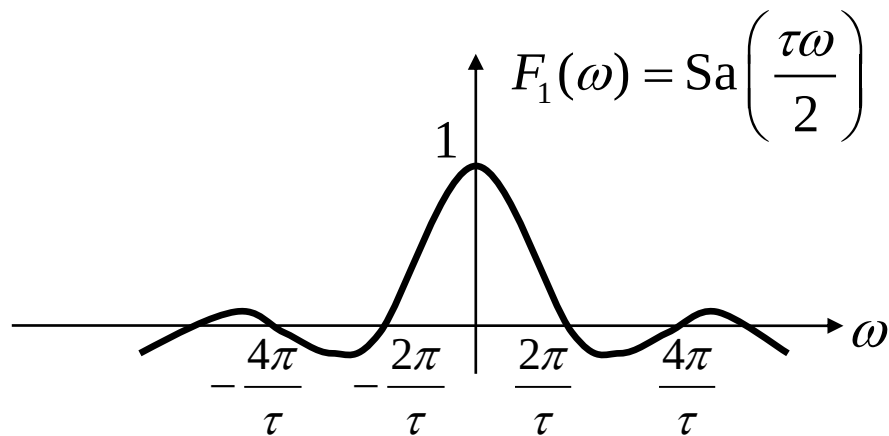
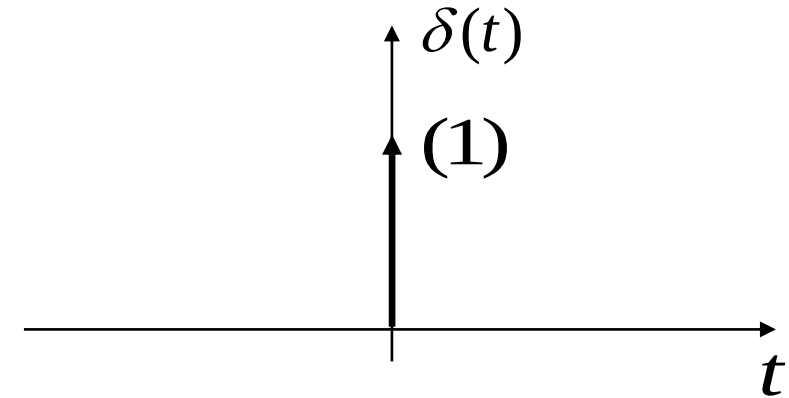
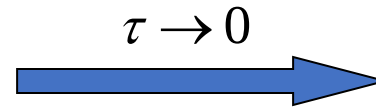
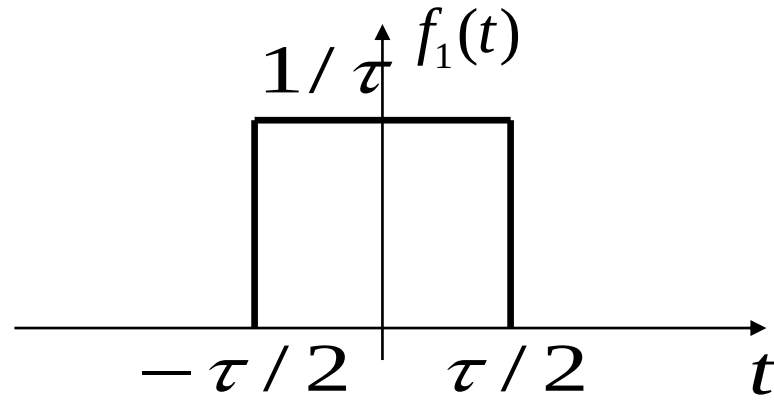
The spectrum of the unit impulse function is a constant, meaning the spectrum is uniform across the entire frequency range. This type of spectrum is often called a "**uniform spectrum**" or "**white spectrum**".



3.6 Fourier Transform of Impulse and Step Functions



Deriving Impulse Function's Fourier Transform from Rectangular Pulse



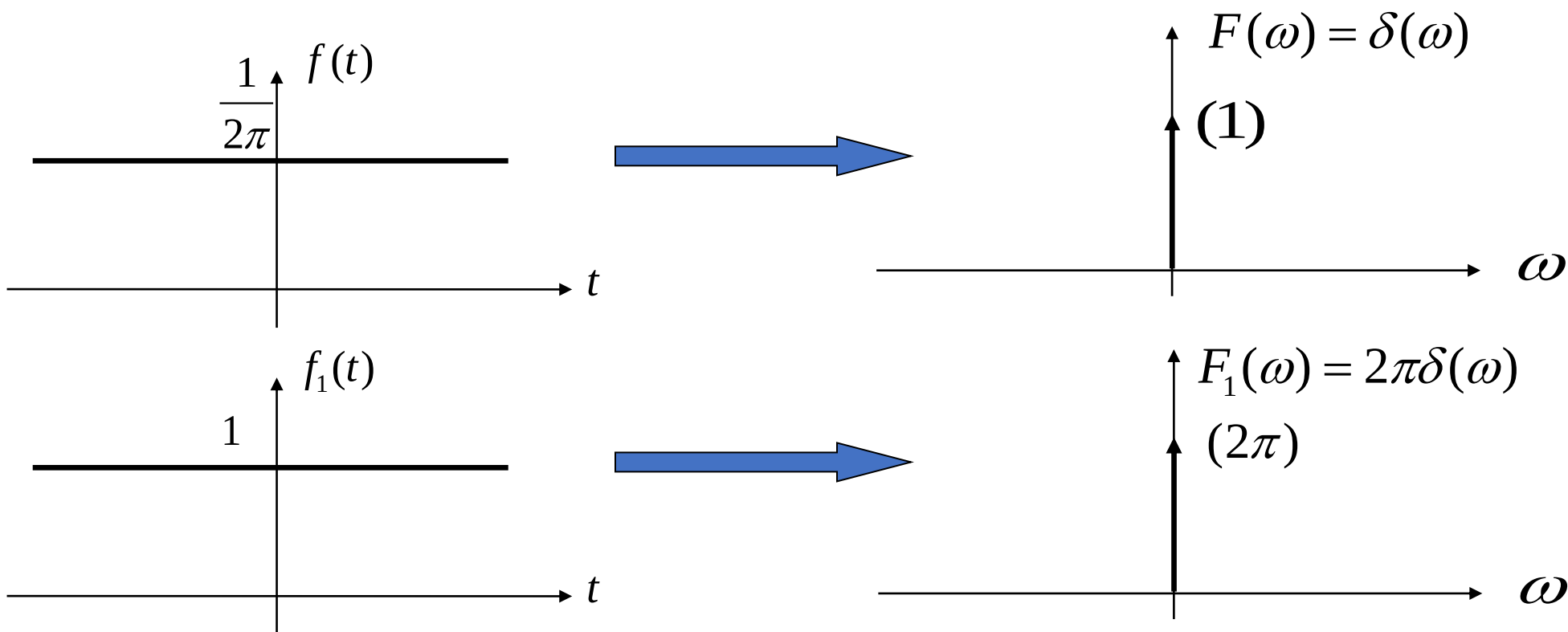
3.6 Fourier Transform of Impulse and Step Functions



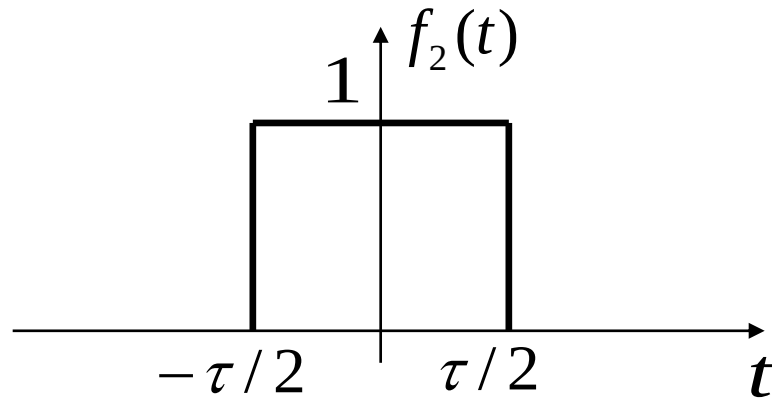
2. Inverse Fourier Transform of Impulse Function

$$f(t) = F^{-1}[\delta(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega) e^{j\omega t} d\omega = \frac{1}{2\pi}$$

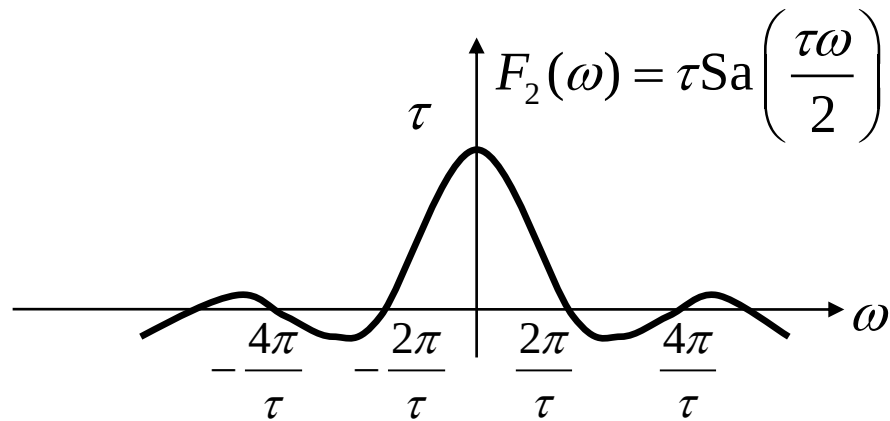
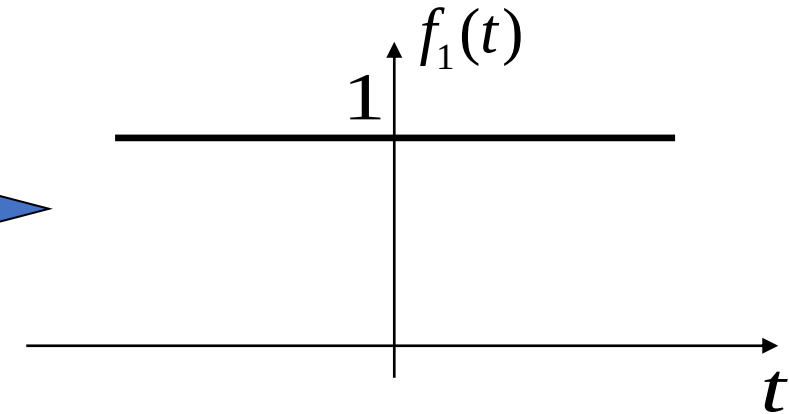
$$\text{或 } F\left[\frac{1}{2\pi}\right] = \delta(\omega), \quad F[1] = 2\pi\delta(\omega)$$



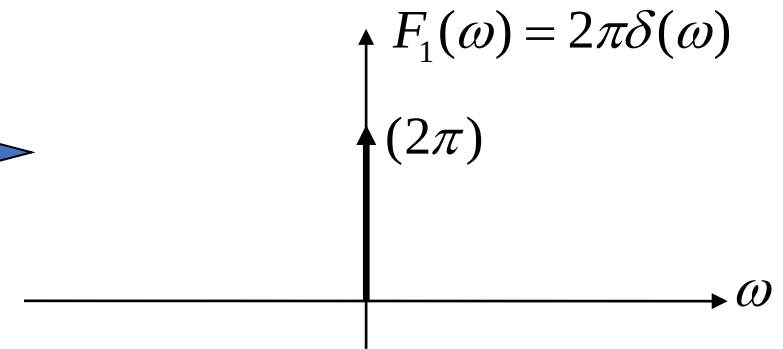
3.6 Fourier Transform of Impulse and Step Functions



$\tau \rightarrow \infty$



$\tau \rightarrow \infty$



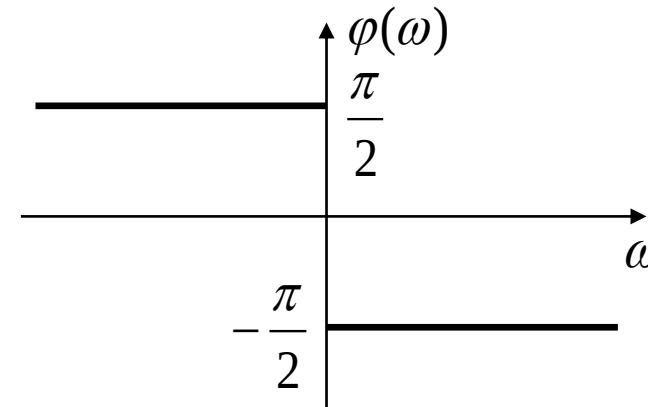
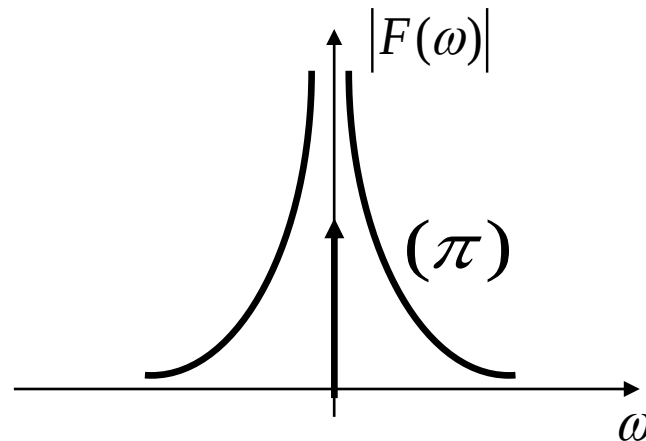
3. Fourier Transform of Step Function

$$\because u(t) = \frac{1}{2} + \frac{1}{2} \text{sgn}(t)$$

$$\begin{aligned} \therefore F(\omega) &= \mathcal{F}[u(t)] = \mathcal{F}\left[\frac{1}{2}\right] + \mathcal{F}\left[\frac{1}{2} \text{sgn}(t)\right] \\ &= \pi\delta(\omega) + \frac{1}{j\omega} \end{aligned}$$

corresponding to
DC component

other frequency components caused by
the jump at $t=0$



- Repeat the examples in the lecture notes.
- 3.54(b) and 3.55(a)(b) in the textbook.

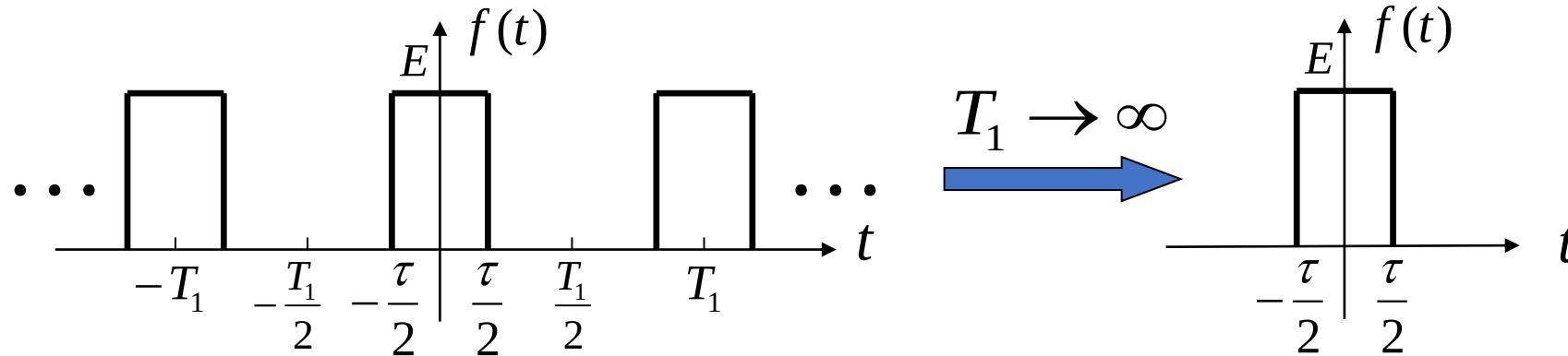
Review of Last Lecture

3.4 Fourier Transform

3.5 Fourier Transform of Typical Aperiodic Signals

3.6 Fourier Transform of Impulse and Step Functions

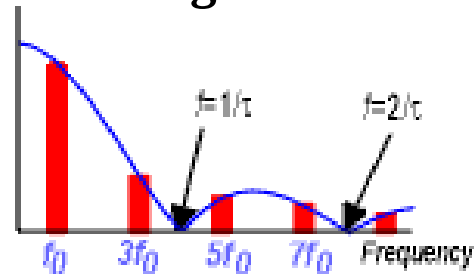
3.4 Fourier Transform



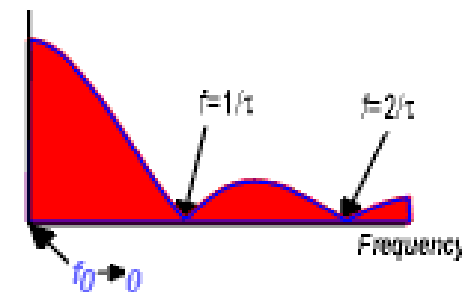
$T_1 \uparrow \longrightarrow \omega_1 = \frac{2\pi}{T_1} \downarrow \longrightarrow$ Spectral line spacing (proportional to ω_1) $\downarrow$

$T_1 \rightarrow \infty \longrightarrow \omega_1 = \frac{2\pi}{T_1} \rightarrow 0 \longrightarrow$ Spectral line spacing $\rightarrow 0$

Discrete Spectrum of Periodic Signals



Continuous Spectrum of Aperiodic Signals



3.4 Fourier Transform



Due to $T_1 \rightarrow \infty$, $F_n = \frac{1}{T_1} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) e^{-jn\omega_1 t} dt \rightarrow 0$

Spectral Density Function: $\lim_{T_1 \rightarrow \infty} F_n T_1 = \lim_{T_1 \rightarrow \infty} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) e^{-jn\omega_1 t} dt$

When $T_1 \rightarrow \infty$, discrete frequency $n\omega_1 \rightarrow$ continuous frequency ω .

Then $\lim_{T_1 \rightarrow \infty} F_n T_1 = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$

Let **$F(\omega)$** denote the **Fourier Transform of aperiodic signal $f(t)$** .

$$F(\omega) = \mathbf{F} [f(t)] = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

Inverse Fourier Transform

$$f(t) = \mathbf{F}^{-1} [F(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

$$F(\omega) = |F(\omega)| e^{j\varphi(\omega)}$$

$|F(\omega)|$: **Amplitude Spectrum**

$\varphi(\omega)$: **Phase Spectrum**

3.4 Fourier Transform



Physical Meaning of Fourier Transform

It can be proved,

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega = \int_0^{\infty} \underbrace{\frac{|F(\omega)|}{\pi}}_{\text{Amplitude}} d\omega \cos[\omega t + \varphi(\omega)] = \lim_{\Delta\omega \rightarrow 0} \sum_{n=0}^{\infty} \underbrace{\frac{|F(n\Delta\omega)|}{\pi}}_{\text{Amplitude}} \Delta\omega \cos[n\Delta\omega t + \varphi(n\Delta\omega)]$$

$f(t)$ is the sum of infinitely many continuous cosine signals with infinitesimal amplitude, $\left(\frac{1}{\pi}|F(\omega)|d\omega\right)$ covering frequencies from 0 to ∞ .

Aperiodic signals can be expressed as a **weighted integral of sinusoidal signals**, containing all **continuous frequency components from zero to infinitely high frequencies**.

For energy-limited signals (e.g., single-pulse signals), the amplitude of each frequency component, $\left(\frac{1}{\pi}|F(\omega)|d\omega\right)$ tends to be infinitely small. Thus, the spectrum can no longer be represented by amplitude, but by the **spectral density function** instead.

Sufficient but Not Necessary Condition for the Existence of Fourier Transform

$$\int_{-\infty}^{\infty} |f(t)| dt < \infty \quad \text{i.e., } f(t) \text{ is absolutely integrable.}$$

Proof: $\int_{-\infty}^{\infty} |f(t)e^{-j\omega t}| dt = \int_{-\infty}^{\infty} |f(t)| dt < \infty$

Most **energy signals** satisfy this condition.

If a signal does not satisfy the above condition, Fourier transform may still exist, and may be conducted by introducing the concept of **impulse function** $\delta(t)$.

3.4 Fourier Transform



Relationship Between Fourier Series & Transform

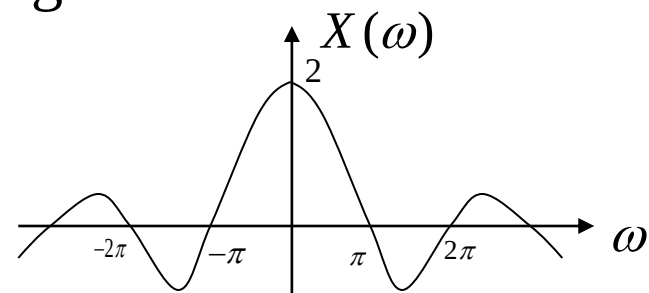
Feature	Fourier Series	Fourier Transform
Expression	$f(t) = \sum_{n=-\infty}^{+\infty} F_n e^{jn\omega_1 t}$	$f(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\omega) e^{j\omega t} d\omega$
Spectrum Type	Discrete Spectrum $F_n = \frac{1}{T_1} \int_{-T_1/2}^{T_1/2} f(t) e^{-jn\omega_1 t} dt$ $F_n = F_n e^{j\varphi_n}$	Continuous Spectrum (Spectral Density) $F(\omega) = \int_{-\infty}^{+\infty} f(t) e^{-j\omega t} dt$ $F(\omega) = F(\omega) e^{j\varphi(\omega)}$
Existence Condition	Dirichlet conditions within one period $\int_0^T f(t) dt < \infty$	Dirichlet conditions over $(-\infty, \infty)$ $\int_{-\infty}^{\infty} f(t) dt < \infty$

3.5 Fourier Transform of Typical Aperiodic Signals

Common Functions and Their Fourier Transforms

Name	Time Function	Spectral Function
Unit Impulse	$\delta(t)$	1
Unit Step	$u(t)$	$\pi\delta(\omega) + \frac{1}{j\omega}$
Signum Function	$\text{sgn}(t) = u(t) - u(-t)$	$\frac{2}{j\omega}$
Unit DC Signal	1	$2\pi\delta(\omega)$
Single-sided Exponential Signal	$e^{-\alpha t}u(t) (\alpha > 0)$	$\frac{1}{\alpha + j\omega}$
Odd-symmetric Two-sided Exponential Signal	$e^{-\alpha t}u(t) - e^{\alpha t}u(-t) (\alpha > 0)$	$\frac{-2j\omega}{\alpha^2 + \omega^2}$
Even-symmetric Two-sided Exponential Signal	$e^{-\alpha t } (\alpha > 0)$	$\frac{2\alpha}{\alpha^2 + \omega^2}$
Rectangular Pulse (Gate Function)	$E \left[u\left(t + \frac{\tau}{2}\right) - u\left(t - \frac{\tau}{2}\right) \right]$	$E\tau \text{Sa}\left(\frac{\omega\tau}{2}\right)$

Given that a signal $x(t)$ has a Fourier transform of the Sa function as shown in the figure, the time-domain expression of this signal is



A

$$x(t) = u(t+1) - u(t-1)$$

B

$$x(t) = 2[u(t+1) - u(t-1)]$$

C

$$x(t) = u(t+1/2) - u(t-1/2)$$

D

$$x(t) = 2[u(t+1/2) - u(t-1/2)]$$

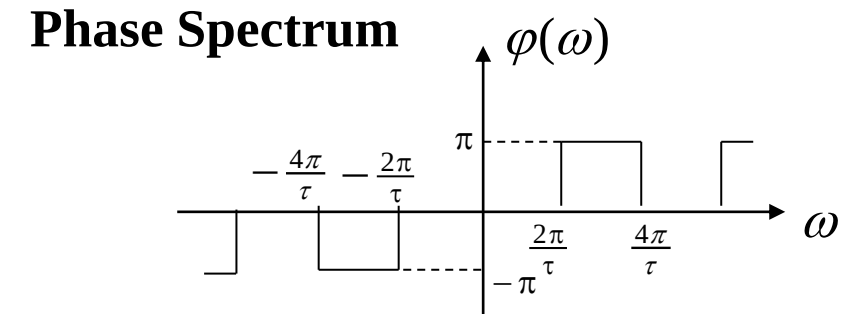
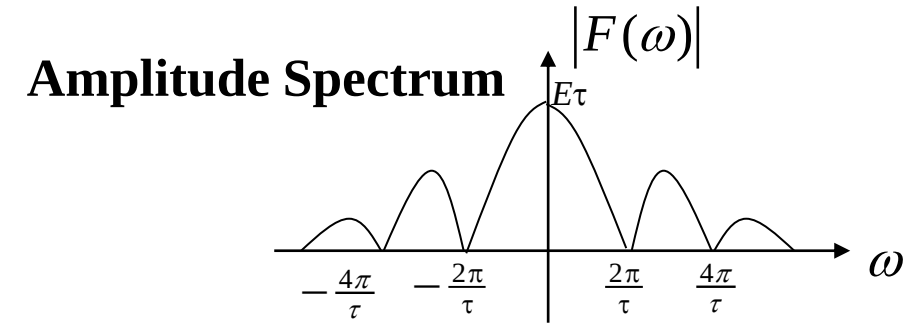
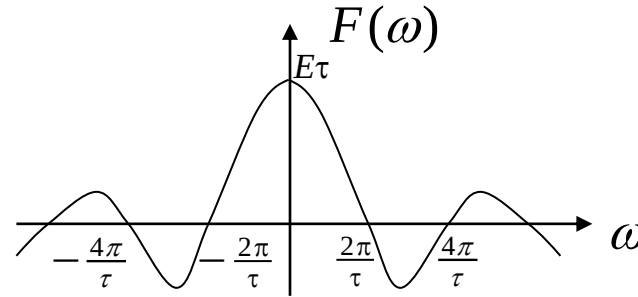
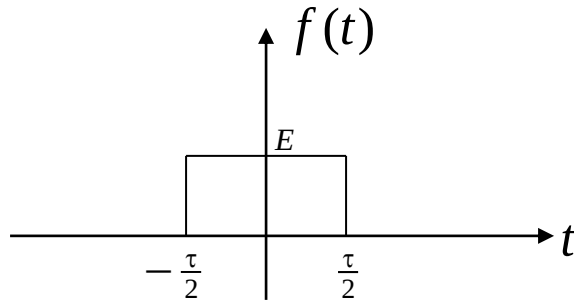
提交

3.5 Fourier Transform of Typical Aperiodic Signals



Rectangular Pulse Signal (Gate Function)

$$E[u(t + \frac{\tau}{2}) - u(t - \frac{\tau}{2})] \xleftrightarrow{FT} E\tau \text{Sa}(\omega \frac{\tau}{2})$$



Equivalent Bandwidth

It can be seen from the above spectrum diagram of the rectangular wave that the signal energy is mainly concentrated within the frequency range $0 \sim 2\pi / \tau$. This range, $B\omega$ is called the **equivalent bandwidth** of the rectangular wave.

$$B_{\omega} = \frac{2\pi}{\tau} \text{ (rad/s) or } \frac{1}{\tau} \text{ (Hz)}$$

Contents of This Lecture

3.7 Basic Properties of Fourier Transform

Learning Objectives of This Lecture

Master the application of various properties of Fourier Transform to calculate the Fourier Transform of complex signals (**Important!**).

3.7 Basic Properties of Fourier Transform

- A time function and its spectral function (if exists) have a unique relationship.
- The basic properties of Fourier Transform allow us to derive the Fourier Transform of complex-time signals from that of typical signals.
- Conversely, we can infer the time-domain changes from frequency-domain operations.
- **Notes:**
 - **Dirichlet's condition is a sufficient but not necessary condition.**
 - For any given signal $f(t)$, regardless of whether it satisfies the absolute integrability condition, Fourier analysis theory does not restrict the use of any method to obtain a formal transform $F(\omega)$. However, some transforms may lack physical meaning.
 - Engineering practice shows:
 - Fourier Transform exists for most **energy signals**;
 - **Generalized Fourier Transform** exists for most **power signals** (e.g., step signal, signum signal);
 - Generalized Fourier Transform may not exist for non-power, non-energy signals.

$$\text{Energy signal: } \int_{-\infty}^{\infty} |f(t)|^2 dt < \infty \quad \text{Power signal: } \lim_{T_1 \rightarrow \infty} \frac{1}{T_1} \int_{-T_1/2}^{T_1/2} |f(t)|^2 dt < \infty$$

3.7 Basic Properties of Fourier Transform



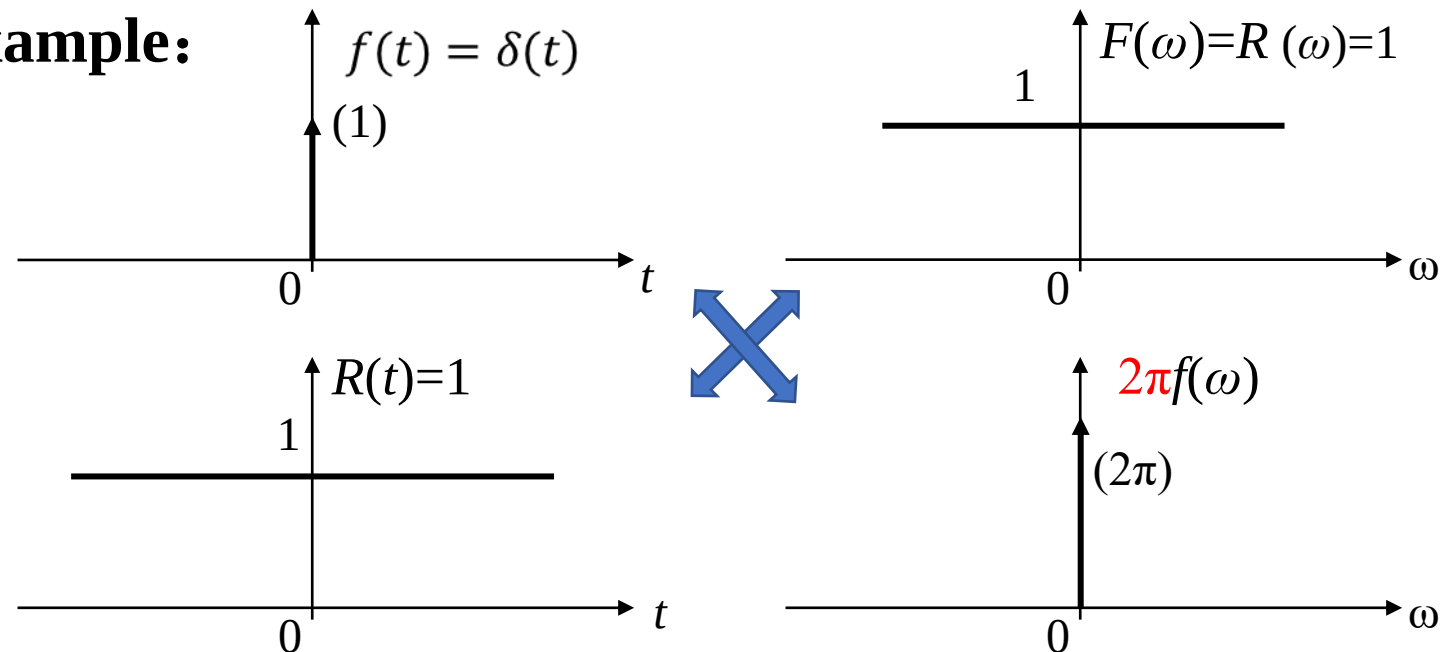
3.7.1 Linearity

if $F[f_1(t)] = F_1(\omega)$, $F[f_2(t)] = F_2(\omega)$,

then $F[a_1 f_1(t) + a_2 f_2(t)] = a_1 F_1(\omega) + a_2 F_2(\omega)$

3.7.2 Symmetry

Example:



3.7 Basic Properties of Fourier Transform



Using the symmetry of Fourier Transform, the problem of finding the inverse Fourier Transform can be converted into finding the Fourier Transform.

$$\text{if } \mathcal{F}[f(t)] = F(\omega) \quad \text{then } \mathcal{F}[F(t)] = 2\pi f(-\omega)$$

$$f(t) \leftrightarrow F(\omega)$$

$$F(t) \leftrightarrow 2\pi f(\omega) \quad (f \text{ is an even function})$$

$$F(t) \leftrightarrow -2\pi f(\omega) \quad (f \text{ is an odd function})$$

$$f(t) \leftrightarrow F(\omega)$$

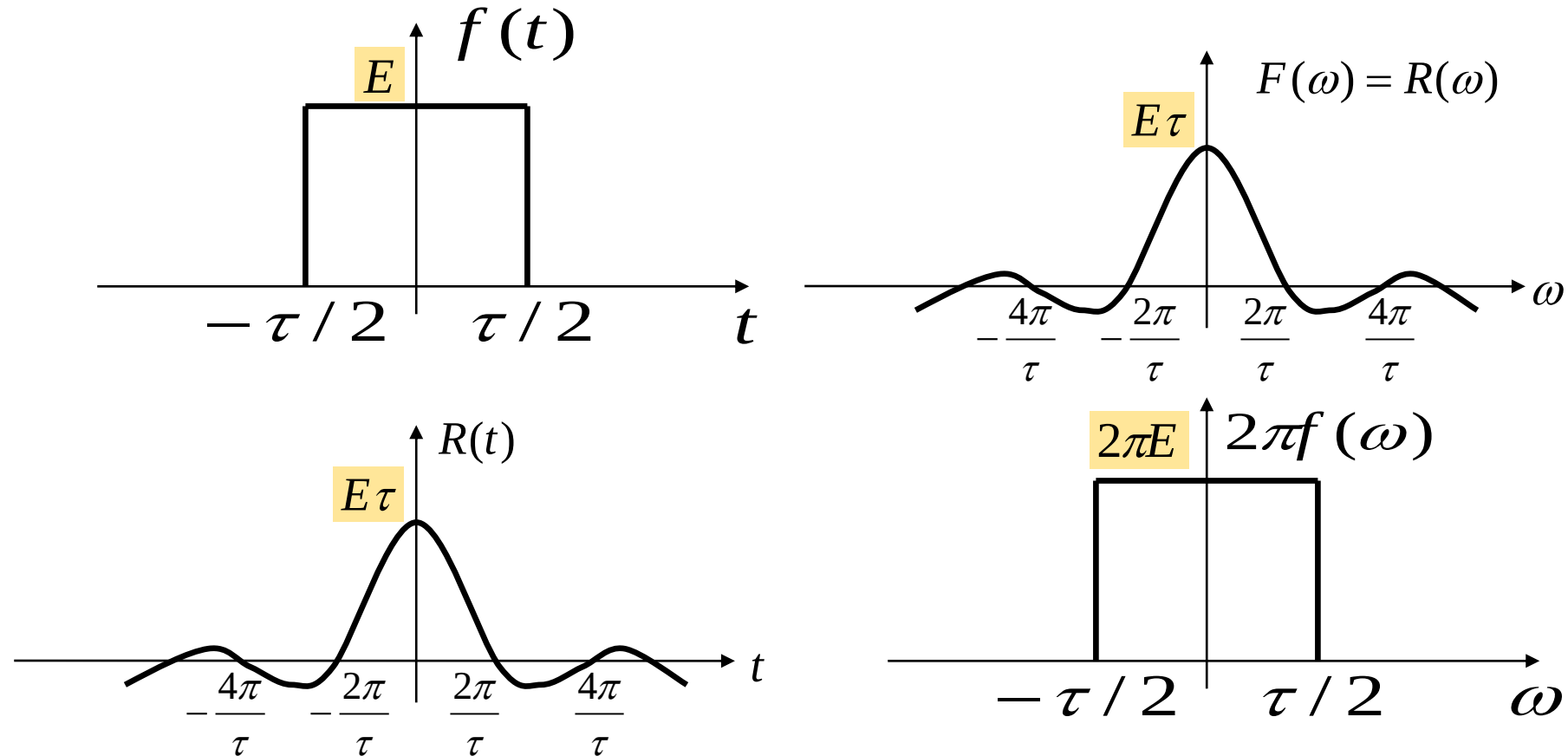
$$\frac{F(t)}{2\pi} \leftrightarrow f(\omega) \quad (f \text{ is an even function})$$

$$\frac{-F(t)}{2\pi} \leftrightarrow f(\omega) \quad (f \text{ is an odd function})$$

3.7 Basic Properties of Fourier Transform



Symmetry Property Example: Gate Function & Sampling Function



$$f(t) = E\left[u\left(t + \frac{\tau}{2}\right) - u\left(t - \frac{\tau}{2}\right)\right] \rightarrow R(\omega) = E\tau \text{Sa}\left(\frac{\omega\tau}{2}\right)$$

$$R(t) = E\tau \text{Sa}\left(\frac{t\tau}{2}\right) \rightarrow F[R(t)] = 2\pi E\left[u\left(\omega + \frac{\tau}{2}\right) - u\left(\omega - \frac{\tau}{2}\right)\right]$$

Given $F[\text{sgn}(t)] = \frac{2}{j\omega}$, find $F^{-1}[j\pi \text{sgn}(\omega)]$

A $\frac{1}{t}$

B $\frac{2\pi}{t}$

C $-\frac{2\pi}{t}$

D $-\frac{1}{t}$

提交

3.7 Basic Properties of Fourier Transform



Example 3-2: Calculate $\mathcal{F}^{-1}[j\pi \operatorname{sgn}(\omega)]$.

Solution: $\because F(t) = j\pi \operatorname{sgn}(t)$

$$\mathcal{F}[F(t)] = j\pi \frac{2}{j\omega} = \frac{2\pi}{\omega}$$

$$\therefore \mathcal{F}^{-1}[j\pi \operatorname{sgn}(\omega)] = \frac{1}{2\pi} \mathcal{F}[F(t)] \Big|_{\omega=-t} = \frac{1}{2\pi} \left[\frac{2\pi}{\omega} \right] \Big|_{\omega=-t} = -\frac{1}{t}$$

$$j\pi \operatorname{sgn}(t) \leftrightarrow \frac{2\pi}{\omega}$$

$$\frac{-1}{2\pi} \left(\frac{2\pi}{t} \right) \leftrightarrow j\pi \operatorname{sgn}(\omega)$$

$$-\frac{1}{t} \leftrightarrow j\pi \operatorname{sgn}(\omega)$$

$\operatorname{sgn}(t)$ is an odd function.

3.7.3 Parity, Realness, and Imaginariness

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt = \int_{-\infty}^{\infty} f(t)\cos(\omega t)dt - j \int_{-\infty}^{\infty} f(t)\sin(\omega t)dt = R(\omega) + jX(\omega)$$

$$\left\{ \begin{array}{l} R(\omega) = \int_{-\infty}^{\infty} f(t)\cos(\omega t)dt \\ X(\omega) = -\int_{-\infty}^{\infty} f(t)\sin(\omega t)dt \end{array} \right. \quad \begin{array}{l} \text{If } f(t) \text{ is a real-valued function, then:} \\ \text{➤ The real part of } F(\omega) \text{ (} R(\omega) \text{) is even;} \\ \text{➤ The imaginary part of } F(\omega) \text{ (} X(\omega) \text{) is odd.} \end{array}$$

Case 1: $f(t)$ is real and even ($f(-t) = f(t)$)

$\sin(\omega t)$ is odd, so $X(\omega) = 0$; **$F(\omega) = R(\omega)$ (real, even).**

Example: $f(t) = e^{-\alpha|t|} \rightarrow F(\omega) = \frac{2\alpha}{\alpha^2 + \omega^2}$.

Case 2: $f(t)$ is real and odd ($f(-t) = -f(t)$)

$\cos(\omega t)$ is even, so $R(\omega) = 0$; **$F(\omega) = jX(\omega)$ (purely imaginary, odd).**

Example: $f(t) = \text{sgn}(t) \rightarrow F(\omega) = \frac{2}{j\omega}$.

3.7 Basic Properties of Fourier Transform

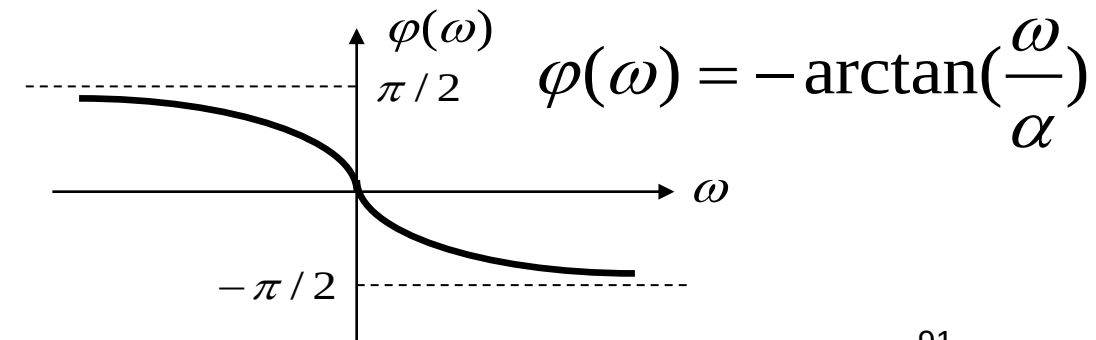
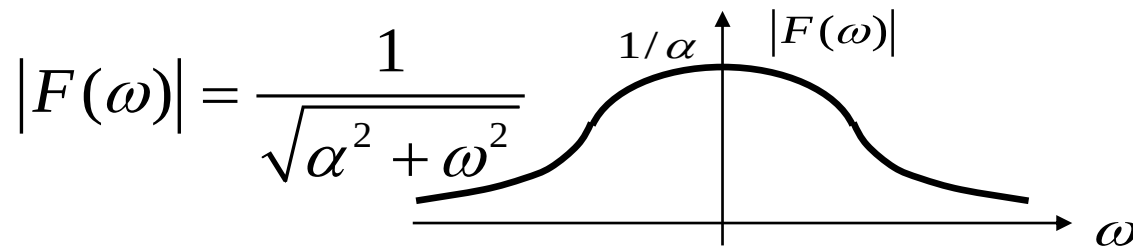
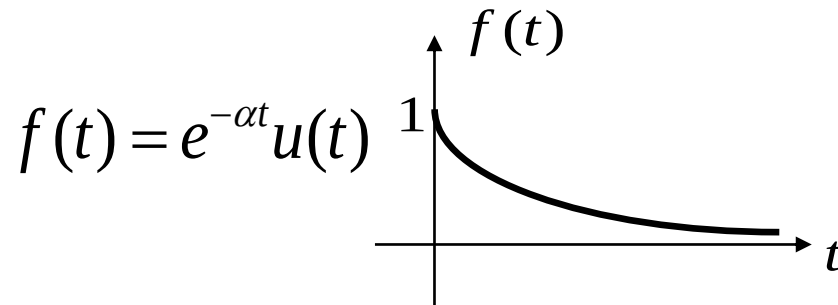


$$F(\omega) = |F(\omega)| e^{j\varphi(\omega)} = R(\omega) + jX(\omega)$$

$$|F(\omega)| = \sqrt{R^2(\omega) + X^2(\omega)}, \quad \varphi(\omega) = \arctan \frac{X(\omega)}{R(\omega)}$$

If $f(t)$ is real or purely imaginary ($f(t) = jg(t)$, $g(t)$ real), then $|F(\omega)|$ is even, and $\varphi(\omega)$ is odd.

Example:



3.7.4 Shift Property

(1) Time-Shift Property

If $F(\omega) = \mathcal{F}[f(t)]$

$$\begin{aligned}\text{Then } \mathcal{F}[f(t - t_0)] &= \int_{-\infty}^{\infty} f(t - t_0) e^{-j\omega t} dt = e^{-j\omega t_0} \int_{-\infty}^{\infty} f(t - t_0) e^{-j\omega(t - t_0)} d(t - t_0) \\ &= F(\omega) e^{-j\omega t_0} = |F(\omega)| e^{j\phi(\omega)} e^{-j\omega t_0}\end{aligned}$$

**Amplitude spectrum remains unchanged.
Phase spectrum has a phase shift of $-\omega t_0$
(proportional to frequency).**

Similarly, $\mathcal{F}[f(t + t_0)] = F(\omega) e^{j\omega t_0}$

3.7 Basic Properties of Fourier Transform

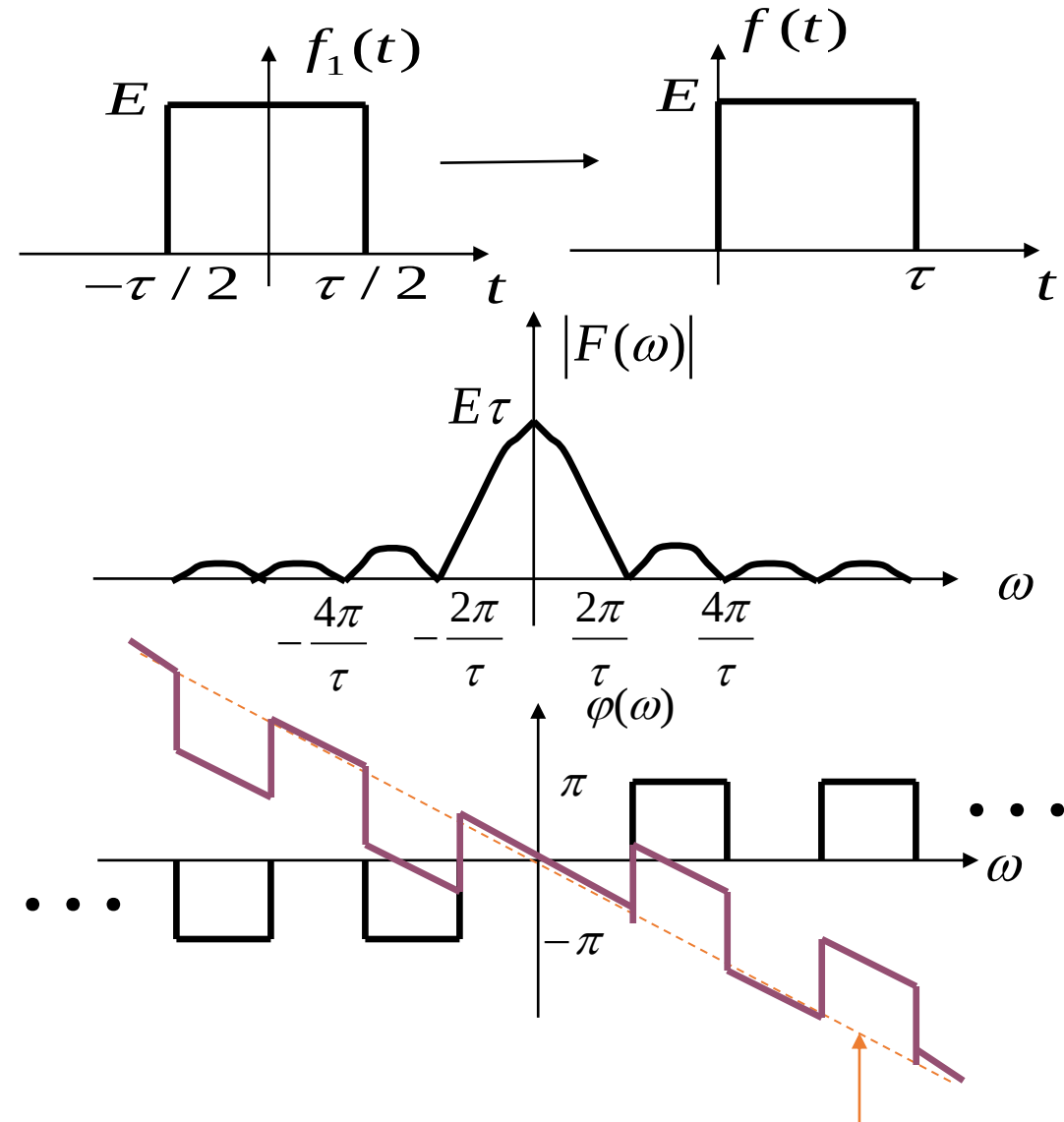
Example 3-3: Find the spectral function of the single-sided rectangular pulse signal $f(t)$ shown in the figure.

Solution: $f(t)$ is the right time shift of the symmetric gate function $f_1(t)$ by $\tau/2$.

$$F_1(\omega) = E\tau \text{Sa}\left(\frac{\omega\tau}{2}\right)$$

$$F(\omega) = E\tau \text{Sa}\left(\frac{\omega\tau}{2}\right)e^{-j\omega\frac{\tau}{2}}$$

The amplitude spectrum remains unchanged. The phase spectrum has a phase shift of $-\omega\tau/2$.



Find the Fourier Transform of $x(t) = e^{-t+2}u(t-2)$

A $e^{-j2\omega} \frac{1}{1+j2\omega}$

B $e^{-j2\omega} \frac{1}{1+j\omega}$

C $e^{-j\omega} \frac{1}{1+j2\omega}$

D $2e^{-j\omega} \frac{1}{1+j\omega}$

提交

3.7 Basic Properties of Fourier Transform



Solution:

Step 1: Fourier Transform of Basic Exponential Signal

Let $x_1(t) = e^{-t}u(t)$. Its Fourier Transform is:

$$\mathcal{F}[x_1(t)] = \frac{1}{1 + j\omega}$$

Step 2: Express $x(t)$ as Time-Shifted $x_1(t)$

$$x(t) = e^{-(t-2)}u(t-2) = x_1(t-2)$$

(Shift $x_1(t)$ to the right by $t_0 = 2$.)

Step 3: Apply Time-Shift Property

$$\mathcal{F}[x(t)] = \mathcal{F}[x_1(t-2)] = \mathcal{F}[x_1(t)] \cdot e^{-j\omega \cdot 2} = \frac{e^{-j2\omega}}{1 + j\omega}$$

(2) Frequency-Shift Property (Modulation Principle)

If $\mathcal{F}[f(t)] = F(\omega)$, then:

$$\begin{aligned}\mathcal{F}[f(t)e^{j\omega_0 t}] &= F(\omega - \omega_0) \\ \mathcal{F}[f(t)e^{-j\omega_0 t}] &= F(\omega + \omega_0)\end{aligned}$$

Proof:

$$\mathcal{F}[f(t)e^{j\omega_0 t}] = \int_{-\infty}^{+\infty} f(t)e^{j\omega_0 t} \cdot e^{-j\omega t} dt = \int_{-\infty}^{+\infty} f(t)e^{-j(\omega - \omega_0)t} dt = F(\omega - \omega_0)$$

Example 3-4: Find the Fourier transforms of $e^{j\omega_0 t}$, $\cos \omega_0 t$ and $\sin \omega_0 t$ by the frequency-shift property.

Solution: $F[1] = 2\pi\delta(\omega)$ $F[e^{j\omega_0 t}] = 2\pi\delta(\omega - \omega_0)$

$$\because \cos \omega_0 t = \frac{1}{2}[e^{j\omega_0 t} + e^{-j\omega_0 t}] \quad \therefore F[\cos \omega_0 t] = \pi[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$$

$$\because \sin \omega_0 t = \frac{1}{2j}[e^{j\omega_0 t} - e^{-j\omega_0 t}] \quad \therefore F[\sin \omega_0 t] = j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$$

3.7 Basic Properties of Fourier Transform



Modulation (Key Application)

If $F(\omega) = \mathcal{F}[f(t)]$

Then $\mathcal{F}[f(t)e^{j\omega_0 t}] = F(\omega - \omega_0)$ $\mathcal{F}[f(t)e^{-j\omega_0 t}] = F(\omega + \omega_0)$

$$\because \cos \omega_0 t = \frac{1}{2}[e^{j\omega_0 t} + e^{-j\omega_0 t}] \quad \sin \omega_0 t = \frac{1}{2j}[e^{j\omega_0 t} - e^{-j\omega_0 t}]$$

$$\therefore \mathcal{F}[f(t) \cos \omega_0 t] = \frac{1}{2}[F(\omega + \omega_0) + F(\omega - \omega_0)]$$

$$\mathcal{F}[f(t) \sin \omega_0 t] = \frac{j}{2}[F(\omega + \omega_0) - F(\omega - \omega_0)]$$

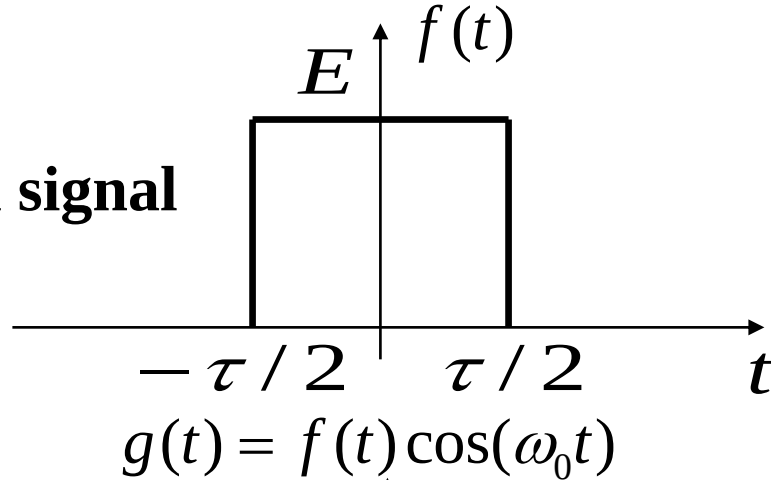
Physical Meaning: The spectrum of $f(t)$ is shifted to $\pm\omega_0$ (**modulation shifts the baseband spectrum to the carrier frequency band**).

3.7 Basic Properties of Fourier Transform

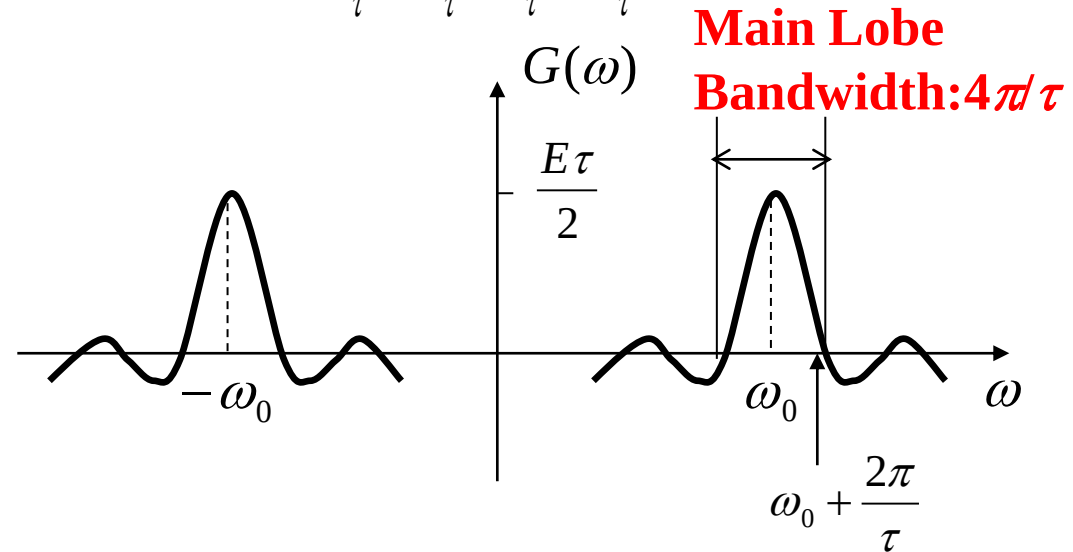
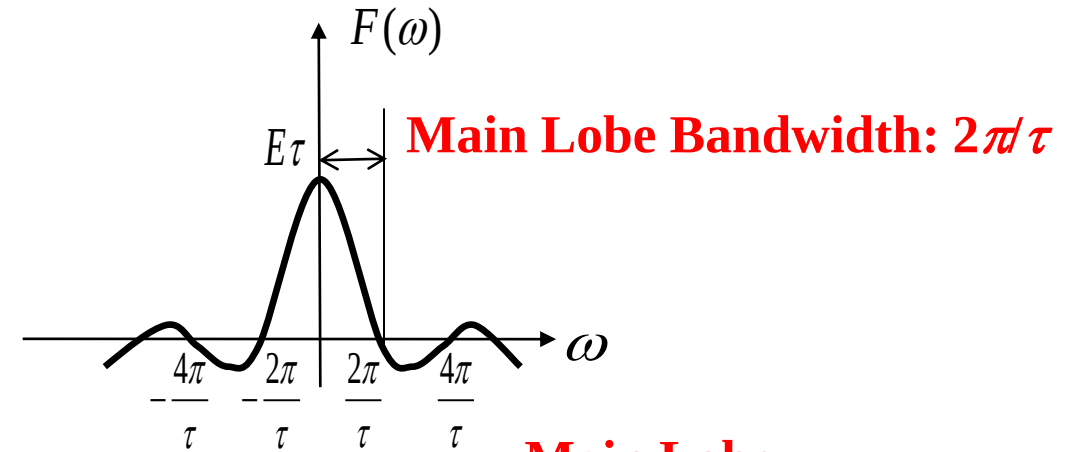
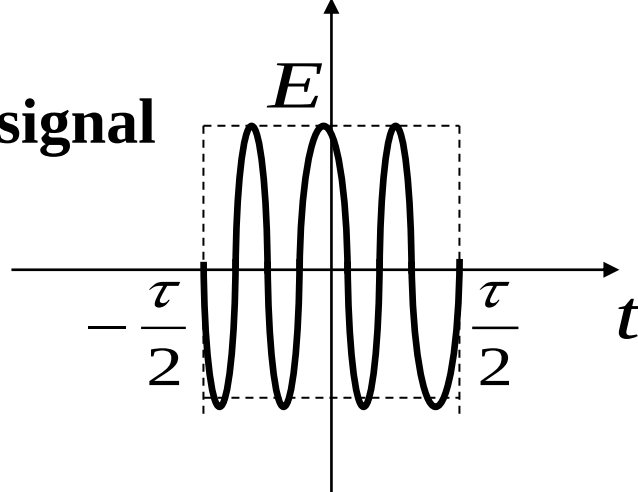
Example 3-5: Find the Fourier transform of a rectangular pulse Amplitude Shift Keying (ASK) signal and draw the spectrum, with pulse amplitude E , pulse width τ and carrier frequency ω_0 .

Solution:

Baseband signal



Modulated signal



3.7 Basic Properties of Fourier Transform



$$F(\omega) = E\tau \text{Sa}\left(\frac{\omega\tau}{2}\right)$$

$$\begin{aligned} G(\omega) &= \frac{1}{2}[F(\omega + \omega_0) + F(\omega - \omega_0)] \\ &= \frac{E\tau}{2} \left\{ \text{Sa}\left[(\omega + \omega_0)\frac{\tau}{2}\right] + \text{Sa}\left[(\omega - \omega_0)\frac{\tau}{2}\right] \right\} \end{aligned}$$

Conclusion: During modulation, (1) bandwidth doubles; (2) spectral density halves.

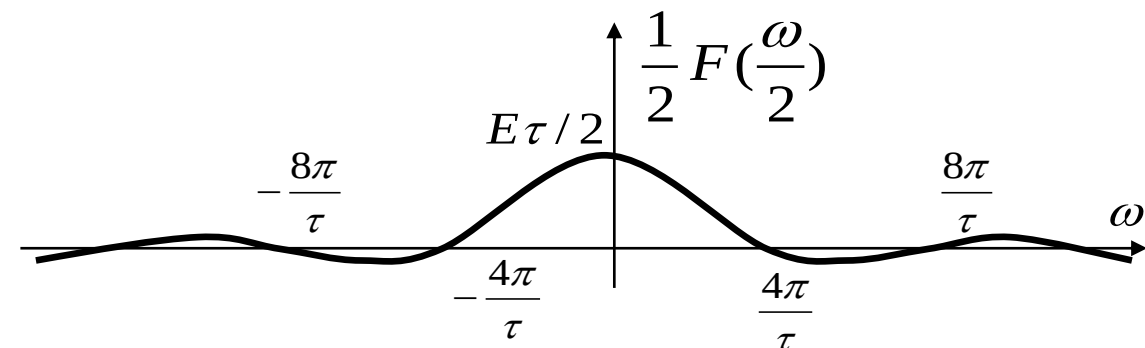
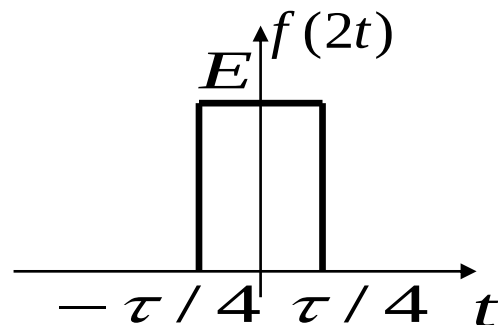
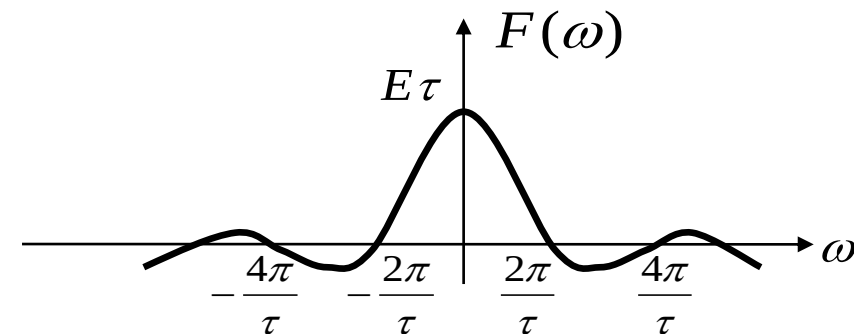
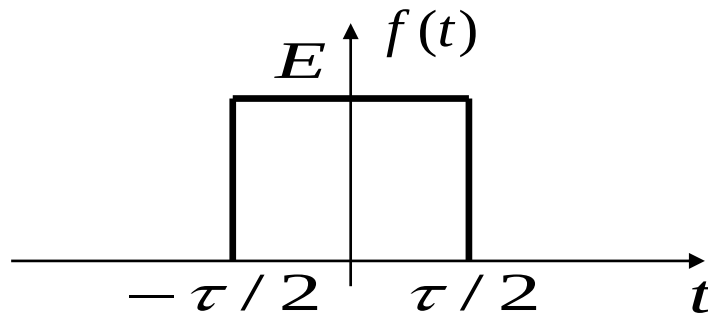
3.7 Basic Properties of Fourier Transform



3.7.5 Scaling Property

$$\text{If } F(\omega) = \mathcal{F}[f(t)] \quad \text{Then } \mathcal{F}[f(at)] = \frac{1}{|a|} F\left(\frac{\omega}{a}\right)$$

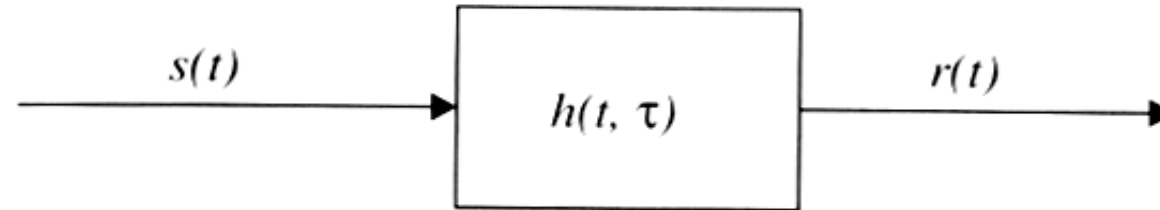
Compressing a signal in time domain by a ($a > 1$) times (signal changes faster) is equivalent to **expanding** the spectrum by a times in frequency domain (more high-frequency components) and reducing the spectral amplitude by $1/a$ (energy conservation). Similarly, **time expansion is equivalent to frequency compression**.



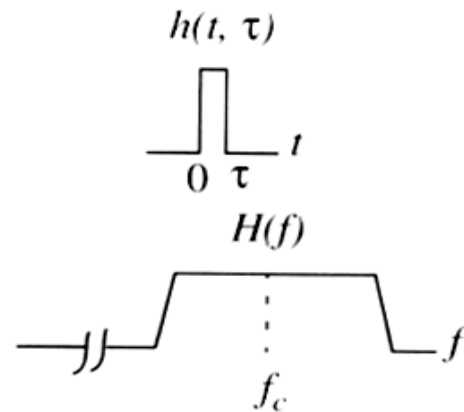
3.7 Basic Properties of Fourier Transform



Example: Time Expansion and Frequency Compression of Communication Channels

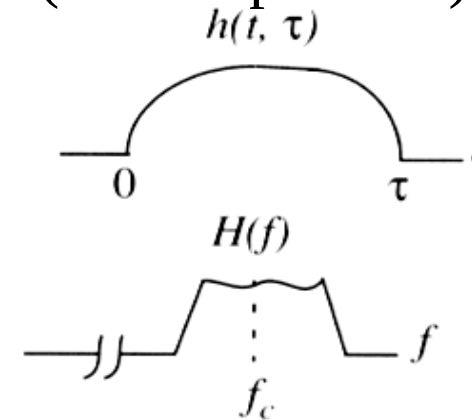


Small channel delay



Large channel bandwidth

larger channel delay
(time expansion)



Smaller bandwidth
(frequency compression)

3.7 Basic Properties of Fourier Transform



Special case: $F[f(-t)] = F(-\omega)$ (**time reversal, $a = -1$**)

Combined Property: Scaling + Time Shift

$$F[f(at - t_0)] = \frac{1}{|a|} F\left(\frac{\omega}{a}\right) e^{-j\frac{\omega t_0}{a}}$$

Proof:

	$f(t) \leftrightarrow F(\omega)$
Scaling	$f(at) \leftrightarrow \frac{1}{ a } F\left(\frac{\omega}{a}\right)$
Time shift	$f\left[a\left(t - \frac{t_0}{a}\right)\right] \leftrightarrow \frac{1}{ a } F\left(\frac{\omega}{a}\right) e^{-j\frac{\omega t_0}{a}}$

	$f(t) \leftrightarrow F(\omega)$
Time shift	$f(t - t_0) \leftrightarrow F(\omega) e^{-j\omega t_0}$
Scaling	$f(at - t_0) \leftrightarrow \frac{1}{ a } F\left(\frac{\omega}{a}\right) e^{-j\frac{\omega t_0}{a}}$

3.7.6 Differentiation and Integration Properties

(1) Time-Domain Differentiation Property

If $F(\omega) = \mathcal{F}[f(t)]$

Then $\mathcal{F}\left[\frac{df(t)}{dt}\right] = j\omega F(\omega), \quad \mathcal{F}\left[\frac{d^n f(t)}{dt^n}\right] = (j\omega)^n F(\omega)$

Proof:
$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

Differentiate both sides with respect to t:

$$\frac{d}{dt} f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) (j\omega) e^{j\omega t} d\omega$$

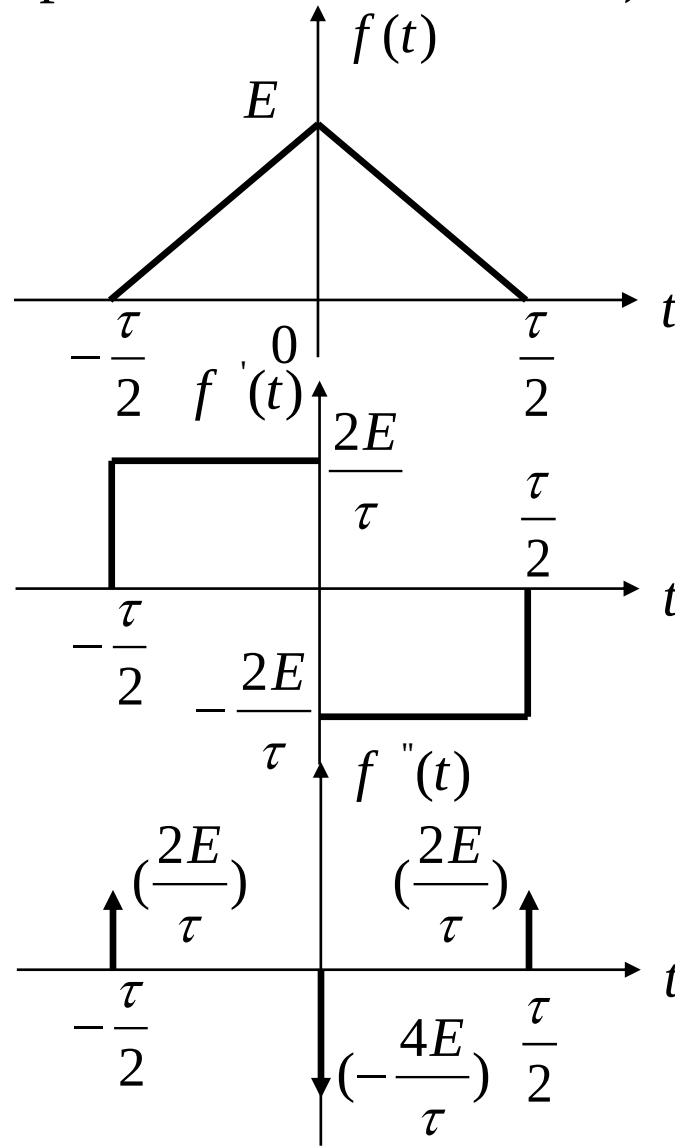
$$\mathcal{F}[f'(t)] = j\omega F(\omega)$$

Example: $\mathcal{F}[\delta(t)] = 1 \quad \mathcal{F}[\delta'(t)] = j\omega \quad \mathcal{F}[\delta^{(n)}(t)] = (j\omega)^n$

3.7 Basic Properties of Fourier Transform



Example 3-6: Find the Fourier Transform of the triangular pulse $f(t)$ shown in the figure (amplitude E , base width τ).



Solution:

$$f_1(t) = f'(t) = \begin{cases} \frac{2E}{\tau} & -\frac{\tau}{2} < t < 0 \\ -\frac{2E}{\tau} & 0 < t < \frac{\tau}{2} \\ 0 & |t| > \frac{\tau}{2} \end{cases}$$

$$f_2(t) = f''(t) = \frac{2E}{\tau} [\delta(t + \frac{\tau}{2}) + \delta(t - \frac{\tau}{2}) - 2\delta(t)]$$

Take Fourier transform of both sides,

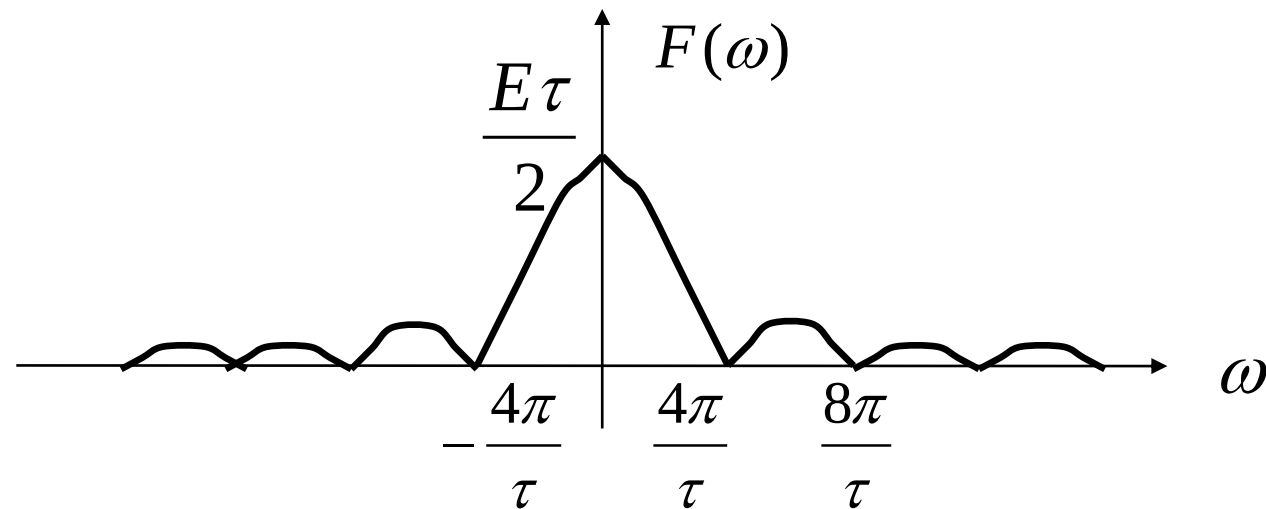
$$\begin{aligned} F_2(\omega) &= \frac{2E}{\tau} (e^{j\omega\frac{\tau}{2}} + e^{-j\omega\frac{\tau}{2}} - 2) = \frac{4E}{\tau} \left[\cos\left(\omega\frac{\tau}{2}\right) - 1 \right] \\ &= -\frac{\omega^2 E \tau}{2} \text{Sa}^2\left(\frac{\omega\tau}{4}\right) \end{aligned}$$

3.7 Basic Properties of Fourier Transform



$$F_2(\omega) = (j\omega)^2 F(\omega) = -\frac{\omega^2 E\tau}{2} \text{Sa}^2\left(\frac{\omega\tau}{4}\right)$$

$$\therefore F(\omega) = \frac{E\tau}{2} \text{Sa}^2\left(\frac{\omega\tau}{4}\right)$$



(2) Time-Domain Integration Property

$$\text{If } F(\omega) = \mathbf{F}[f(t)]$$

$$\text{Then } \mathbf{F}\left[\int_{-\infty}^t f(\tau)d\tau\right] = \frac{F(\omega)}{j\omega} + \pi F(0)\delta(\omega)$$

$$\text{where } F(0) = F(\omega)|_{\omega=0} \quad \text{DC Component}$$

$$\text{If } F(0) = 0 \quad \text{no DC Component}$$

$$\text{Then } \mathbf{F}\left[\int_{-\infty}^t f(\tau)d\tau\right] = \frac{F(\omega)}{j\omega}$$

(3) Frequency-Domain Differentiation Property

$$\text{If } F(\omega) = \mathcal{F}[f(t)]$$

$$\text{Then } \mathcal{F}[(-jt)f(t)] = \frac{dF(\omega)}{d\omega},$$

$$\mathcal{F}[(-jt)^n f(t)] = \frac{d^n F(\omega)}{d\omega^n}$$

$$\mathcal{F}[t^n f(t)] = j^n \frac{d^n F(\omega)}{d\omega^n}$$

Example: $\because \mathcal{F}[1] = 2\pi\delta(\omega)$

$$\therefore \mathcal{F}[t] = j2\pi\delta'(\omega)$$

$$\mathcal{F}[t^n] = 2\pi j^n \delta^{(n)}(\omega)$$

(4) Frequency-Domain Integration Property

$$\text{If } F(\omega) = \mathbf{F}[f(t)]$$

$$\text{Then } \mathbf{F}^{-1}\left[\int_{-\infty}^{\omega} F(u)du\right] = \frac{f(t)}{-jt} + \pi f(0)\delta(t)$$

$$\text{If } f(0) = 0$$

$$\text{Then } \mathbf{F}^{-1}\left[\int_{-\infty}^{\omega} F(u)du\right] = \frac{f(t)}{-jt}$$

- Repeat the examples in the lecture notes.
- 3.58(a)(b) and 3.62(c) in the textbook.

Review of Last Lecture

3.7 Basic Properties of Fourier Transform

3.7 Basic Properties of Fourier Transform



3.7.1 Linearity

if $F[f_1(t)] = F_1(\omega)$, $F[f_2(t)] = F_2(\omega)$,

then $F[a_1 f_1(t) + a_2 f_2(t)] = a_1 F_1(\omega) + a_2 F_2(\omega)$

3.7.2 Symmetry

if $F[f(t)] = F(\omega)$

then $F[F(t)] = 2\pi f(-\omega)$

$$f(t) \leftrightarrow F(\omega)$$

$$\frac{F(t)}{2\pi} \leftrightarrow f(\omega) \quad (f \text{ is an even function})$$

$$\frac{-F(t)}{2\pi} \leftrightarrow f(\omega) \quad (f \text{ is an odd function})$$

$$f(t) \leftrightarrow F(\omega)$$

$$F(t) \leftrightarrow 2\pi f(\omega) \quad (f \text{ is an even function})$$

$$F(t) \leftrightarrow -2\pi f(\omega) \quad (f \text{ is an odd function})$$

3.7.3 Parity, Realness, and Imaginariness

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt = \int_{-\infty}^{\infty} f(t) \cos(\omega t) dt - j \int_{-\infty}^{\infty} f(t) \sin(\omega t) dt = R(\omega) + jX(\omega)$$

$$\left\{ \begin{array}{l} R(\omega) = \int_{-\infty}^{\infty} f(t) \cos(\omega t) dt \\ X(\omega) = - \int_{-\infty}^{\infty} f(t) \sin(\omega t) dt \end{array} \right. \quad \begin{array}{l} \text{If } f(t) \text{ is a real-valued function, then:} \\ \text{➤ The real part of } F(\omega) \text{ (} R(\omega) \text{) is even;} \\ \text{➤ The imaginary part of } F(\omega) \text{ (} X(\omega) \text{) is odd.} \end{array}$$

Case 1: $f(t)$ is real and even, $F(\omega) = R(\omega)$ (real, even).

Case 2: $f(t)$ is real and odd, $F(\omega) = jX(\omega)$ (purely imaginary, odd).

$$|F(\omega)| = \sqrt{R^2(\omega) + X^2(\omega)}, \quad \varphi(\omega) = \arctan \frac{X(\omega)}{R(\omega)}$$

If $f(t)$ is real or purely imaginary ($f(t) = jg(t)$, $g(t)$ real), then $|F(\omega)|$ is even, and $\varphi(\omega)$ is odd.

3.7.4 Shift Property

(1) Time-Shift Property

If $F(\omega) = \mathcal{F}[f(t)]$ **Then** $\mathcal{F}[f(t - t_0)] = F(\omega)e^{-j\omega t_0} = |F(\omega)|e^{j\varphi(\omega)}e^{-j\omega t_0}$

Amplitude spectrum remains unchanged. Phase spectrum has a phase shift of $-\omega t_0$.

Similarly, $\mathcal{F}[f(t + t_0)] = F(\omega)e^{j\omega t_0}$

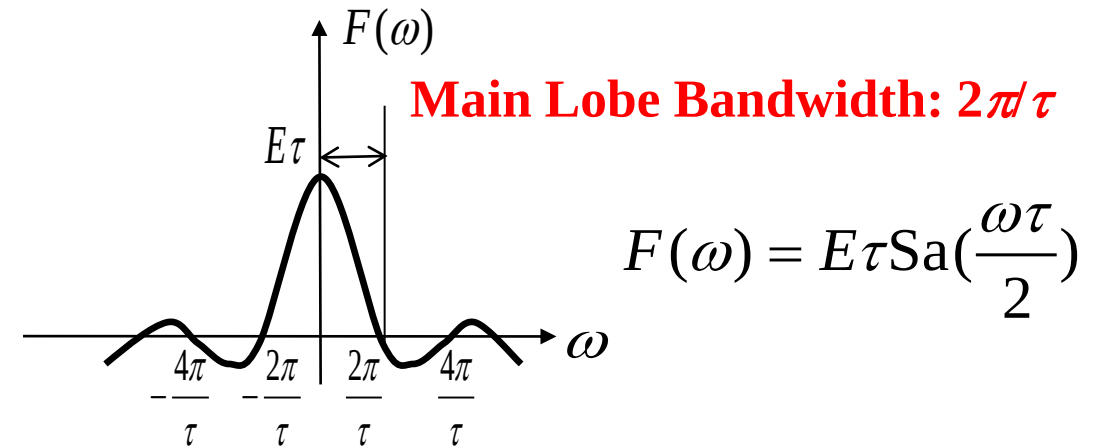
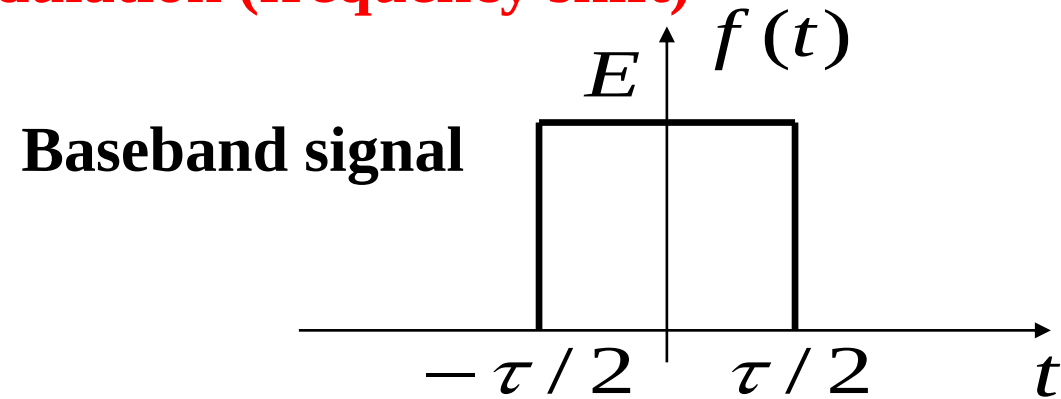
(2) Frequency-Shift Property (Modulation Principle)

If $\mathcal{F}[f(t)] = F(\omega)$, then $\mathcal{F}[f(t)e^{j\omega_0 t}] = F(\omega - \omega_0)$, $\mathcal{F}[f(t)e^{-j\omega_0 t}] = F(\omega + \omega_0)$

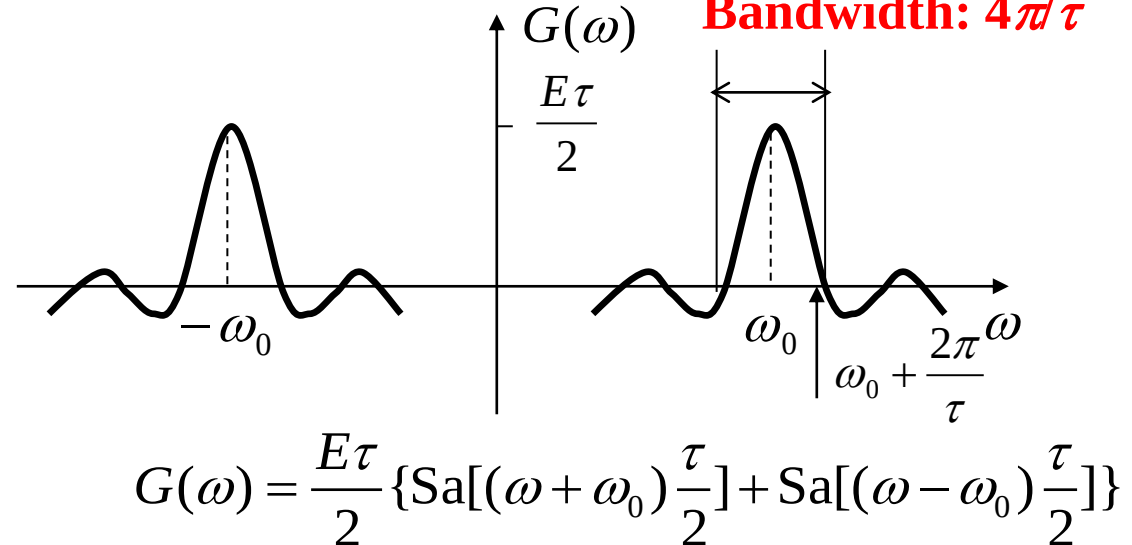
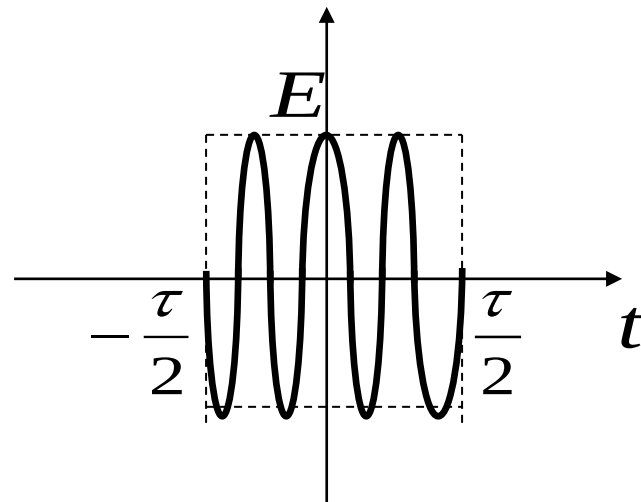
$$\mathcal{F}[f(t)\cos\omega_0 t] = \frac{1}{2}[F(\omega + \omega_0) + F(\omega - \omega_0)] \quad \mathcal{F}[f(t)\sin\omega_0 t] = \frac{j}{2}[F(\omega + \omega_0) - F(\omega - \omega_0)]$$

3.7 Basic Properties of Fourier Transform

Modulation (frequency shift)



ASK signal $g(t) = f(t) \cos(\omega_0 t)$



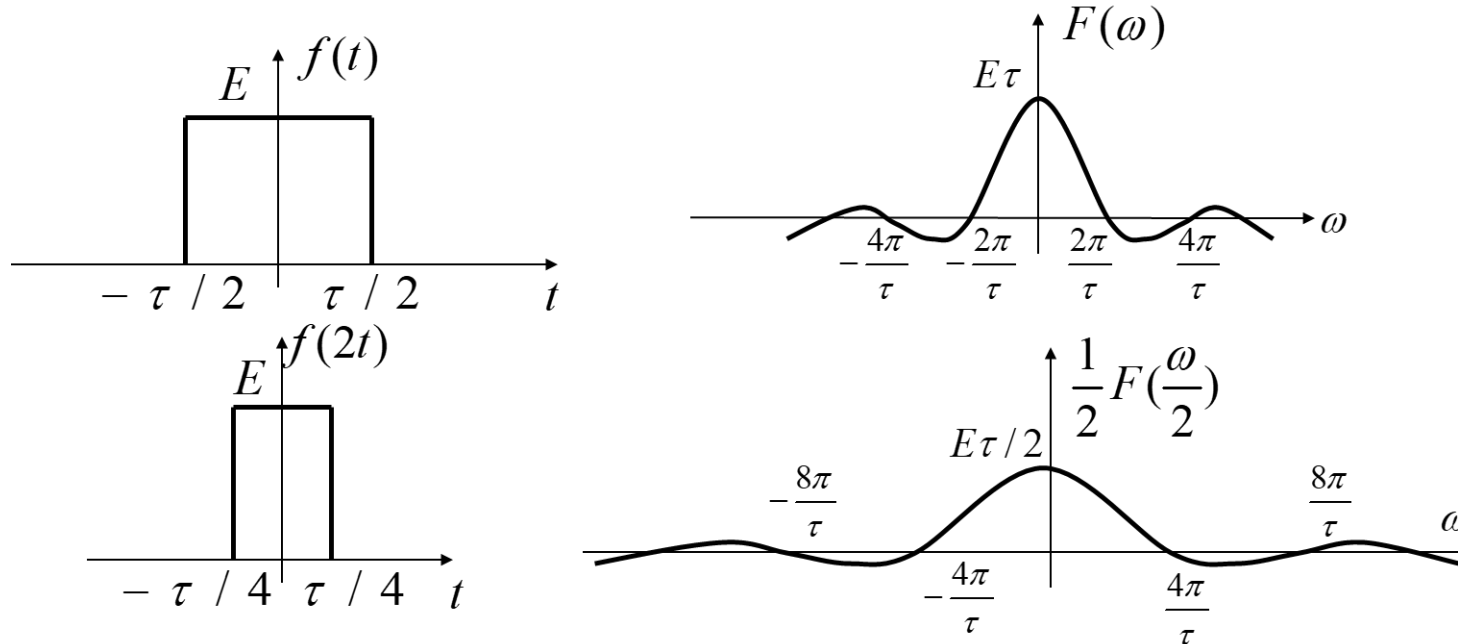
During modulation, (1) bandwidth doubles; (2) spectral density halves.

3.7 Basic Properties of Fourier Transform

3.7.5 Scaling Property

If $F(\omega) = \mathcal{F}[f(t)]$ Then $\mathcal{F}[f(at)] = \frac{1}{|a|} F\left(\frac{\omega}{a}\right)$

Compressing a signal in time domain by a ($a > 1$) times (signal changes faster) is equivalent to **expanding** the spectrum by a times in frequency domain (more high-frequency components) and reducing the spectral amplitude by $1/a$ (energy conservation). Similarly, **time expansion is equivalent to frequency compression**.



Special case: $\mathcal{F}[f(-t)] = F(-\omega)$
(**time reversal, $a = -1$**)

Combined Property:
Scaling + Time Shift

$$\mathcal{F}[f(at - t_0)] = \frac{1}{|a|} F\left(\frac{\omega}{a}\right) e^{-j\frac{\omega t_0}{a}}$$

3.7 Basic Properties of Fourier Transform



3.7.6 Differentiation and Integration Properties

(1) Time-Domain Differentiation Property

$$\text{If } F(\omega) = \mathcal{F}[f(t)]$$

$$\text{Then } \mathcal{F}\left[\frac{df(t)}{dt}\right] = j\omega F(\omega), \quad \mathcal{F}\left[\frac{d^n f(t)}{dt^n}\right] = (j\omega)^n F(\omega)$$

(2) Time-Domain Integration Property

$$\text{If } F(\omega) = \mathcal{F}[f(t)]$$

$$\text{Then } \mathcal{F}\left[\int_{-\infty}^t f(\tau) d\tau\right] = \frac{F(\omega)}{j\omega} + \pi F(0)\delta(\omega)$$

DC Component

A red arrow points from the text 'DC Component' to the term $F(0)$ in the equation above.

(3) Frequency-Domain Differentiation Property

If $F(\omega) = \mathbf{F}[f(t)]$

Then $\mathbf{F}[t^n f(t)] = j^n \frac{d^n F(\omega)}{d\omega^n}$

(4) Frequency-Domain Integration Property

If $F(\omega) = \mathbf{F}[f(t)]$

Then $\mathbf{F}^{-1}\left[\int_{-\infty}^{\omega} F(u) du\right] = \frac{f(t)}{-jt} + \pi f(0)\delta(t)$

Contents of This Lecture

3.8 Convolution Theorem

3.9 Fourier Transform of Periodic Signals

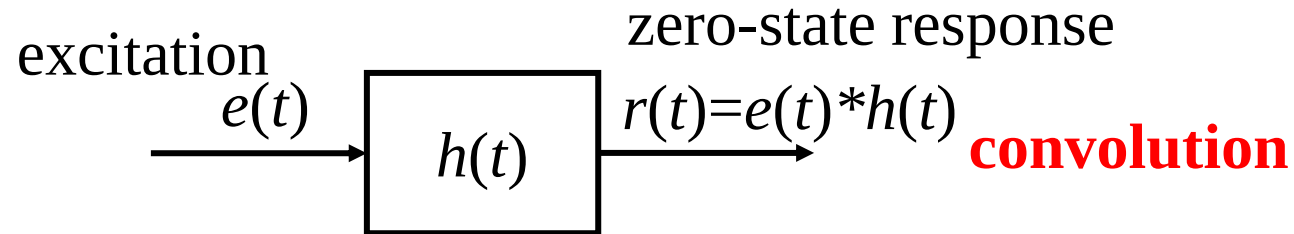
3.10 Fourier Transform of Sampled Signals

3.11 Sampling Theorem

Learning Objectives of This Lecture

1. Master proficiently the **Time-domain Convolution Theorem**, the **Frequency-domain Convolution Theorem** and their applications;
2. Be familiar with the relationship between the **Fourier Transform of periodic signals** and the Fourier Series;
3. Master proficiently the **Sampling Theorem** and its applications.

Convolution is a core technique in system analysis. The **zero-state response** of a system is the **convolution** of the **excitation** and the **impulse response**.



- **Four Steps** of Convolution Operation:

1. Time Reversal: $h(\tau) \rightarrow h(-\tau)$ **2. Time Shifting:** $h(-\tau) \rightarrow h(t - \tau)$ $\begin{cases} t < 0, & \text{left shift by } t \\ t > 0, & \text{right shift by } t \end{cases}$

3. Multiplication: $e(\tau)h(t - \tau)$ **4. Integration:** $e(t) * h(t) = \int_{-\infty}^{\infty} e(\tau)h(t - \tau)d\tau$ **Computation is complex!**

- **Convolution Theorem**—Convert complex time-domain computations into simple frequency-domain computations, and vice versa.
- **Sampling Theorem**—An application of the Convolution Theorem.

To convert a time-continuous piece of music into digital music in a CD, it is necessary to perform sampling (time discretization) on the analog signal. Which of the following sampling frequencies (unit: Hz, representing the number of samples per second) do you think achieves the highest music quality?

- ☐ A 5.5 kHz
- ☐ B 11 kHz
- ☐ C 22 kHz
- ☒ D 44 kHz

To convert a time-continuous piece of music into digital music in a CD, it is necessary to perform sampling (time discretization) on the analog signal. Which of the following sampling frequencies (unit: Hz, representing the number of samples per second) do you think achieves the highest music quality?

44 kHz Sampling Frequency 

22 kHz Sampling Frequency 

11 kHz Sampling Frequency 

5.5 kHz Sampling Frequency 

3.8 Convolution Theorem

(1) Time-domain Convolution Theorem

The time-domain convolution of two signals is equivalent to the multiplication of their spectra in the frequency domain.

$$\text{if } F_1(\omega) = \mathcal{F}[f_1(t)], \quad F_2(\omega) = \mathcal{F}[f_2(t)],$$

then

$$\mathcal{F}[f_1(t) * f_2(t)] = F_1(\omega)F_2(\omega)$$

(2) Frequency-domain Convolution Theorem

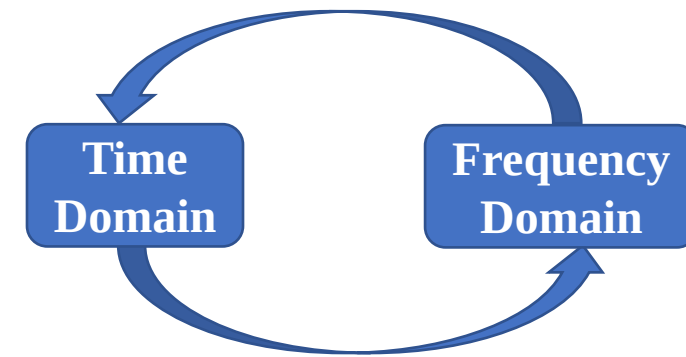
The time-domain convolution of two signals is equivalent to the multiplication of their spectra in the frequency domain.

$$\text{if } F_1(\omega) = \mathcal{F}[f_1(t)], \quad F_2(\omega) = \mathcal{F}[f_2(t)],$$

then

$$\mathcal{F}[f_1(t) \cdot f_2(t)] = \frac{1}{2\pi} F_1(\omega) * F_2(\omega)$$

$$\text{where: } F_1(\omega) * F_2(\omega) = \int_{-\infty}^{\infty} F_1(u)F_2(\omega - u)du$$



3.8 Convolution Theorem



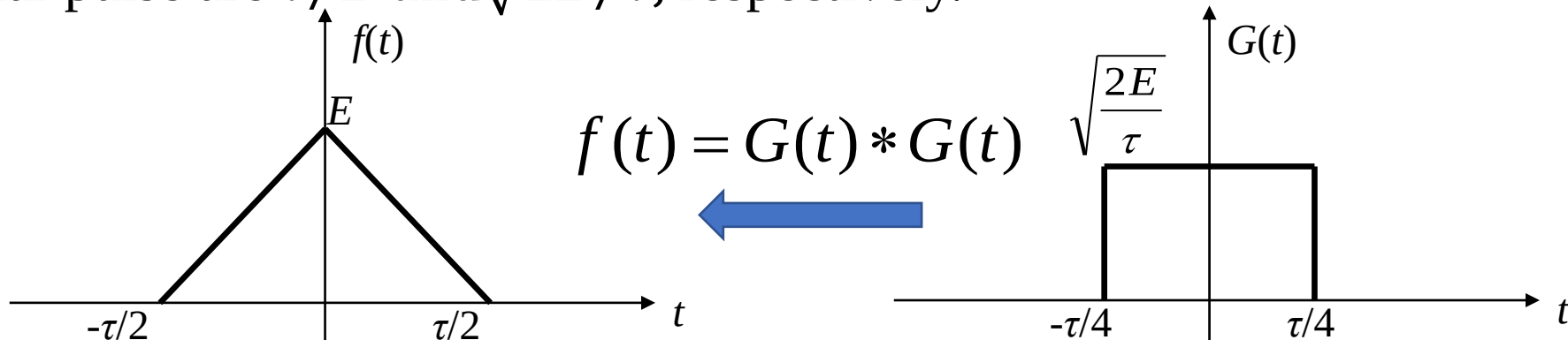
Example 3-7: Use the Time-domain Convolution Theorem to find the spectrum of a triangular pulse:

$$f(t) = \begin{cases} E(1 - \frac{2|t|}{\tau}) & |t| \leq \frac{\tau}{2} \\ 0 & |t| > \frac{\tau}{2} \end{cases}$$

Solution:

Step 1: Treat the triangular pulse as **the convolution of two identical rectangular pulses**.

The width of the triangular pulse is the sum of the widths of two rectangular pulses. The peak amplitude of the triangular pulse is the product of the square of the amplitudes of the two rectangular pulses and their width. Therefore, the width and amplitude of each rectangular pulse are $\tau/2$ and $\sqrt{2E/\tau}$, respectively.

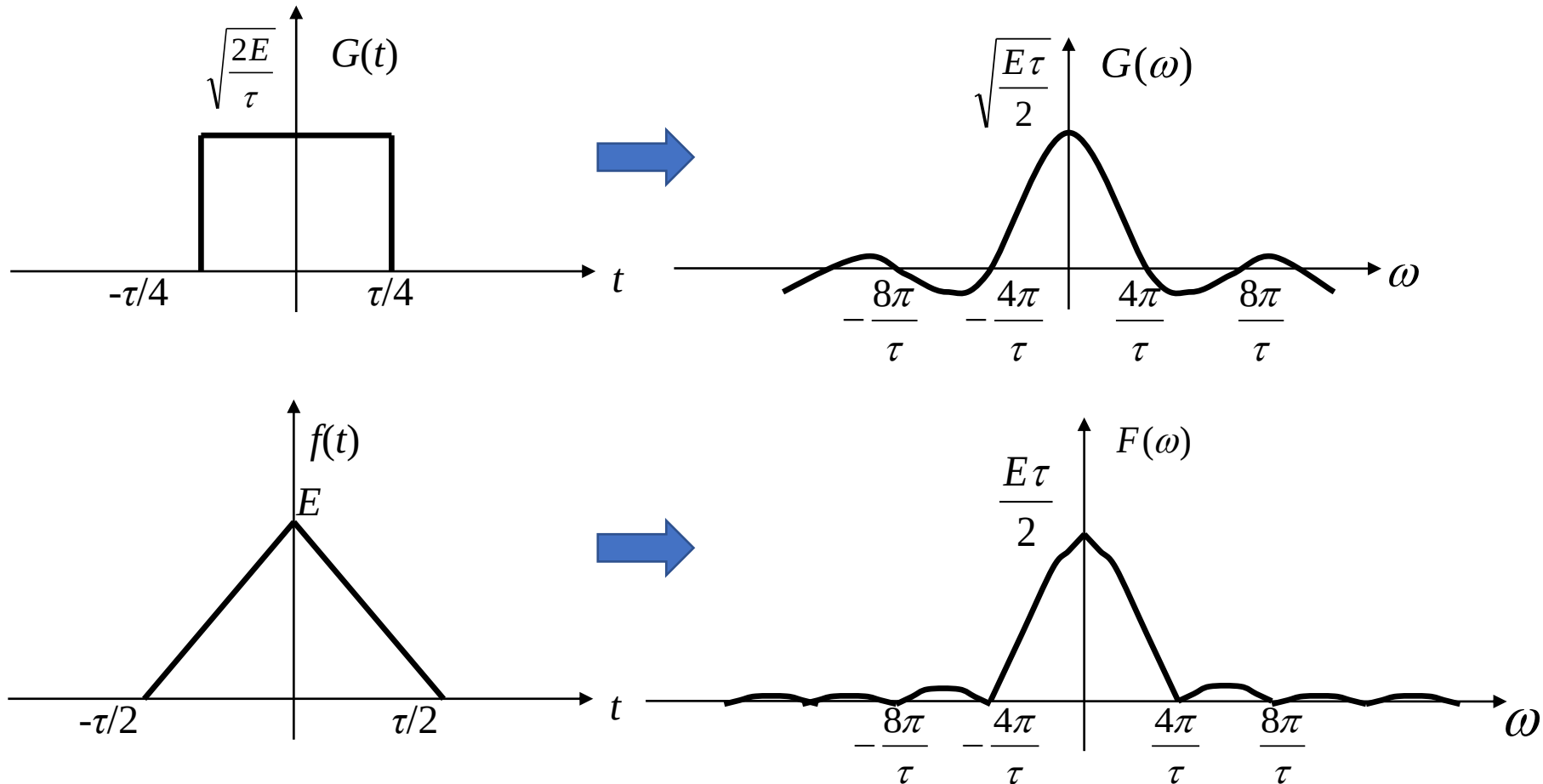


3.8 Convolution Theorem



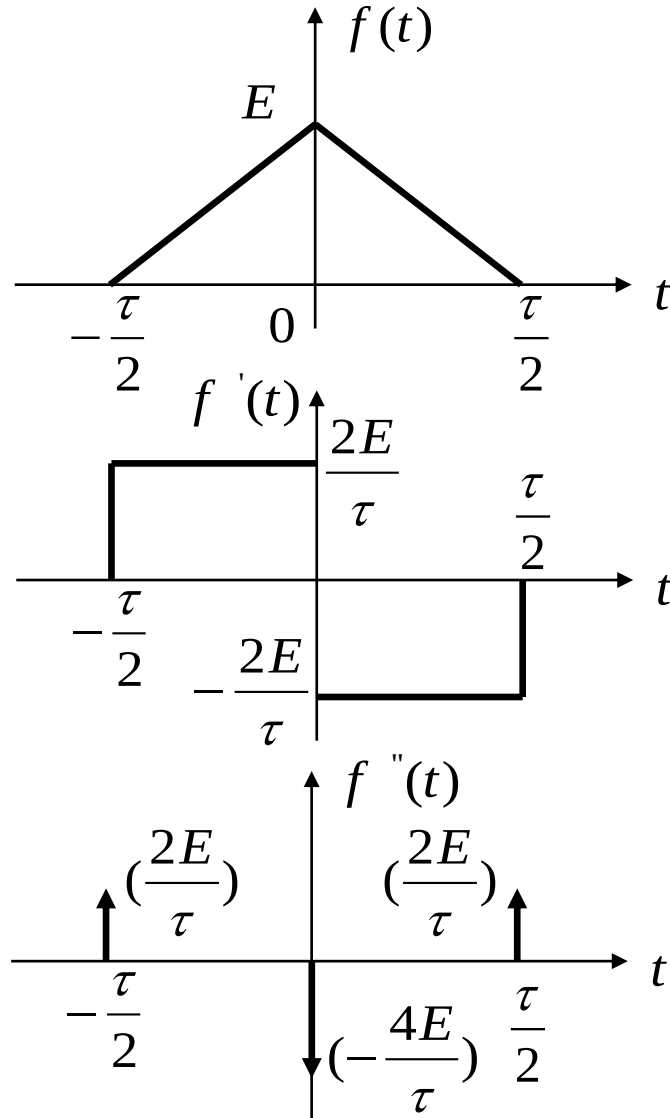
Step 2: Apply **the Time-domain Convolution Theorem**.

$$G(\omega) = \sqrt{\frac{2E}{\tau}} \cdot \frac{\tau}{2} \text{Sa}\left(\frac{\omega\tau}{4}\right) = \sqrt{\frac{E\tau}{2}} \text{Sa}\left(\frac{\omega\tau}{4}\right) \quad \therefore F(\omega) = G^2(\omega) = \frac{E\tau}{2} \text{Sa}^2\left(\frac{\omega\tau}{4}\right)$$



3.8 Convolution Theorem

Review **Example 3-6:** Use **the Time-domain Differentiation Property** to find the Fourier Transform of the Triangular Pulse Signal.

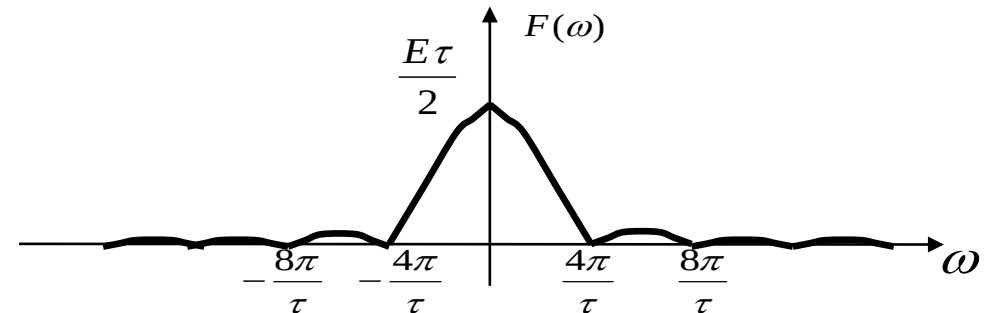


Solution: $f''(t) = \frac{2E}{\tau} \left[\delta\left(t + \frac{\tau}{2}\right) + \delta\left(t - \frac{\tau}{2}\right) - 2\delta(t) \right]$

Take the Fourier Transform on both sides of the above equation:

$$\begin{aligned} (j\omega)^2 F(\omega) &= \frac{2E}{\tau} (e^{j\omega\frac{\tau}{2}} + e^{-j\omega\frac{\tau}{2}} - 2) \\ &= -\frac{\omega^2 E \tau}{2} \text{Sa}^2\left(\frac{\omega\tau}{4}\right) \end{aligned}$$

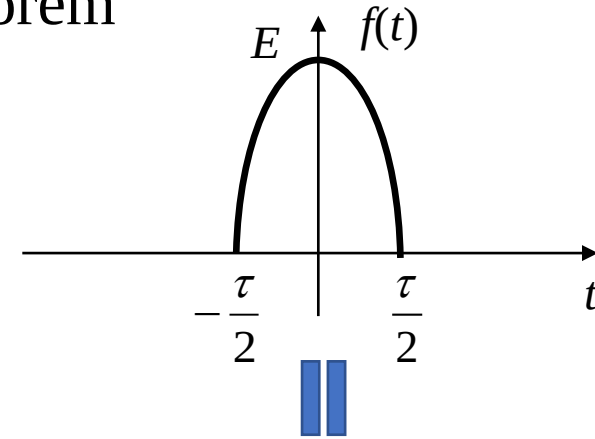
$$\therefore F(\omega) = \frac{E\tau}{2} \text{Sa}^2\left(\frac{\omega\tau}{4}\right)$$



3.8 Convolution Theorem

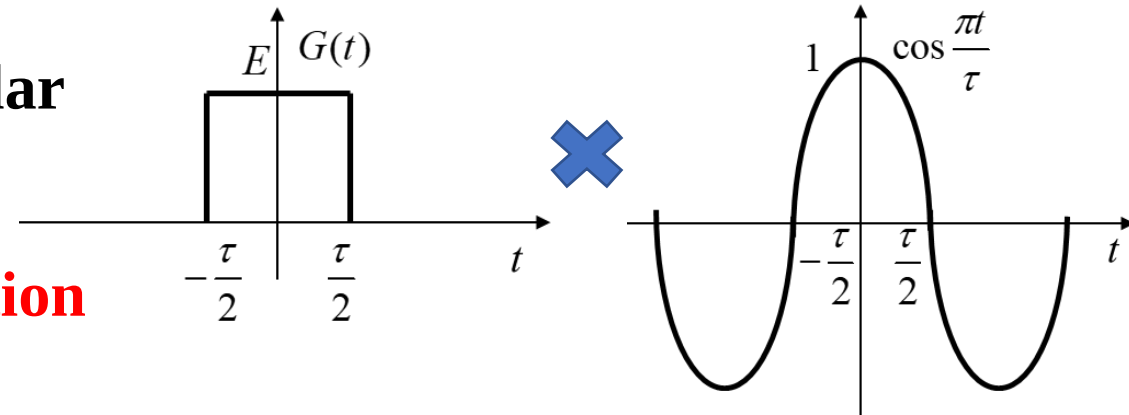
Example 3-7: Use the Frequency-domain Convolution Theorem to Find the Spectrum of a Cosine Pulse.

$$f(t) = \begin{cases} E \cos \frac{\pi t}{\tau} & |t| \leq \frac{\tau}{2} \\ 0 & |t| > \frac{\tau}{2} \end{cases}$$



Solution:

Step 1: Treat $f(t)$ as the **product** of a rectangular pulse $G(t)$ and a cosine signal.



Step 2: Apply the **Frequency-domain Convolution Theorem**.

$$G(\omega) = E\tau \text{Sa}\left(\frac{\omega\tau}{2}\right)$$

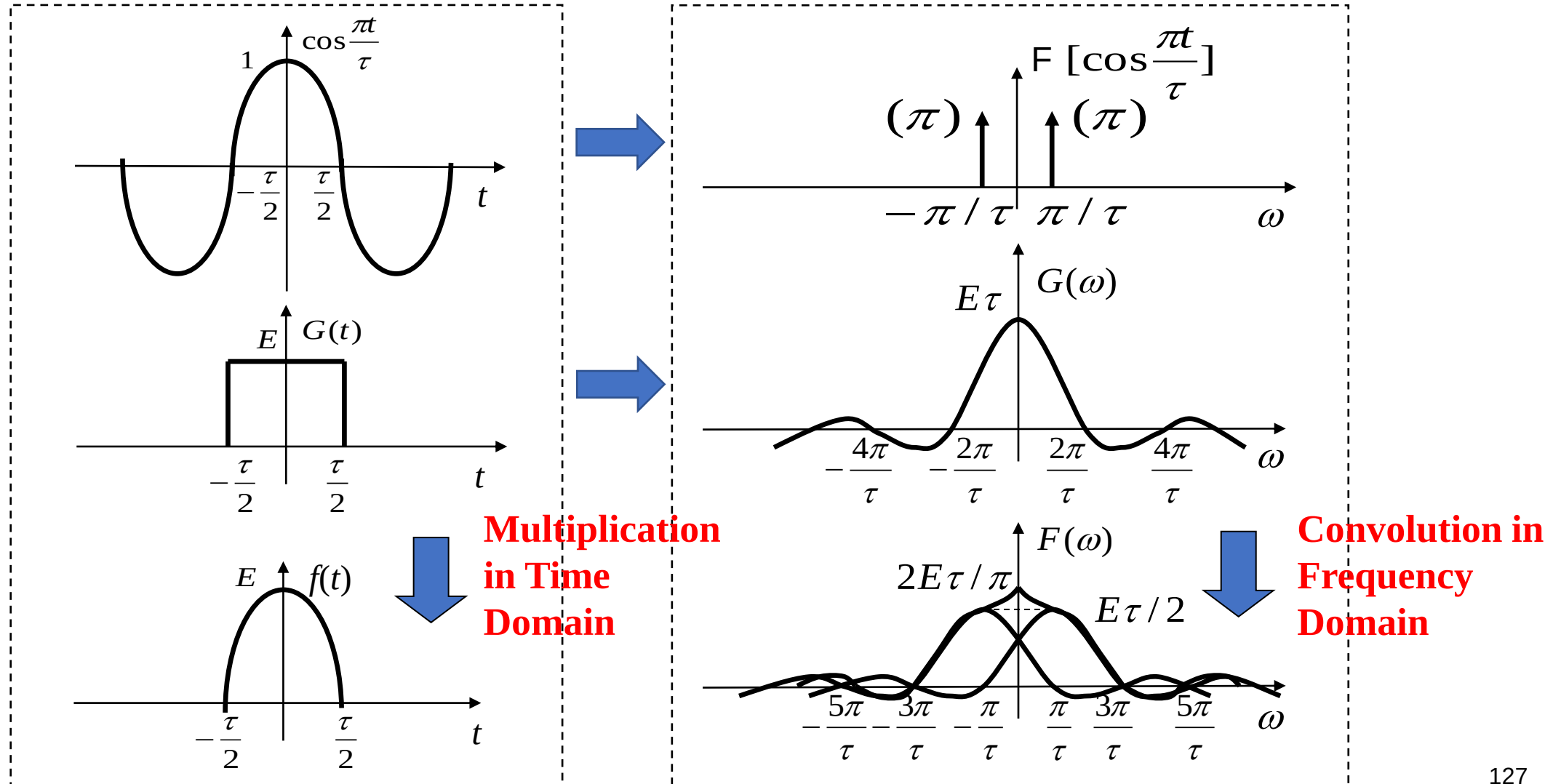
$$\mathcal{F}\left[\cos \frac{\pi t}{\tau}\right] = \pi\delta\left(\omega + \frac{\pi}{\tau}\right) + \pi\delta\left(\omega - \frac{\pi}{\tau}\right)$$

$$F(\omega) = \frac{1}{2\pi} G(\omega) * \pi\left[\delta\left(\omega + \frac{\pi}{\tau}\right) + \delta\left(\omega - \frac{\pi}{\tau}\right)\right] = \frac{E\tau}{2} \left\{ \text{Sa}\left[\left(\omega + \frac{\pi}{\tau}\right)\frac{\tau}{2}\right] + \text{Sa}\left[\left(\omega - \frac{\pi}{\tau}\right)\frac{\tau}{2}\right] \right\}$$

3.8 Convolution Theorem



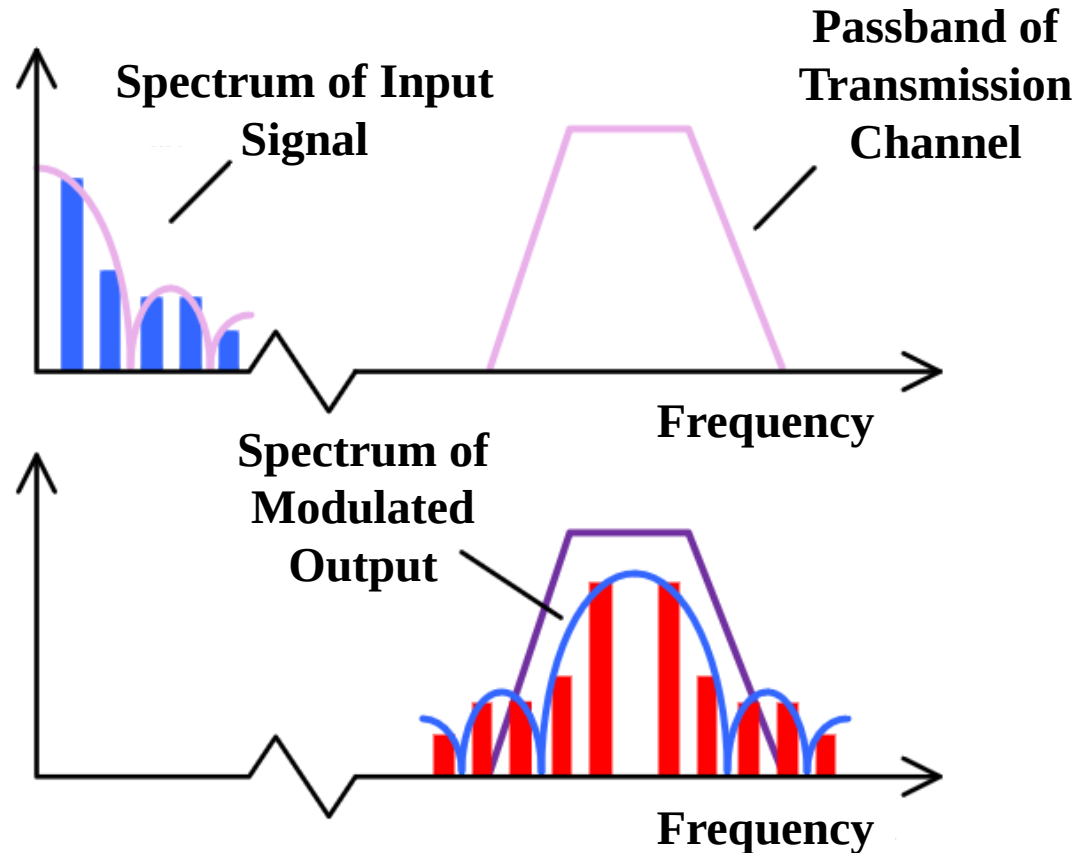
$$F(\omega) = \frac{E\tau}{2} \left\{ \text{Sa}\left[\left(\omega + \frac{\pi}{\tau}\right)\frac{\tau}{2}\right] + \text{Sa}\left[\left(\omega - \frac{\pi}{\tau}\right)\frac{\tau}{2}\right] \right\} = 2E\tau \cos\left(\frac{\omega\tau}{2}\right) \bigg/ \left[\pi \left(1 - \frac{\tau^2\omega^2}{\pi^2} \right) \right]$$



3.8 Convolution Theorem



Application of the Time-domain Convolution Theorem in Communications



$$x(t) * h(t) \xrightarrow{F} X(\omega)H(\omega)$$

Time-domain Convolution Theorem

- The channel bandwidth is limited.
- If some main frequency components of the signal fall outside the channel bandwidth, the signal output by the channel will be distorted.

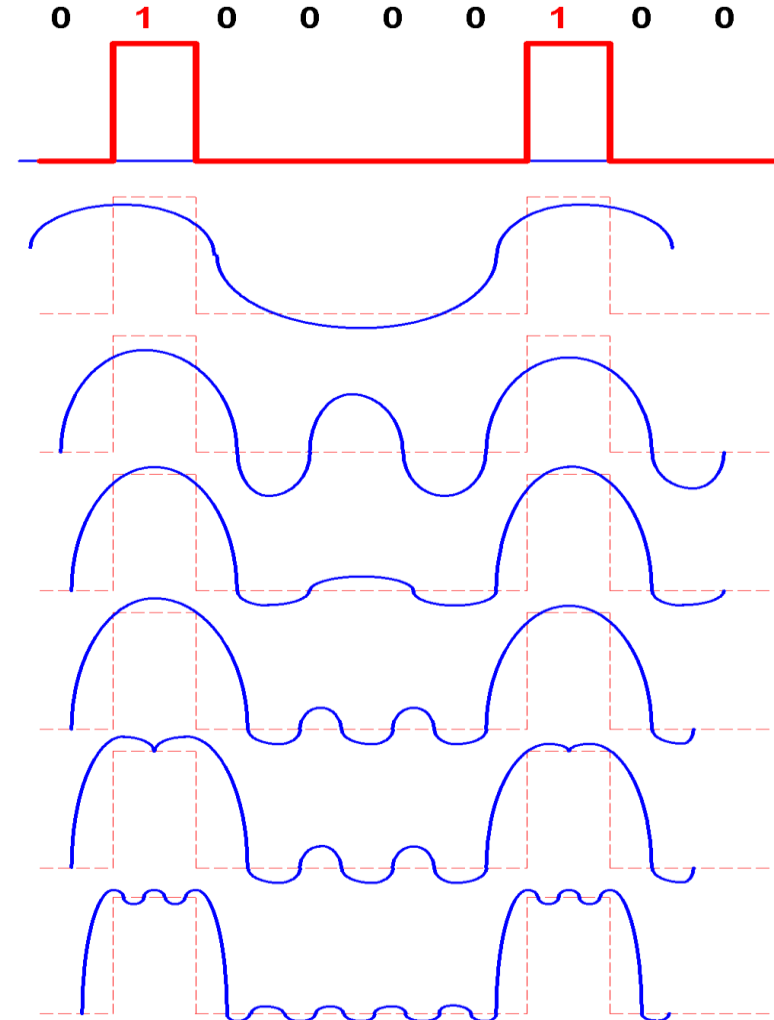
3.8 Convolution Theorem



Application of the Time-domain Convolution Theorem in Communications

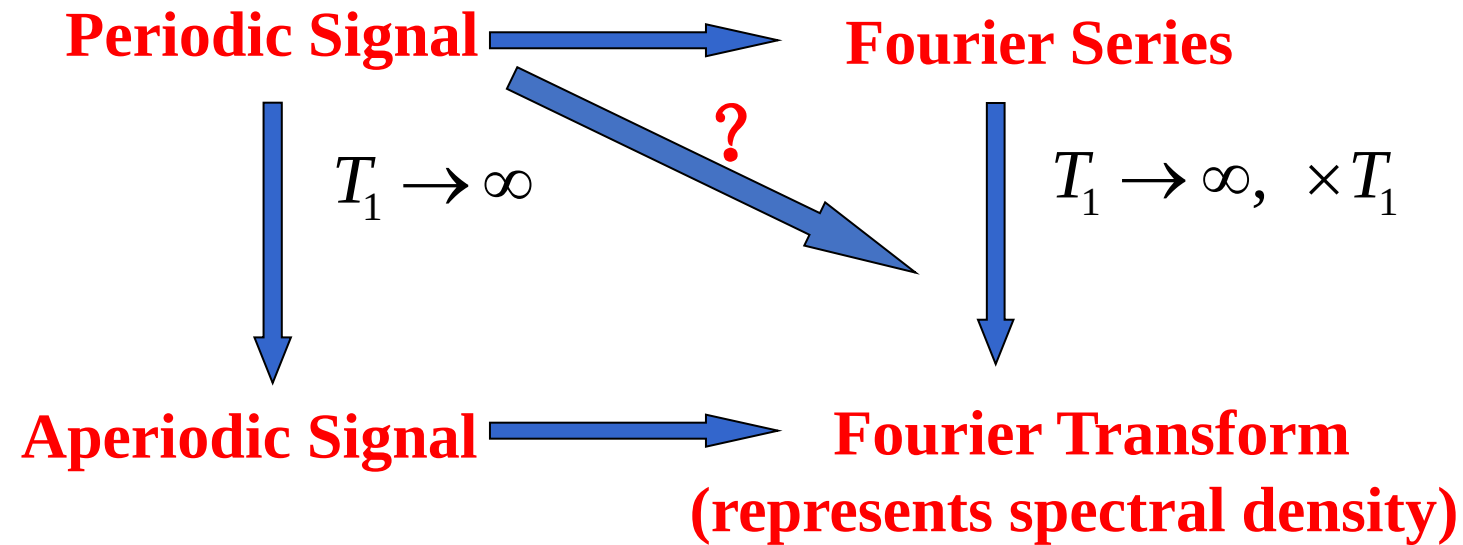
- Signal bandwidth:
2000 Hz

→ Bit rate:
2000 bits/s



- Output signals of the channel under different bandwidths:

3.9 Fourier Transform of Periodic Signals



Try to find the Fourier Transform of the signal $f(t) = 1 + 2\cos t$

- ☐ A $F(\omega) = \delta(\omega) + 2\pi[\delta(\omega - 1) + \delta(\omega + 1)]$
- ☐ B $F(\omega) = 2\pi\delta(\omega) + \delta(\omega - 1) + \delta(\omega + 1)$
- ☐ C $F(\omega) = \delta(\omega) + \delta(\omega - 1) + \delta(\omega + 1)$
- ☒ D $F(\omega) = 2\pi[\delta(\omega) + \delta(\omega - 1) + \delta(\omega + 1)]$

提交

3.9 Fourier Transform of Periodic Signals



3.9.1 Fourier Transform of Sine and Cosine Signals

$$\cos(\omega_0 t) = \frac{1}{2} (e^{j\omega_0 t} + e^{-j\omega_0 t})$$

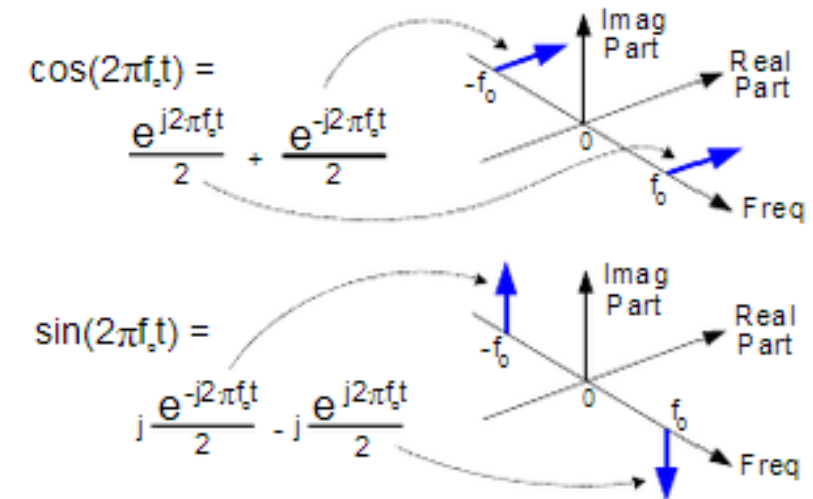
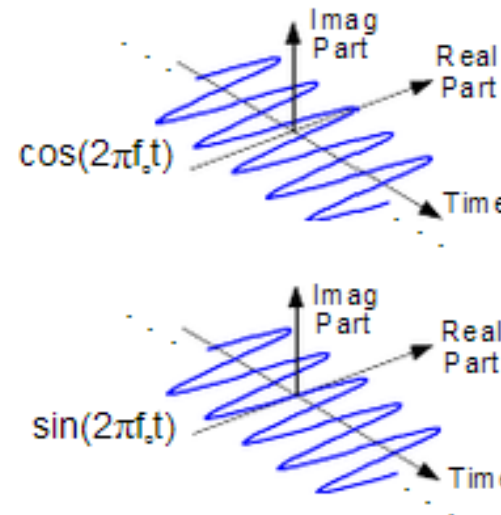
$$\sin(\omega_0 t) = \frac{1}{2j} (e^{j\omega_0 t} - e^{-j\omega_0 t})$$

$$F[e^{j\omega_0 t}] = 2\pi\delta(\omega - \omega_0)$$

$$F[e^{-j\omega_0 t}] = 2\pi\delta(\omega + \omega_0)$$

$$F[\cos \omega_0 t] = \pi[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$$

$$F[\sin \omega_0 t] = j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$$



3.9 Fourier Transform of Periodic Signals



3.9.2 Fourier Transform of General Periodic Signals

Let the period of the periodic signal $f(t)$ be T_1 , and the angular frequency be $\omega_1 = 2\pi f_1 = 2\pi/T_1$. Its Fourier series is:

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t} \quad (1)$$

where: $F_n = \frac{1}{T_1} \int_{-T_1/2}^{T_1/2} f(t) e^{-jn\omega_1 t} dt$

Take the Fourier transform on both sides of equation (1):

$$F(\omega) = \mathcal{F}[f(t)] = \mathcal{F}\left[\sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}\right] = \sum_{n=-\infty}^{\infty} F_n \mathcal{F}[e^{jn\omega_1 t}] = 2\pi \sum_{n=-\infty}^{\infty} F_n \delta(\omega - n\omega_1)$$

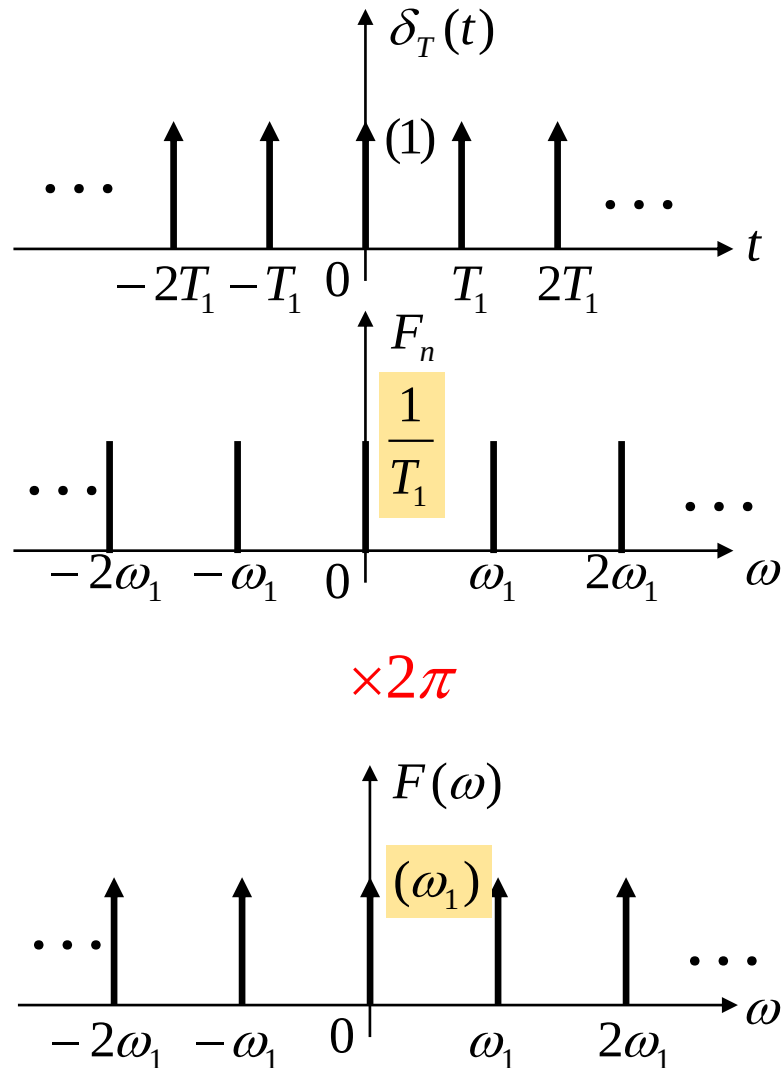
that is, $F(\omega) = 2\pi \sum_{n=-\infty}^{\infty} F_n \delta(\omega - n\omega_1)$

Using the Frequency-shift Property

The Fourier transform of the periodic signal $f(t)$ is composed of a series of impulse functions, which are located at the harmonic frequencies of the signal ($0, \pm\omega_1, \pm2\omega_1, \dots$). The strength of each impulse is equal to 2π times the corresponding coefficient F_n of the Fourier series of $f(t)$.

3.9 Fourier Transform of Periodic Signals

Example 3-9: Find the Fourier Transform of the Periodic Unit Impulse Sequence.



$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_1)$$

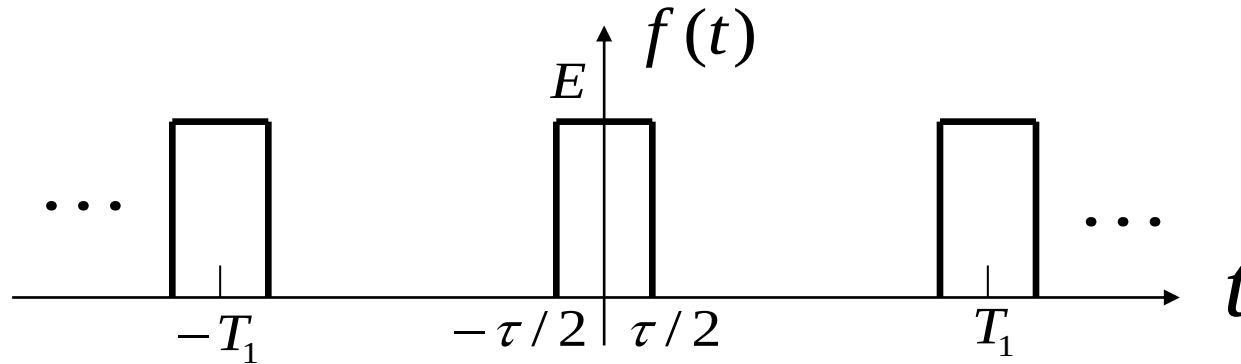
$$F_n = \frac{1}{T_1} \int_{-T_1/2}^{T_1/2} \delta(t) e^{-jn\omega_1 t} dt = \frac{1}{T_1} \int_{-T_1/2}^{T_1/2} \delta(t) dt = \frac{1}{T_1}$$

$$\delta_T(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t} = \frac{1}{T_1} \sum_{n=-\infty}^{\infty} e^{jn\omega_1 t}$$

$$F(\omega) = \frac{2\pi}{T_1} \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_1) = \omega_1 \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_1) \quad (2)$$

3.9 Fourier Transform of Periodic Signals

Example 3-10: Find the Fourier Series and Fourier Transform of the Periodic Rectangular Pulse Signal. Given that the periodic rectangular pulse signal $f(t)$ has an amplitude E , pulse width τ , period T_1 , and angular frequency $\omega_1 = 2\pi/T_1$.



Solution: The complex amplitude of the Fourier series is:

$$F_n = \frac{E\tau}{T_1} \text{Sa}\left(\frac{n\omega_1\tau}{2}\right)$$

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t} = \frac{E\tau}{T_1} \sum_{n=-\infty}^{\infty} \text{Sa}\left(\frac{n\omega_1\tau}{2}\right) e^{jn\omega_1 t}$$

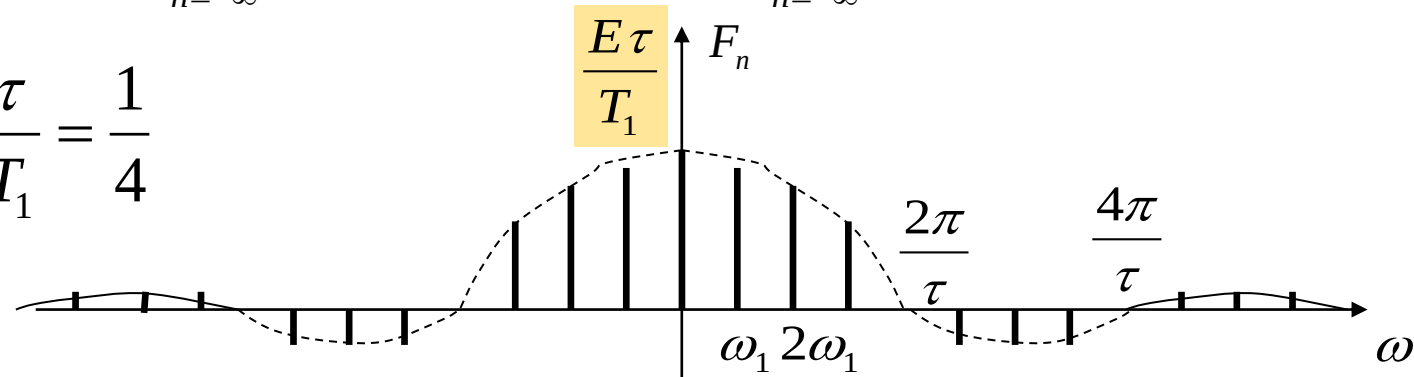
3.9 Fourier Transform of Periodic Signals



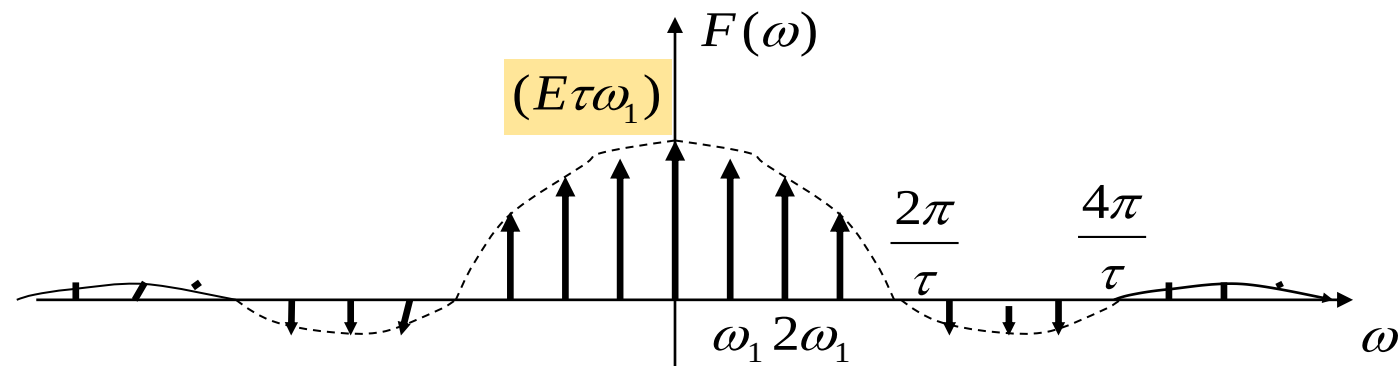
$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t} = \frac{E\tau}{T_1} \sum_{n=-\infty}^{\infty} \text{Sa}\left(\frac{n\omega_1\tau}{2}\right) e^{jn\omega_1 t}$$

$$F(\omega) = 2\pi \sum_{n=-\infty}^{\infty} F_n \delta(\omega - n\omega_1) = E\tau\omega_1 \sum_{n=-\infty}^{\infty} \text{Sa}\left(\frac{n\omega_1\tau}{2}\right) \delta(\omega - n\omega_1)$$

E.g., $\frac{\tau}{T_1} = \frac{1}{4}$



$\times 2\pi$

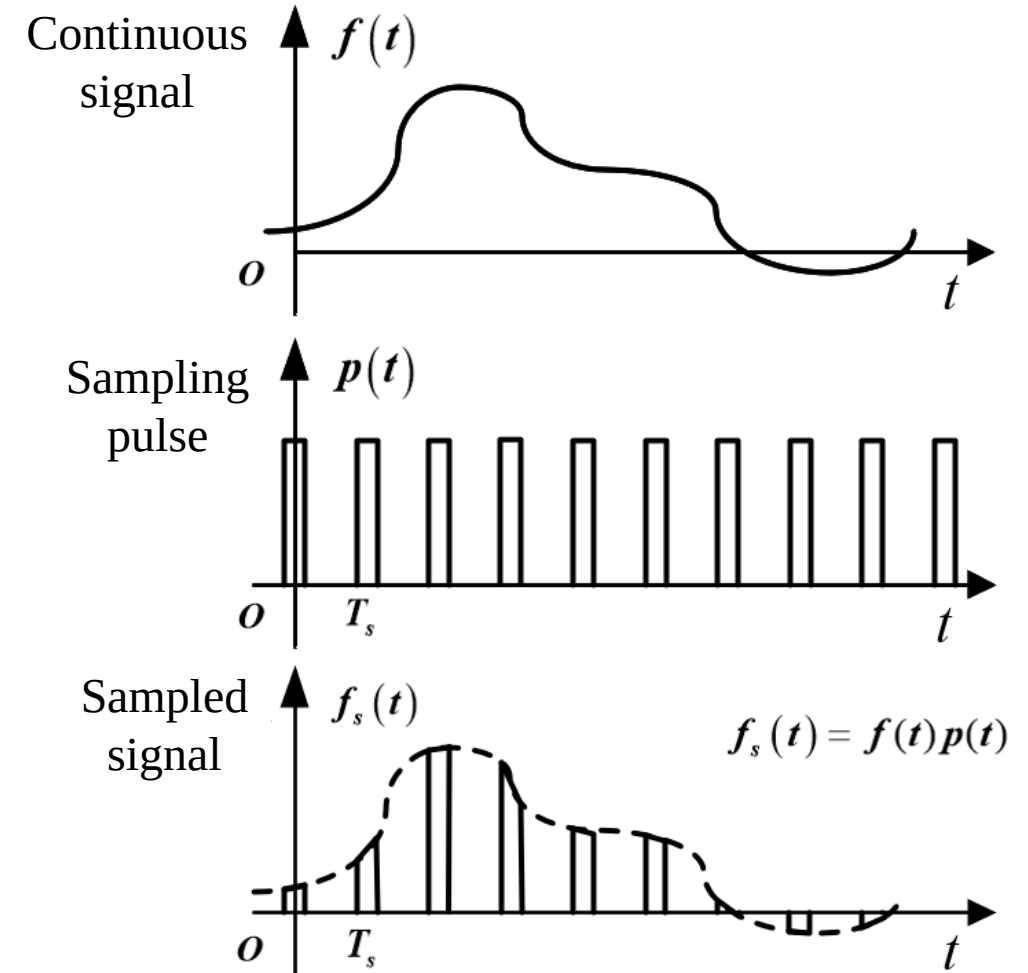
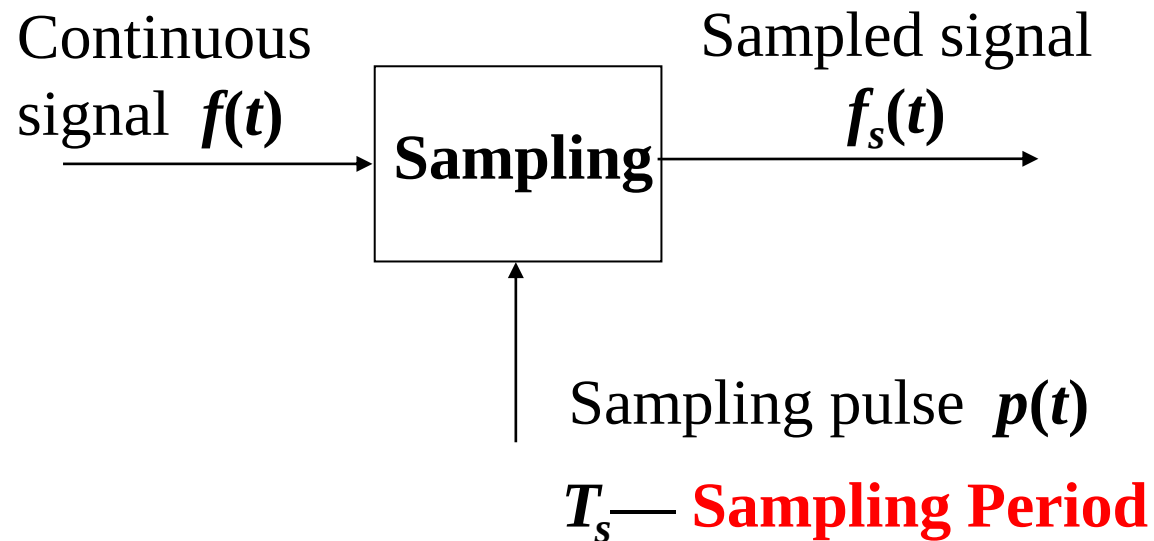


3.10 Fourier Transform of Sampled Signals

3.10.1 Sampling of Signals

Sampling -- Use a sampling pulse sequence to "sample" a series of discrete samples from a continuous signal.

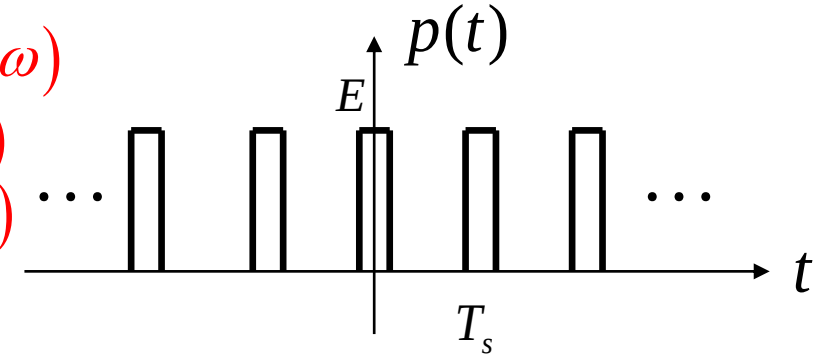
This discrete signal is called a **sampled signal**.



3.10 Fourier Transform of Sampled Signals

3.10.2 Fourier Transform of Sampled Signals

Let the Fourier transform of the continuous signal $f(t)$ be $F(\omega)$
 the Fourier transform of the sampling pulse $p(t)$ be $P(\omega)$
 the Fourier transform of the sampled signal $f_s(t)$ be $F_s(\omega)$



$$P(\omega) = 2\pi \sum_{n=-\infty}^{\infty} P_n \delta(\omega - n\omega_s)$$

where: $P_n = \frac{1}{T_s} \int_{-\frac{T_s}{2}}^{\frac{T_s}{2}} p(t) e^{-jn\omega_s t} dt$

$$f_s(t) = f(t)p(t)$$

$$F_s(\omega) = \frac{1}{2\pi} F(\omega) * P(\omega) = \frac{1}{2\pi} F(\omega) * 2\pi \sum_{n=-\infty}^{\infty} P_n \delta(\omega - n\omega_s)$$

so $F_s(\omega) = \sum_{n=-\infty}^{\infty} P_n \cdot F[(\omega - n\omega_s)]$

$$\omega_s = 2\pi f_s = \frac{2\pi}{T_s}$$

Sampling Angular Frequency **Sampling Frequency**

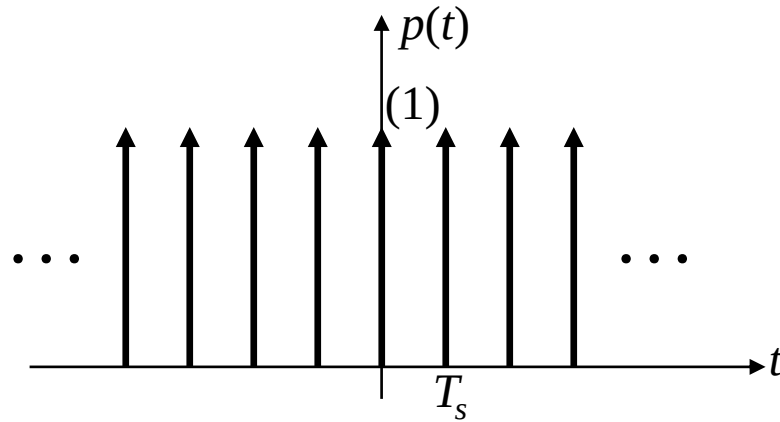
Frequency-domain Convolution Theorem

3.10 Fourier Transform of Sampled Signals



Impulse Sampling

If the sampling pulse $p(t)$ is an impulse sequence, this is called "**impulse sampling**" or "**ideal sampling**".



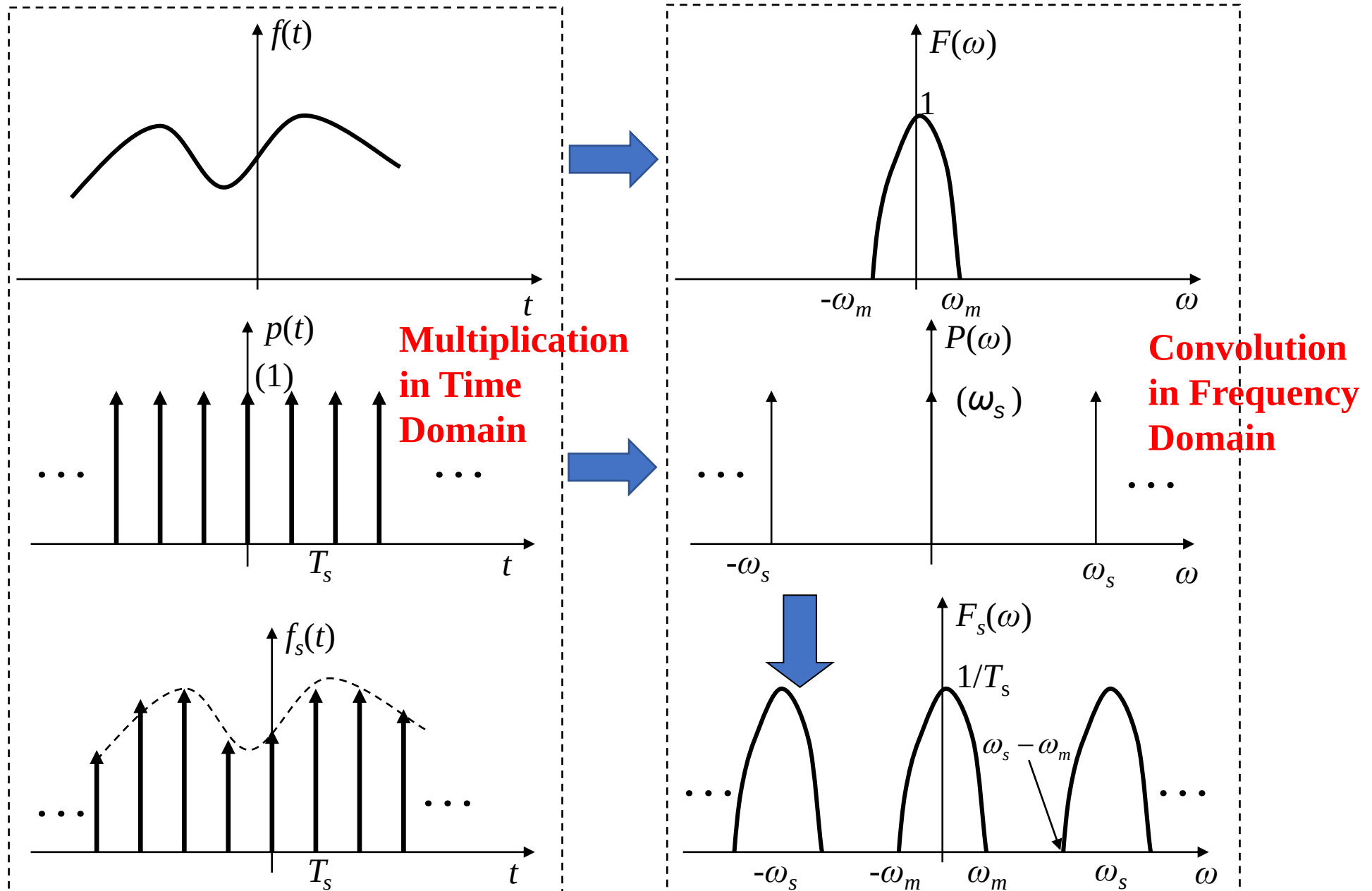
$$p(t) = \delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

$$F_s(\omega) = \sum_{n=-\infty}^{\infty} P_n \cdot F(\omega - n\omega_s)$$

$$= \frac{1}{T_s} \sum_{n=-\infty}^{\infty} F(\omega - n\omega_s) \quad \text{Periodically extended in the frequency domain with period } \omega_s$$

$$\text{where } P_n = \frac{1}{T_s} \int_{-\frac{T_s}{2}}^{\frac{T_s}{2}} \delta(t) e^{-jn\omega_s t} dt = \frac{1}{T_s}$$

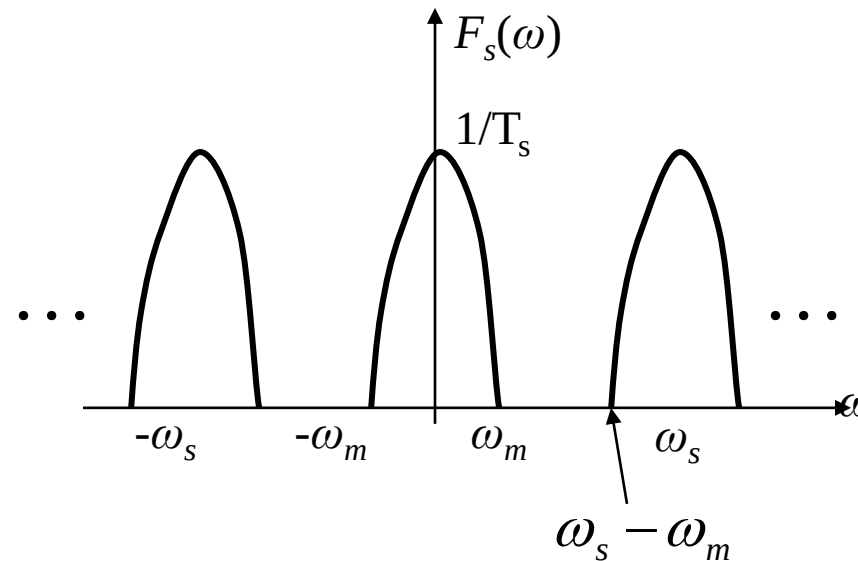
3.10 Fourier Transform of Sampled Signals



3.11 Sampling Theorem



When sampling a continuous signal with sampling pulses, **how to select the sampling frequency and sampling period?** And **how to recover the original continuous signal from the sampled signal?**



It can be seen from the above figure that only when $\omega_s \geq 2\omega_m$ is satisfied, will $F_s(\omega)$ not produce spectral aliasing, that is, $f_s(t)$ retains all the information of the original continuous-time signal.

At this time, as long as $f_s(t)$ is applied to the **ideal low-pass filter**, the original signal $f(t)$ can be recovered.

3.11 Sampling Theorem

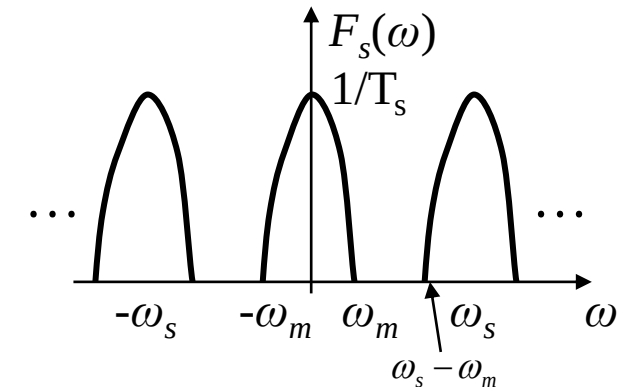
Time-domain Sampling Theorem: A band-limited signal $f(t)$, if its spectrum is confined to the range $-\omega_m \sim \omega_m$, can be uniquely represented by its equally spaced samples when the sampling interval is no larger than $\frac{1}{2f_m}$ (where $\omega_m = 2\pi f_m$) or the sampling frequency is greater than or equal to $2f_m$.

Nyquist Sampling Frequency: The minimum allowable sampling frequency.

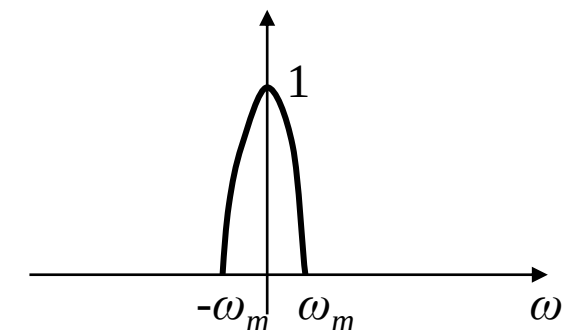
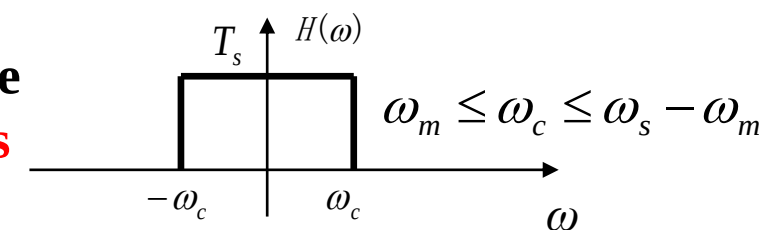
$$f_{s \min} = 2 f_m$$

Nyquist Sampling Interval: The maximum allowable sampling interval.

$$T_{s \max} = \frac{1}{f_{s \min}} = \frac{1}{2 f_m}$$



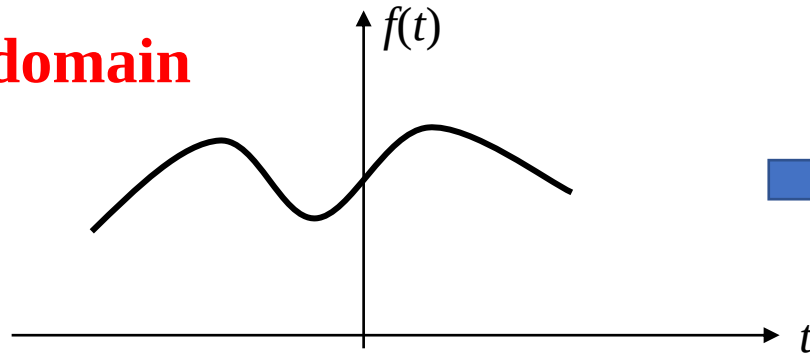
Frequency Response of the Ideal Low-pass Filter



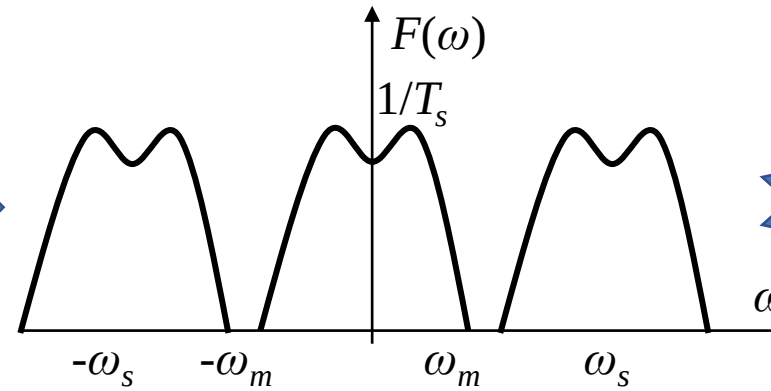
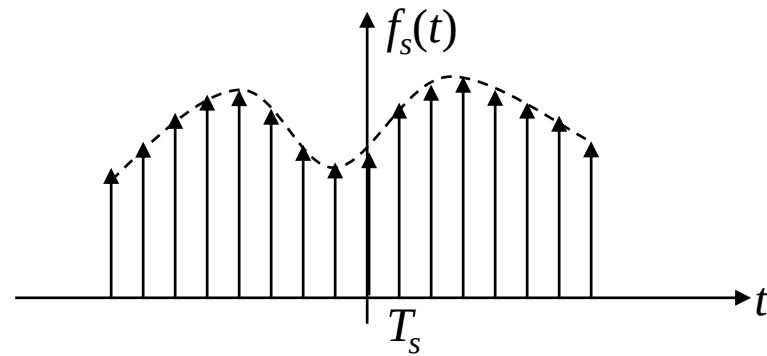
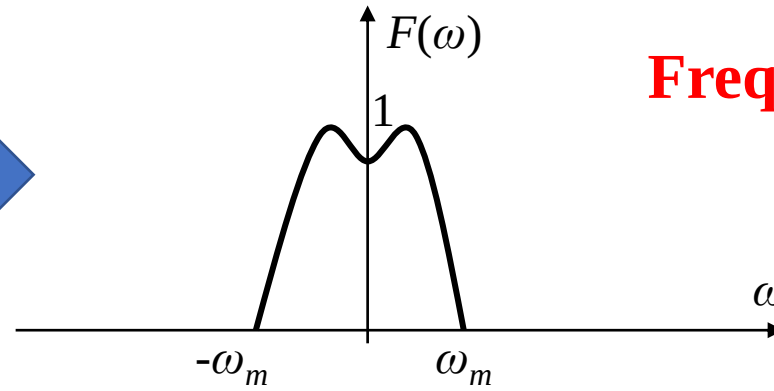
3.11 Sampling Theorem



Time domain

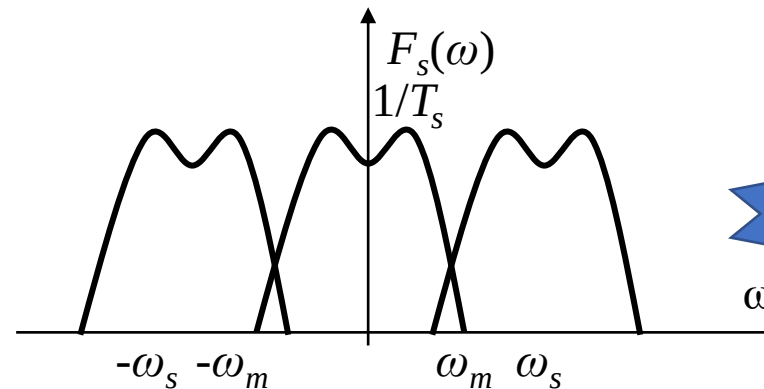
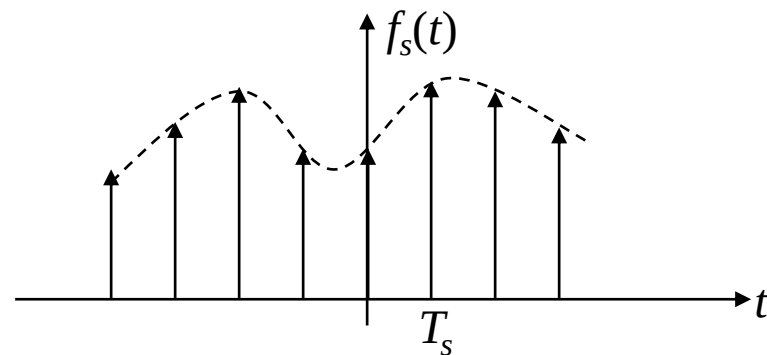


Frequency domain



$$\omega_s > 2\omega_m$$

Oversampling



$$\omega_s < 2\omega_m$$

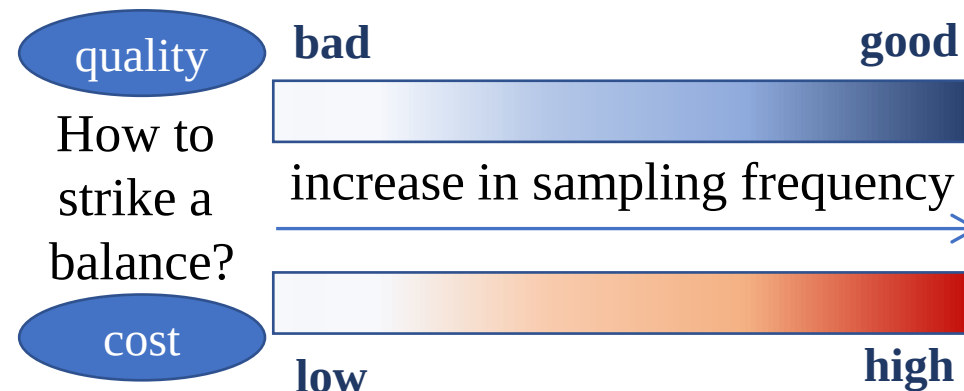
Undersampling

3.11 Sampling Theorem



Example 3-11: CDs usually adopt a sampling frequency of 44 kHz. Please explain the reason. Is a higher sampling frequency always better?

- Most human ears can perceive frequencies ranging from 20 Hz to 16 kHz. To test your audible frequency range, you can visit: <https://www.bilibili.com/video/av40724545/>
- The corresponding Nyquist sampling frequency is 32 kHz.
- A pre-alias filter can be used to remove frequency components above 16 kHz.
- For high-quality audio signals, we generally use sampling frequencies higher than the Nyquist sampling frequency, such as 44 kHz, 48 kHz, 96 kHz, or 128 kHz.
- The selection of sampling frequency needs to strike a balance between sound quality and cost.



Selection of Sampling Frequency in Engineering

- If the highest frequency ω_m of the signal is known, generally take $\omega_s \geq (5 \sim 10)\omega_m$
- If the bandwidth ω_B of the signal is known, generally take $\omega_s \geq (5 \sim 10)\omega_B$
- If the highest frequency of the interference signal is ω_f , it should satisfy $\omega_s \geq 2\omega_f$

Empirical Data of Sampling Period in Industrial Control Systems:

Flow signal: $T_s = 1 \sim 5$ seconds;

Pressure signal: $T_s = 3 \sim 10$ seconds;

Liquid level signal: $T_s = 6 \sim 8$ seconds ; Temperature signal: $T_s = 15 \sim 20$ seconds

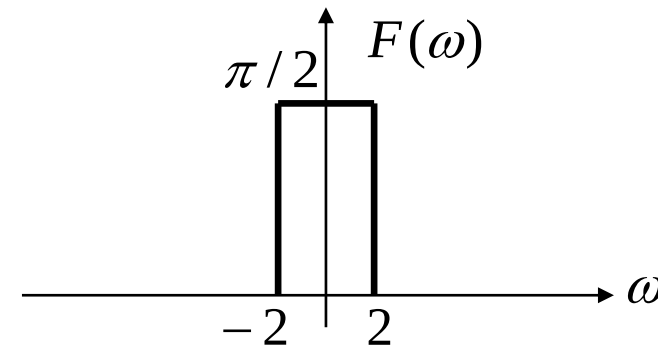
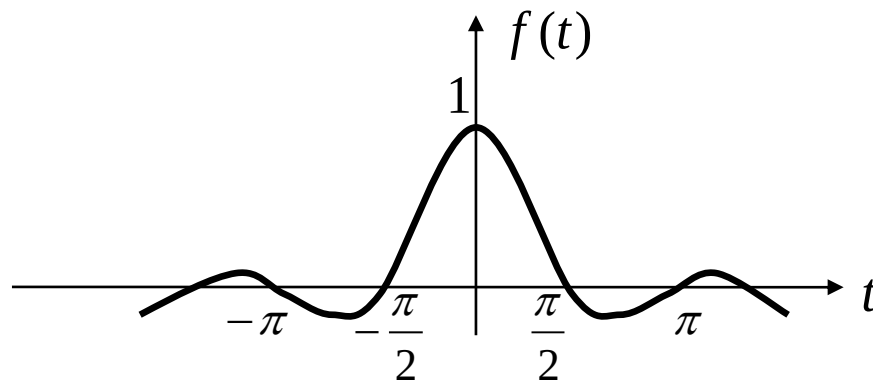
Component signal: $T_s = 15 \sim 20$ seconds

Frequency-domain Sampling Theorem: If the signal $f(t)$ is a time-limited signal concentrated in the time range $-t_m \sim +t_m$, and if its spectrum $F_1(\omega)$ is sampled in the frequency domain with a frequency interval no larger than $\frac{1}{2t_m}$, then the sampled spectrum $F_s(\omega)$ can uniquely represent the original signal.

3.11 Sampling Theorem

- Example 3-12:** Given the signal $f(t) = \text{Sa}(2t)$, sample it with $\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$,
- (1) Determine the Nyquist sampling frequency;
 - (2) If $\omega_s = 6\omega_m$, find the sampled signal $f_s(t) = f(t)\delta_T(t)$, and draw its waveform;
 - (3) Find $F_s(\omega) = \mathcal{F}[f_s(t)]$, and draw its spectrum;
 - (4) Determine the cutoff frequency ω_c of the low-pass filter.

Solution: (1) $\because f(t) = \text{Sa}(2t) \quad \therefore F(\omega) = \frac{\pi}{2} [u(\omega + 2) - u(\omega - 2)]$



Nyquist sampling frequency:

$$f_{s \text{ min}} = 2 f_m = 2 \times \frac{2}{2\pi} = \frac{2}{\pi} \text{ Hz}$$

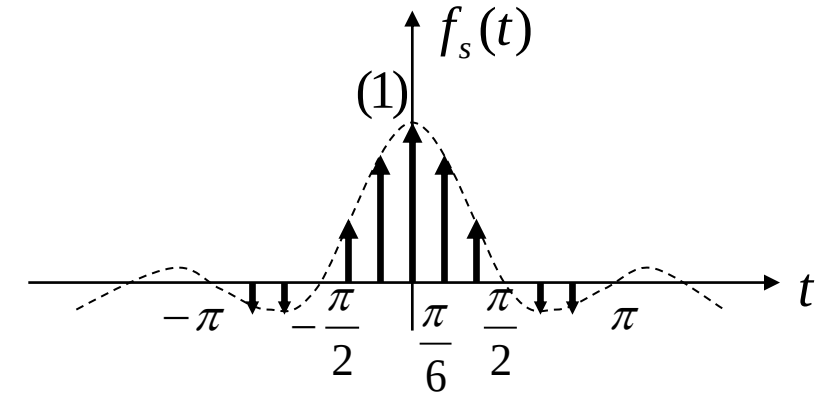
3.11 Sampling Theorem



$$(2) \because \omega_s = 6\omega_m = 12 \text{ rad/s} \quad \therefore T_s = \frac{2\pi}{\omega_s} = \frac{2\pi}{12} = \frac{\pi}{6} \text{ s}$$

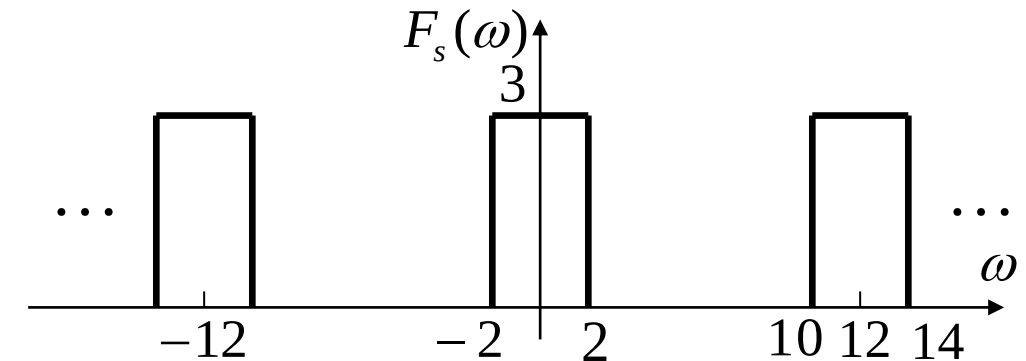
$$f_s(t) = f(t)\delta_T(t) = \sum_{n=-\infty}^{\infty} f(nT_s)\delta(t - nT_s)$$

$$= \sum_{n=-\infty}^{\infty} \text{Sa}(2t)\big|_{t=nT_s} \delta(t - nT_s) = \sum_{n=-\infty}^{\infty} \text{Sa}\left(\frac{n\pi}{3}\right) \delta(t - nT_s)$$



$$(3) \quad F_s(\omega) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} F(\omega - n\omega_s) = \frac{6}{\pi} \sum_{n=-\infty}^{\infty} F(\omega - 12n)$$

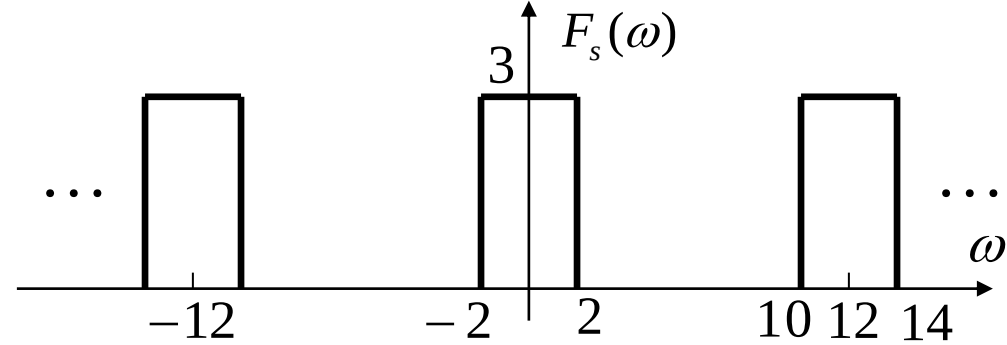
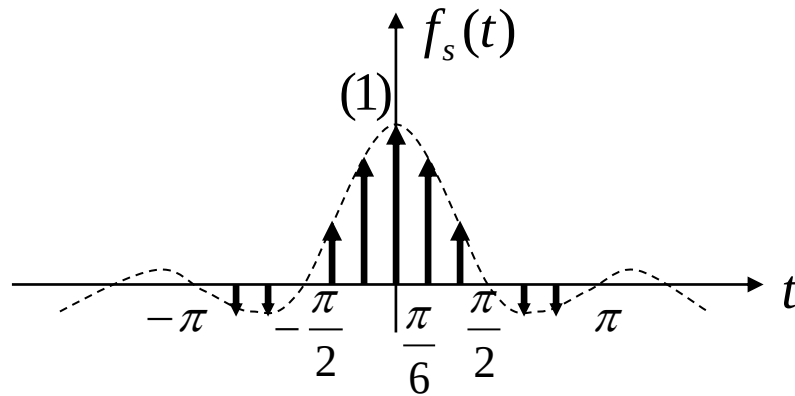
$$= 3 \sum_{n=-\infty}^{\infty} [u(\omega + 2 - 12n) - u(\omega - 2 - 12n)]$$



3.11 Sampling Theorem



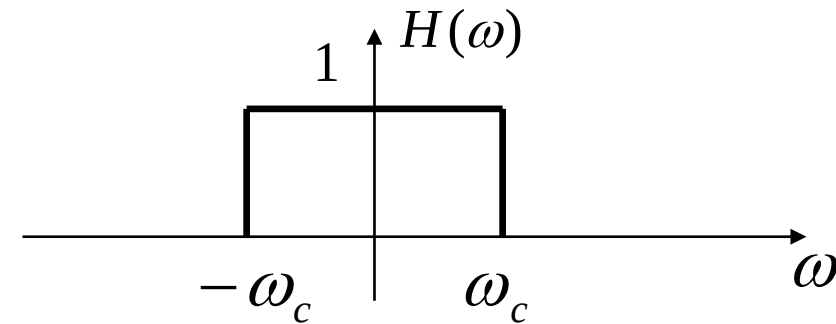
(4)



The cutoff frequency ω_c of the low-pass filter should satisfy the following formula:

$$\omega_m \leq \omega_c \leq \omega_s - \omega_m$$

then $2 \leq \omega_c \leq 10$



- Repeat the examples in the lecture notes.
- 3.58(g), 3.59(f), 4.16(d) and 4.29(a) in the textbook.